SPATIAL AND TEMPORAL PATTERNS OF TREE REGENERATION IN THE ACADIAN FOREST OF CENTRAL MAINE

By

Matthew G. Olson

A.A.S. Paul Smiths College, 1999

B.S. University of Maine, 2001

M.S. University of Tennessee, 2003

A THESIS

Submitted in Partial Fulfillment of the

Requirements for the Degree of

Doctor of Philosophy

(in Forest Resources)

The Graduate School

University of Maine

December 2009

Advisory Committee:

Robert G. Wagner, Henry W. Saunders Distinguished Professor in Forestry, Advisor

Jeremy A. Wilson, Irving Chair for Forest Ecosystem Management

William A. Halteman, Professor of Statistics

Michael R. Saunders, Assistant Professor of Silviculture, Purdue University

John C. Brissette, Research Forester and Research Work Unit Leader, Northern Research Station, USDA Forest Service

DISSERTATION

ACCEPTANCE STATEMENT

On behalf of the Graduate Committee for _____, I affirm that this manuscript is the final and accepted thesis. Signatures of all committee members are on file with the Graduate School at the University of Maine, 42 Stodder Hall, Orono, Maine.

_____

Committee chair's signature, name, and title (Date)

LIBRARY RIGHTS STATEMENT

In presenting this thesis in partial fulfillment of the requirements for an advanced degree at the University of Maine, I agree that the Library shall make it freely available for inspection. I further agree that permission for "fair use" copying of this thesis for scholarly purposes may be granted by the Librarian. It is understood that any copying or publication of this thesis for financial gain shall not be allowed without my written permission.

Signature:

Date:

SPATIAL AND TEMPORAL PATTERNS OF TREE REGENERATION IN THE ACADIAN FOREST OF CENTRAL MAINE

By Matthew G. Olson

Thesis Advisor: Dr. Robert G. Wagner

An Abstract of the Thesis Presented
in Partial Fulfillment of the Requirements for the
Degree of Doctor of Philosophy
(in Forest Resources)
December 2009

Relatively few studies of forest regeneration have attempted to integrate the analysis of spatial and temporal pattern. In order to fill this gap in our understanding of forest regeneration, I used data from two long-term silviculture studies on the Penobscot Experimental Forest (PEF) in Bradley, ME. Multivariate analysis was used to retrace compositional development of permanent sample plots in a USDA Forest Service silviculture experiment over five decades (1953-2005) based on both pre-treatment forest composition (i.e., plot-level tree composition when the experiment was established) and harvest disturbance history. This study revealed that different harvest treatments initiated an array of developmental pathways at the plot level that were linked to both factors. Hardwood dominance increased in plots with a history of heavy and infrequent cutting, while pre-treatment composition largely was maintained in plots where cutting was lighter and more frequent. Balsam fir dominance was maintained across a wide range of harvest intensities, which contrasted with the ubiquitous decline in cedar importance over the experiment. The Acadian Forest Ecosystem Research Program (AFERP) experiment

was used to evaluate spatial responses of forest regeneration to gap harvesting and variability in local biotic and abiotic conditions. First, spatial pattern analysis was used to describe the spatial patterns within stands containing harvest gaps of various sizes (0.01 to 0.22 ha) and over a range of spatial scales. This study indicated that stands that have received gap-harvest treatments have unique spatial patterns in understory tree density and canopy openness, and supported the hypothesis that gap harvesting created a coarse-grained spatial pattern in both factors. Second, spatial and non-spatial models were developed to identify plot-level factors affecting the species richness of tree regeneration. Negative associations were found between: 1) richness of small regeneration (≥ 0.1 m and < 0.75 m tall) and balsam fir regeneration density and 2) richness of large regeneration (≥ 0.75 m and < 1.4 m tall) and overstory basal area. Both small and large regeneration richness were positively associated with percent canopy openness. Collectively, the findings of this investigation highlighted the importance of local environmental and historical factors (i.e., neighborhood-scale factors) in determining the spatial and temporal patterns of natural regeneration in the Acadian Forest.

DEDICATION

To Liz, family, friends, and Hank Williams:

Thanks for everything

ACKNOWLEDGEMENTS

I would like to thank my advisor, Dr. Robert "Bob" Wagner, for providing insights on good science and thoughtful mentorship during my Ph.D. program. I would like to thank Dr. Mike Saunders, fellow AFERP guru and committee member, for both his time and intellectual investment in helping me develop as a project manager, Ph.D. student, and forest scientist. I also would like to thank my other committee members, Drs. Jeremy Wilson, Bill Halteman, and John Brissette, for their guidance while developing my Ph.D. project and their thoughtful reviews of my dissertation.

I would like to thank my field assistants, Griff Lambert (summer 2006), C.J. Kanozak (summer 2006), and Andrew Choptiany (summer 2007), for their dedication in helping me implement the field-based portion of this project and their overall intellectual curiosity in my study. I also would like to thank the 2005-2007 AFERP crews for their effort in maintaining our long-term experiments without which this project would never have been finished.

I also would like to thank Andrew Nelson for his help with AFERP experiments, Rick Morrill for his help with GIS, and Matt Russell for his help in acquiring data from the USFS silviculture experiment's long-term database.

Financial support for this project was provided by the Maine Agricultural and Forest Experiment Station and Northeastern States Research Cooperative.

TABLE OF CONTENTS

DEDICATION iii

ACKNOWLEDGEMENTS iv

LIST OF TABLES ix

LIST OF FIGURES xi

PROLOGUE xiii

Chapter

1. LONG-TERM COMPOSITIONAL DYNAMICS OF ACADIAN MIXEDWOODSTANDS UNDER DIFFERENT SILVICULTURAL REGIMES 1

1.1. Abstract 1

1.2. Introduction 2

1.3. Methods 4

1.3.1. Study Site 4

1.3.2 Long-Term Silviculture Experiment 6

1.3.3. Study Approach 7

1.3.4. Statistical Analysis 9

1.4. Results 11

1.4.1. Cluster Analysis and Classification 11

1.4.2. Basal Area and Density Development 13

1.4.3. Compositional Development 14

1.5. Discussion 20

2. EFFECT OF GAP HARVESTING ON UNDERSTORY SPATIAL PATTERN OF TREE REGENERATION AND CANOPY OPENNESS IN MIXEDWOOD STANDS OF CENTRAL MAINE 26

2.1. Abstract 26

2.2. Introduction 27

2.3. Methods 32

2.3.1. Study Site 32

2.3.2. Experimental Design 34

2.3.3. Understory Vegetation and Light Sampling 35

2.3.4. Variables and Spatial Analysis 37

2.4. Results 44

2.4.1. Non-Spatial Pattern 44

2.4.2. Univariate Spatial Pattern 45

2.4.3. Association Between Regeneration Density and Canopy Openness 51

2.5. Discussion 52

2.5.1. Non-Spatial Pattern 52

2.5.2. Treatment Effects on Univariate Spatial Patterns 53

2.5.3. Regeneration Density and Canopy Openness Associations 56

2.5.4. Understory Dynamics Inferred from Spatial Pattern 58

2.5.5. Overlay of Factors Structuring Spatial Pattern of Regeneration 61

2.5.6. Spatial Pattern Analysis in Silviculture Research 63

2.6. Conclusions 64

3. FACTORS AFFECTING SPECIES RICHNESS OF TREE REGENERATION IN MATURE, ACADIAN MIXEDWOODS OF CENTRAL MAINE 66

3.1. Abstract 66

3.2. Introduction 67

3.3. Methods 69

3.3.1. Study Site 69

3.3.2. Experimental Design 71

3.3.3. Vegetation and Light Sampling 72

3.3.4. Statistical Analysis 75

3.4. Results 78

3.4.1. Tree Regeneration Richness across Stands and Treatments 78

3.4.2. Models of Plot-Level Richness 80

3.5. Discussion 84

3.5.1. Experiment-Wide Patterns of Tree Regeneration Richness 84

3.5.2. Ecological Correlates of Tree Regeneration Richness 86

3.5.3. Influence of Correlates on Tree Recruitment and Diversity 90

3.6. Conclusions 91

EPILOGUE 92

Strengths and Limitations 93

Recommendations 97

Future Directions 99

BIBLIOGRAPHY 100

APPENDICES 111

Appendix A. Distribution of Sample Plots by HDC 112

Appendix B. Variography Results 113

Appendix C. Distribution of Balsam Fir Regeneration 115

Appendix D. Model Selection Results 116

Appendix E. Species List of Tree Regeneration 117

Appendix F. LME Slopes for Nested Stand-Level Models 118

BIOGRAPHY OF THE AUTHOR 120

LIST OF TABLES

Table 1.1. Comparison of pre-treatment clusters based on mean importance values of species.. 12

Table 1.2. Comparison of harvest disturbance clusters based on mean and median harvest disturbance variables.. 13

Table 1.3. Results of Mann-Whitney tests comparing median importance values at T0 and T5 by species for each sub-cluster..... 18

Table 2.1. Treatment comparison of mean density and stocking of regeneration and mean and coefficient of variation of canopy openness for the AFERP study. 44

Table 2.2. Treatment comparisons of mean structural strength of theoretical variogram models of regeneration density and canopy openness for the AFERP study.. 49

Table 2.3. Partial Mantel correlations and test results by stand and gap-harvest treatment.. 52

Table 3.1. Description of response and explanatory variables. 76

Table 3.2. Model details and slope estimates for closest approximating models of (a) small and (b) large regeneration richness. 81

Table A.1. Percent breakdown of harvest disturbance clusters according to plot membership by silvicultural treatments. 112

Table B.1. Comparison of theoretical variogram model fits estimated by weighted least squares for both tree regeneration density and canopy openness across eight stands of the AFERP study. 113

Table B.2. Nugget, sill, and range parameters from best fit theoretical variogram models estimated by weighted least squares for both tree regeneration density and canopy openness across eight stands. 114

Table C.1. Mean density and percent stocking of balsam fir regeneration centered on 7-quantile means of overstory basal area, depth-to-water table, and percent canopy transmittance. 115

Table D.1. AIC for models of (A) small and (B) large regeneration richness estimated by LME and OLS regression. 116

Table E.1. List of tree species tallied in understory plots along with stands and treatments where they were found for the AFERP study..... 117

LIST OF FIGURES

Figure 1.1. Location of long-term silviculture experiment on the Penobscot Experimental Forest (PEF) with location of the PEF in Maine. 5

Figure 1.2. Changes in mean basal area by species for nine sub-clusters over six time periods from 1953-2005. 15

Figure 1.3. Changes in mean density by species for nine sub-clusters over six time periods from 1953-2005. 16

Figure 1.4. Developmental pathways of nine sub-clusters in NMS space over six time periods from 1953-2005.. 17

Figure 1.5. Changes in mean importance values by species for nine sub-clusters over six time periods from 1953-2005. 19

Figure 2.1. Location of AFERP study on the Penobscot Experimental Forest (PEF) with location of PEF in Maine. 33

Figure 2.2. Depiction of randomly-oriented transects within AFERP RA4 and an illustration of the 4 of 13 RPCS design. 36

Figure 2.3. Spatial correlograms of regeneration density and canopy openness over a 93-meter extent in the understory of gap-harvest treatments of the AFERP study. 46

Figure 2.4. Interpolated maps of regeneration density generated by Ordinary Kriging for eight stands of the AFERP study..... 48

Figure 2.5. Mean ranges of spatial autocorrelation and semivariance for regeneration density and canopy openness along with mean canopy gap diameter. 50

Figure 3.1. Location of AFERP study on the Penobscot Experimental Forest (PEF) with location of PEF in Maine. 70

Figure 3.2. Depiction of randomly-oriented transects within AFERP RA4 and an illustration of the 4 of 13 RPCS design. 73

Figure 3.3. Treatment comparison of tree regeneration richness estimated by sample-based rarefaction 80

Figure F.1. Estimated slope coefficients ± two standard deviations from LME models presented over the treatment gradient of the AFERP study..... 119

PROLOGUE

Confusion exists concerning the term “forest regeneration”, since it is often used interchangeably when referring to both a local grouping of juvenile trees (e.g., seedlings and saplings) and a process (Johnson et al. 2002). In particular, it is the process of forest regeneration that lacks a clear definition. The process of forest regeneration, in its broadest sense, describes the replacement of an older cohort of trees by a younger one, which implies a process of canopy turnover. However, the process-based definition of forest regeneration commonly used in research emphasizes the development of juvenile trees originating from seed, sprout, or other mode of establishment (e.g., fragmentation and layering) and focuses more on the suite of factors influencing juvenile tree development than canopy turnover. Herein, forest regeneration refers to both a process and local groupings of juvenile trees. Furthermore, the process-based definition used herein refers mainly to juvenile tree development, but also includes spatio-temporal aspects of the broader canopy turnover process.

What is clear about forest regeneration is its importance to achieving sustainable forest management. Sustainability of our forest resources ultimately hinges on our ability to successfully regenerate forests that have been harvested or have been disturbed in other ways (Wagner and Colombo 2001). Forest regeneration has always been viewed as the most important task for silviculturists (Smith et al. 1997). The perceived importance of forest regeneration to wood supply and, in particular, the need to regenerate desirable tree species in a timely manner had a major influence on the emergence of silviculture (Puettmann et al. 2009). Forest regeneration continues to play an important role in the evolution of silvicultural practices. For example, the difficulty in regenerating desirable

species (e.g., sugar maple, yellow birch, white ash) following harvesting in northern hardwood forests impacted by beech bark disease has stimulated research to identify forest vegetation management approaches that mitigate the negative effects of prolific beech regeneration on the development of stands dominated by desirable species in north-central Maine (Andrew Nelson, unpublished MS thesis).

Despite the history of productive research addressing the regeneration ecology of plant communities (e.g., individualistic concept [Gleason 1926], gap-phase dynamics [Watt 1947], initial floristic composition [Egler 1954], safe sites [Harper et al. 1961], recruitment limitation [Clark et al. 1998], neighborhood dynamics [Canham and Uriarte 2006]), regeneration still is widely regarded as a complex ecological process (Rees et al. 2001, Gratzer et al. 2004, Miina and Heinonen 2008). Because of this complexity, models of natural forest regeneration are usually fraught with weak predictive power, which limits their usefulness in forecasting future forest composition (Miina and Heinonen 2008). Levin (1992) suggested that some of the difficulty that arises when trying to understand a complex ecological process may be an artifact of the observer's failure to capture the process over appropriate spatial and temporal scales. The forest ecology literature is replete with studies on temporal aspects of forest regeneration and stand development, while the spatial aspects have received far less attention (Tilman and Karieva 1997, Rees et al. 2001, Gratzer et al. 2004). However, the importance of explicit spatial pattern in forest dynamics has increasingly gained acceptance largely as an outgrowth of improved statistical methods for testing spatially explicit hypotheses (see Tilman and Karieva [1997] for a review of spatial ecology). More recently, natural forest regeneration has been viewed as a neighborhood-scale process whereby the establishment

and recruitment of juvenile trees are regulated by spatio-temporal heterogeneity in seed dispersal, safe sites, and resource availability (see Canham and Uriarte [2006] for a review of neighborhood dynamics). The underlying premise in studies of neighbor dynamics is that incorporating methods to evaluate both the spatial and temporal pattern of tree regeneration and subsequent stand development will more fully capture the process of forest regeneration. Although the theory of neighborhood dynamics can contribute to research investigating the influences of forest management practices on forest regeneration, few studies have included both spatial and temporal aspects of tree regeneration at sub-stand scales.

The research presented in this thesis addresses spatial and temporal aspects of natural forest regeneration in the Acadian Forest of central Maine based on data from two long-term studies maintained at the Penobscot Experimental Forest (PEF) in Bradley, ME: U.S. Forest Service (USFS) compartment management experiment and the Acadian Forest Ecosystem Research Program (AFERP) expanding-gap experiment. The primary objective of this research was to gain a deeper understanding about the process of forest regeneration in the Acadian Forest by studying regeneration pattern in both space and time. This objective was accomplished by integrating studies on: 1) the temporal pathways of compositional development and 2) understory spatial pattern of tree regeneration.

This thesis is divided into three chapters and a concluding Epilogue. Chapter 1 uses the USFS experiment to investigate compositional development of Acadian mixedwoods over a period of five decades (1953-2005) in relation to initial tree species composition and harvest disturbance history. Chapters 2 and 3 address understory spatial

patterns of tree regeneration within the AFERP experiment. Chapter 2 uses explicit spatial patterning of tree regeneration and canopy openness as response variables to evaluate the effects of gap harvesting on understory spatial patterns. Chapter 3 investigates the influence of soil drainage, canopy openness, local sapling and overstory structure, and gap harvesting on species richness of tree regeneration using both spatial and non-spatial modeling approaches.

Chapter 1

LONG-TERM COMPOSITIONAL DYNAMICS OF ACADIAN MIXEDWOOD STANDS UNDER DIFFERENT SILVICULTURAL REGIMES

1.1. ABSTRACT

Pathways of compositional development over five decades (1953-2005) were reconstructed from permanent sample plot data for a long-term silviculture experiment in an Acadian mixedwood forest of central Maine. The purpose of this study was to elucidate compositional dynamics at the sample plot level in relation to both initial species composition (i.e., plot-level tree composition when the experiment was established) and harvest disturbance history. Cluster analysis was used to group plots across the experiment, which sorted into three distinct groupings based on pre-treatment composition (PTC) and harvest disturbance history (HDC) yielding nine sub-clusters (i.e., three HDCs nested within each PTC). Non-metric multidimensional scaling (NMS) was used to retrace development of the nine sub-clusters over five decades.

From 1953-2005, silvicultural harvest treatments generated an array of compositional outcomes at the plot level. Hardwood dominance increased in plots with a history of heavy and infrequent cutting, while northern conifer dominance was maintain in plots where cutting was lighter and more frequent. Balsam fir dominance was maintained across a wide range of harvest intensities, suggesting that the development of balsam fir has contributed to the persistence of a northern-conifer component under higher harvest intensities. In contrast to balsam fir, the ubiquitous decline in cedar importance documented in this study suggests that additional treatments may be needed

to enhance cedar recruitment in similar forest types where cedar regeneration is a management objective. Results from this investigation indicated that neighborhood-scale (i.e., plot level) compositional dynamics were associated with variability in harvest disturbance generated by the long-term silviculture experiment overlain on plot-to-plot variability in tree species composition at the time the experiment was established.

1.2. INTRODUCTION

Forest management and silviculture practices are based on understanding the dynamic responses of forests to natural disturbances and silvicultural treatments over long time periods (Irland et al. 2006). Long-term forest inventory data can provide valuable information on regional forest dynamics over the range of time-scales relevant to forest resource management. Where long-term datasets are coupled with silviculture experiments, opportunities exist to advance our understanding about regional forest dynamics through research that addresses the influence of silviculture on compositional and structural dynamics. For example, results from a long-term silviculture experiment maintained by the USDA Forest Service has brought into question the suitability of single-tree selection methods to achieve balanced, uneven-sized structures in northern conifer forests of the Northeastern U.S. (Kenefic et al. 2006); a result afforded by nearly a half century of experimentation.

A major challenge for forest managers relying on natural regeneration is the ability to forecast future forest composition under various levels of harvest intensity (Taylor et al. 2009). In naturally-regenerated, mixed-species stands, uncertainty in forecasting species composition can arise due to the range of successional outcomes

resulting from silvicultural treatments (Abrams et al 1985, Gould et al. 2005). Two common themes of research on secondary succession in mixed-species forests are that local stand conditions (i.e., species composition and stand structure) and disturbance history affect future composition. The role of local stand conditions and disturbance in driving patterns of tree recruitment and successional replacement has been widely appreciated since Watt's (1947) seminal work on the pattern and process of plant community dynamics. More recently, the importance of local stand conditions to forest dynamics has led to a theory of "neighborhood" dynamics, in which fine-scale interactions regulate the demography of component tree species (Canham and Uriarte 2006).

Even when applied with the objective of creating homogeneity at the stand-level, silvicultural treatments often fail to entirely eliminate sub-stand heterogeneity (Puettmann et al. 2009). According to neighborhood dynamics theory, the range of sub-stand conditions produced by silviculture treatment can potentially lead to a range of successional outcomes within a single stand. From an ecological perspective, the range of neighborhood dynamics produced by silviculture experiments can serve as a platform to investigate lines of inquiry beyond the scope of their intended design (Puettmann et al. 2009). For example, when experiments generate within-treatment heterogeneity among sample plots while simultaneously creating overlap in sample plot conditions among treatments, grouping methods (e.g., multivariate cluster analysis) can be used to form groups across treatments based on local plot-level conditions for further analysis of forest dynamics.

This study used data collected over five decades (1953-2005) from a long-term experiment comparing traditional silvicultural regeneration methods in central Maine to document pathways of compositional development in Acadian mixedwoods. Rather than charting the development of silvicultural treatments through plot averaging, we used plot-to-plot variability intrinsic to the long-term experiment to investigate compositional dynamics at the plot-level. Since we assumed that initial species composition and harvest disturbance history influenced plot-level development, we attempted to control for both initial species composition and harvest disturbance history to elucidate developmental pathways.

1.3. METHODS

1.3.1. Study Site

This study took place on the Penobscot Experimental Forest (PEF) near the towns of Bradley and Eddington, Maine (Figure 1.1). The PEF is within the Acadian Forest, a conifer-dominated mixedwood ecosystem covering much of Maine and the Canadian Maritimes (Halliday 1937). The PEF is a 1,680-ha forest owned by the University of Maine Foundation and is jointly managed by the University of Maine and the U.S. Forest Service, Northern Research Station. Climate in central Maine is cool and humid. February (avg. temp. -7.1º C) and July (avg. temp. 20.0º C) are the coldest and warmest months, respectively. Average precipitation is 107 cm. The growing season is approximately 160 days at the PEF. Soils at the PEF are complex resulting from glaciation. The dominant parent material is a fine-textured till derived from sedimentary bedrock which have developed into loams. Contained within the PEF is a broad gradient

of soil drainage classes; however, a large portion of the forest grows on soils classified as somewhat poorly to poorly drained and moderate to low productivity (Sendak et al. 2003).

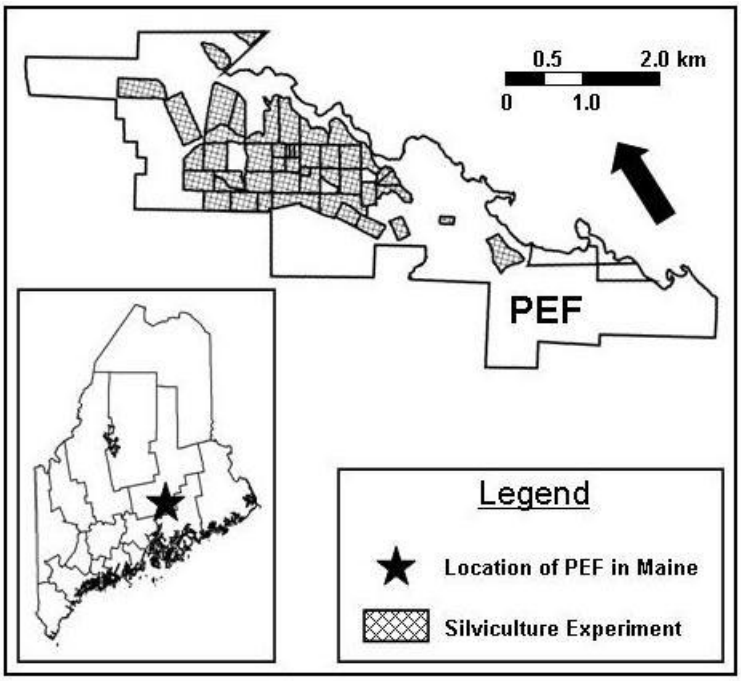

Figure 1.1. Location of long-term silviculture experiment on the Penobscot Experimental Forest (PEF) with location of the PEF in Maine (inset image, bottom left).

Forest types on the PEF are predominantly mixedwood types. The dominant conifer species on the PEF include balsam fir, eastern hemlock (*Tsuga canadensis*), red spruce (*Picea rubens*), white spruce (*Picea glauca*), northern white-cedar (*Thuja occidentalis*), and white pine (*Pinus strobus*), while the common hardwood species include red maple (*Acer rubrum*), paper birch (*Betula papyrifera*), gray birch (*B. populifolia*), trembling aspen (*Populus tremuloides*), and bigtooth aspen (*P. grandidentata*) (Sendak et al. 2003).

The history of the PEF prior to 1950 is not well documented, but the forest structure was probably irregular and uneven-aged as a result of periodic anthropogenic and natural disturbances (Sendak et al. 2003). Throughout much the 1800s, a sawmill operated within the boundaries of PEF, which likely harvested pine and spruce preferentially over much the forest during this period (Saunders and Wagner 2008). The 1913-1919 outbreak of the eastern spruce budworm also had a significant influence on forest structure and composition before the PEF was established in the early 1950s (Seymour 1992, Saunders and Wagner 2008).

1.3.2 Long-Term Silviculture Experiment

The USDA Forest Service (USFS) installed a replicated study between 1952-1957 to investigate the influences of silviculture and exploitative harvesting practices on the composition, structure, growth, and yield of mixed northern conifer stands in the Northeastern U.S. (Sendak et al. 2003). At the start of the long-term experiment, two replicates of eight harvest treatments were randomly assigned to sixteen 7-18 ha treatment units: 5, 10, and 20-year single-tree selection methods, two and three-stage shelterwood methods, fixed and modified diameter-limit harvests and commercial clear-cutting. In the early 1980s, both replicates of the three-stage shelterwood treatment were divided to investigate the influence of pre-commercial thinning (PCT) on mixed northern conifer stand development. As an additional amendment to the original experiment, a natural area, which was established as a control, was eventually divided to create two "pseudo-controls". The addition of the PCT and pseudo-control units to this experiment

brought the number of treatment units from 16 to 20. For more information on treatment prescriptions see Sendak et al. (2003).

Within each compartment, tree inventories have been conducted on a systematic grid of 8-21 permanent sample plots before and after each harvest entry and approximately every five years after harvest. Stem diameter at breast height (DBH = 1.35 m) and condition of all trees > 11.4 cm DBH were recorded by species and species groups within 0.08-ha circular plots; saplings between 1.2 and 11.4 cm DBH were recorded within nested, 0.02-ha circular plots.

1.3.3. Study Approach

The focus of this study was the compositional dynamics of Acadian mixedwoods based on plot-level data from the PEF long-term silviculture experiment. Thus, the sample plot served as the experimental unit for this analysis. Forest composition was estimated from individual tree measurements (stems ≥ 1.2 cm DBH) recorded on 231 permanent sample plots from 16 of 20 treatment units collected over six time periods spanning five decades (1953-2005). Both replicates of the natural area (untreated controls) were excluded since they had not received harvest treatments. The other two units excluded from this analysis, one replicate of the fixed diameter-limit and commercial clear-cutting treatments, were removed since they were not inventoried before the establishment of the original experiment and, therefore, lacked data on pre-treatment forest composition. The six time periods (T) examined were: 1) T0: 1953-1957 (pre-treatment sample), 2) T1: 1960-1964, 3) T2: 1969-1974, 4) T3: 1980-1983, 5) T4:

1990-1994, and 6) T5: 1999-2005. Time periods were selected that captured at least one inventory on all plots and the last inventory before a harvest occurred.

Plot-level importance values (IV) of six tree species and species groups, hereafter referred to as species, were used to represent the relative ecological importance of tree species to forest composition at the plot level (*sensu* Curtis and McIntosh 1951). The six species variables were: fir (balsam fir), cedar (northern white-cedar), hemlock (eastern hemlock), HMWD (hardwood-dominated mixedwood; composed of paper and grey birch, American beech, sugar maple, trembling and bigtooth aspen, white and red pine, and eastern larch), maple (red maple), and spruce (red, black, and white spruce). For each plot at each time period, IV was calculated by species as the mean of two percentages (%): % of total basal area and stem density.

Causes of tree mortality over the entire experiment were not recorded until the 1990s, which meant tree losses due to harvesting could not be specifically quantified in each plot for the full five-decade period. However, sample plots in all compartments that received harvesting were inventoried before and after each harvest entry. Based on these pre- and post-harvest inventories, four variables describing the harvest disturbance were generated using the reduction in basal area (BA) for all plots from harvested units over the five-decade period: 1) total BA reduction (TBAr), calculated as the sum of the differences in plot-level BA between all pre- and post-harvest inventories; 2) mean percent BA reduction (MBAr), calculated as the percent decrease in plot-level BA between pre- and post-harvest inventories divided by the total number of recorded basal area reductions; 3) frequency of BA reduction (FRQr), representing the total number of recorded BA reductions; and 4) total hardwood BA reduction (THDr), calculated as the

sum of the differences in plot-level hardwood BA between all pre- and post-harvest inventories. Since favoring the regeneration of northern conifer species was and continues to be a primary objective of this experiment, THDr was included to account for any bias toward removing hardwoods.

1.3.4. Statistical Analysis

In order to account for the effects of initial forest composition and harvest disturbance on plot-level compositional development, permanent sample plots were grouped according to pre-treatment tree composition and harvest disturbance after five decades of treatment. Two rounds of agglomerative, hierarchical cluster analysis were performed in PC-ORD (Version 4.07; McCune and Mefford 1999) starting with all 231 plots in order to group plots based on IV of species and harvest disturbance variables separately. Combinations of distance measures and linkage methods were tested on both datasets to find an optimal cluster solution for each dataset (i.e., compromise between number of clusters and information retained). Preliminary analysis on results from the first round of cluster analysis indicated that THDr was not useful for distinguishing harvest disturbance clusters; therefore cluster analysis was rerun on harvest disturbance variables without THDr. Pre-treatment clusters (PTC) were formed using correlation distance and flexible beta linkage (beta = -0.85) and harvest disturbance clusters (HDC) were formed using Euclidean distance and flexible beta (beta = -0.65).

Parametric and non-parametric statistics were performed in the R programming environment (Version 2.8.1; R Development Core Team 2008) and significance was assessed at $\alpha = 0.05$ to classify PTC and HDC to enhance ecological interpretability.

PTC were classified according to compositional types using one-way analysis of variance (ANOVA) to test for differences in mean IV within species among clusters. Appropriate transformations were applied to species and harvest disturbance data when necessary. Cluster types for PTC were based on the fewest species needed to sum to > 50% IV within each cluster. HDC were classified using both parametric and non-parametric statistics. ANOVA was used to test for differences in mean TBAr and MBAr among clusters. Mann-Whitney tests (non-parametric) were used to determine if HDC differed based on harvest frequency (FRQr). Means were separated when ANOVA detected significant differences using Tukey's HSD, while Mann-Whitney test results were used to separate median FRQr by HDC. In addition to classifying HDC according to differences in harvest disturbance variables, the percentages of plots within each of the nine silvicultural treatments were summarized for each HDC (Appendix A).

The five-decade development of sub-clusters (i.e., HDCs nested within PTCs) was retraced to document species-level trends in basal area, density, and relative importance. Mann-Whitney tests were used to compare mean IVs at T0 and T5 for each species separately within sub-clusters to determine if species importance changed significantly over the study period ($p < 0.05$). Non-metric multidimensional scaling (NMS) was run in PC-ORD (Version 4.07; McCune and Mefford 1999) to ordinate sub-clusters across the six time periods in compositional space using species' IVs. Prior to NMS, IVs at each inventory period were averaged by sub-cluster. Following the recommendation of McCune and Grace (2002), multiple runs of NMS were performed to find an appropriate solution. The best two-dimensional solution was obtained using the "slow and thorough" autopilot setting with Euclidean distance, which stabilized after 105

iterations. Final stress and instability for the two-dimensional solution was 12.06 and 8.98×10^{-6}, respectively. A Monte Carlo test based on 50 runs with randomized data found a 1.96 % probability that a similar solution could have been obtained by chance. NMS coordinates of sub-clusters at each inventory period were used to retrace compositional development in species space over five decades.

1.4. RESULTS

1.4.1. Cluster Analysis and Classification

Cluster analysis revealed three pre-treatment composition clusters (PTC) and three harvest disturbance clusters (HDC) among the sample plots. Clusters were identified by pruning dendrograms at long stems, which retained 73% and 81% of information from original pre-treatment and disturbance pattern datasets, respectively. Based on these results, nine sub-clusters (i.e., three HDCs nested within each of three PTCs) could be formed to retrace compositional development over five decades.

ANOVA results for comparisons of pre-treatment mean IV within species among clusters are presented in Table 1.1. Mean IV for hemlock (33.9%) was highest in PTC A ($p < 0.05$). Although mean IV of fir was lowest in PTC A, fir on average still accounted for 29.2% of pre-treatment species composition. Species composition of PTC A was dominated by hemlock and fir (i.e., fewest species needed to account for > 50%). PTC B was dominated by fir and cedar (38.2% and 28.5%, respectively), which was the highest IV attained by cedar among PTCs. Fir dominated species composition of PTC C (52.8%), which was the highest mean IV for fir across PTCs. Other species with notably high IV within PTCs were: spruce in PTC A and C (15.1% and 13.9%, respectively) and

hemlock in B (12.2%). Mean IV for maple and HMWD were consistently ranked low within PTCs (IV < 10%) and maple IV was not significantly different among clusters.

Table 1.1. Comparison of pre-treatment clusters based on mean importance values of species. Cluster type identifies the fewest top-ranked species needed to sum to > 50% cumulative importance value.

Species	**Mean Importance Value (%) by Pre-Treatment Cluster**		
	A	**B**	**C**
Fir	29.2 c*	38.2 b	52.8 a
Cedar	5.6 b	28.5 a	6.7 b
Hemlock	33.9 a	12.2 b	11.2 b
HMWD	8.7 a	5.0 b	7.9 ab
Maple	7.5 *ns*	7.8 *ns*	7.5 *ns*
Spruce	15.1 a	8.3 b	13.9 a
Cluster Type	Hemlock-Fir	Fir-Cedar	Fir

* Mean separation of ANOVA comparisons within rows at α = 0.05; *ns* = non-significant

ANOVA and Mann-Whitney tests detected differences ($p < 0.05$) in harvest disturbance variables among HDCs (Table 1.2), which were consistent with the membership of plots within silvicultural treatments (Appendix A). HDC B had lower mean TBAr and MBAr and higher FRQr than the other HDCs (9.3 m^2, 20.2%, and 5, respectively); a pattern consistent with HDC B composed mainly of plots from selection harvest treatments. In contrast, HDC C had the highest mean TBAr and MBAr (25.4 m^2, 60.3%, and 2), which reflected the higher proportion of plots from treatments associated with heavy overstory removal (i.e., commercial clear-cutting, two-stage shelterwood, and both diameter-limit treatments). Harvest disturbance variables for HDC A were intermediate between the other two HDCs (mean TBAr and MBAr and median FRQr of

15.1 m^2, 35.2%, and 3, respectively). The difference in median FRQr between HDC B and C was non-significant. Collectively, plots within HDC A were more of a mixture of even- and uneven-aged silvicultural treatments than the others. For convenience, HDC A, B, and C will be referred to as medium intensity, low intensity, and high intensity, respectively.

Table 1.2. Comparison of harvest disturbance clusters (HDCs) based on mean and median harvest disturbance variables. Cluster type was derived by ranking HDCs according to harvest disturbance variables.

Disturbance Variable	**Variable Estimate by Harvest Disturbance Cluster**		
	A	**B**	**C**
TBAr (m^2)	15.1 b*	9.3 c	25.4 a
MBAr (%)	35.2 b	20.2 c	60.3 a
FRQr	3 b**	5 a	2 b
Cluster Type	Medium Intensity	Low Intensity	High Intensity

* Mean separation results for ANOVA comparisons within rows at $\alpha = 0.05$
** Medians separated based on Mann-Whitney comparisons within rows

1.4.2. Basal Area and Density Development

Five-decade development of basal area and density of the nine sub-clusters are presented in Figures 1.2 and 1.3, respectively. Across PTC types, changes in total density and basal area were greatest in the high-intensity sub-clusters, intermediate in medium-intensity sub-clusters, and least in low-intensity sub-clusters. By the fifth decade, total density was higher and basal area was lower than pre-harvest levels for all sub-clusters. Over the study period, fir had the highest density in nearly all sub-clusters. Although hemlock and spruce basal area decreased over time in all sub-clusters, their densities tended to increase. HMWD and maple densities increased rapidly in high-

intensity sub-clusters. Basal area and density of cedar decreased over time in all sub-clusters, while maple basal area and density tended to increase. Of all species, only cedar density decreased appreciably over the study period.

1.4.3. Compositional Development

Figure 1.4 shows five decades of compositional development of sub-clusters in two-dimensional NMS space, coefficients of determination (r^2) for both axes, and corresponding biplot with Pearson correlations (r) between species importance values and NMS axes. According to r^2 values, NMS axes 1 and 2 accounted for 49.7% and 34.0%, respectively, of compositional variation in the data. NMS axis 1 was strongly correlated with fir and hemlock IV (r = -0.844 and 0.951, respectively), while NMS axis 2 was strongly correlated with cedar, HMWD, and maple IV (r = -0.874, 0.765, and 0.738). Spruce IV was most strongly correlated with axis 1 (r = 0.598). Correlations suggested that axis 1 captured a gradient defined largely by decreasing fir importance and increasing hemlock importance and, to lesser extent, spruce importance moving from left to right in ordination space, while NMS axis 2 captured a gradient defined largely by increasing cedar importance and decreasing HMWD and maple importance moving from top to bottom in ordination space.

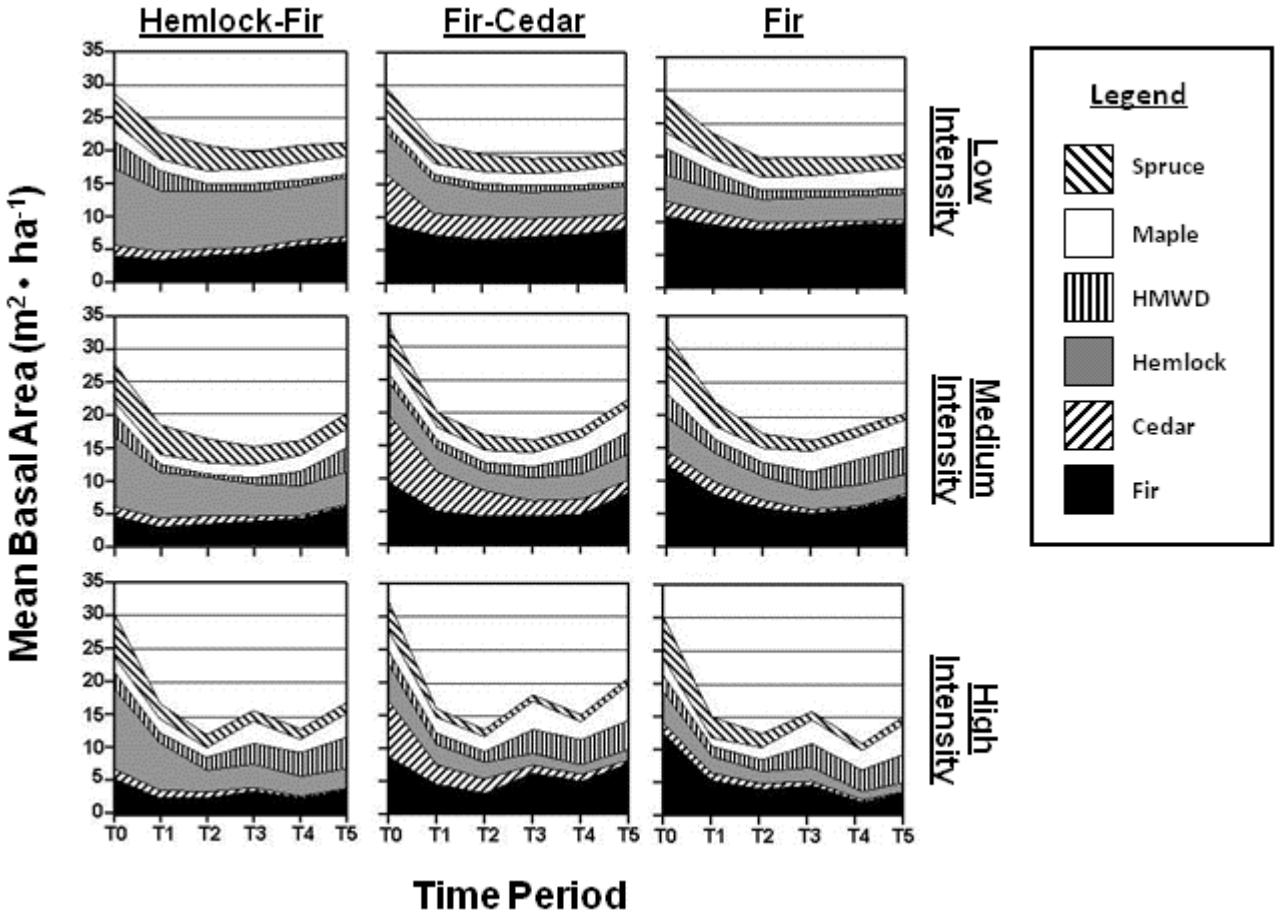

Figure 1.2. Changes in mean basal area by species for nine sub-clusters (three HDC by three PTC combinations) over six time periods from 1953-2005.

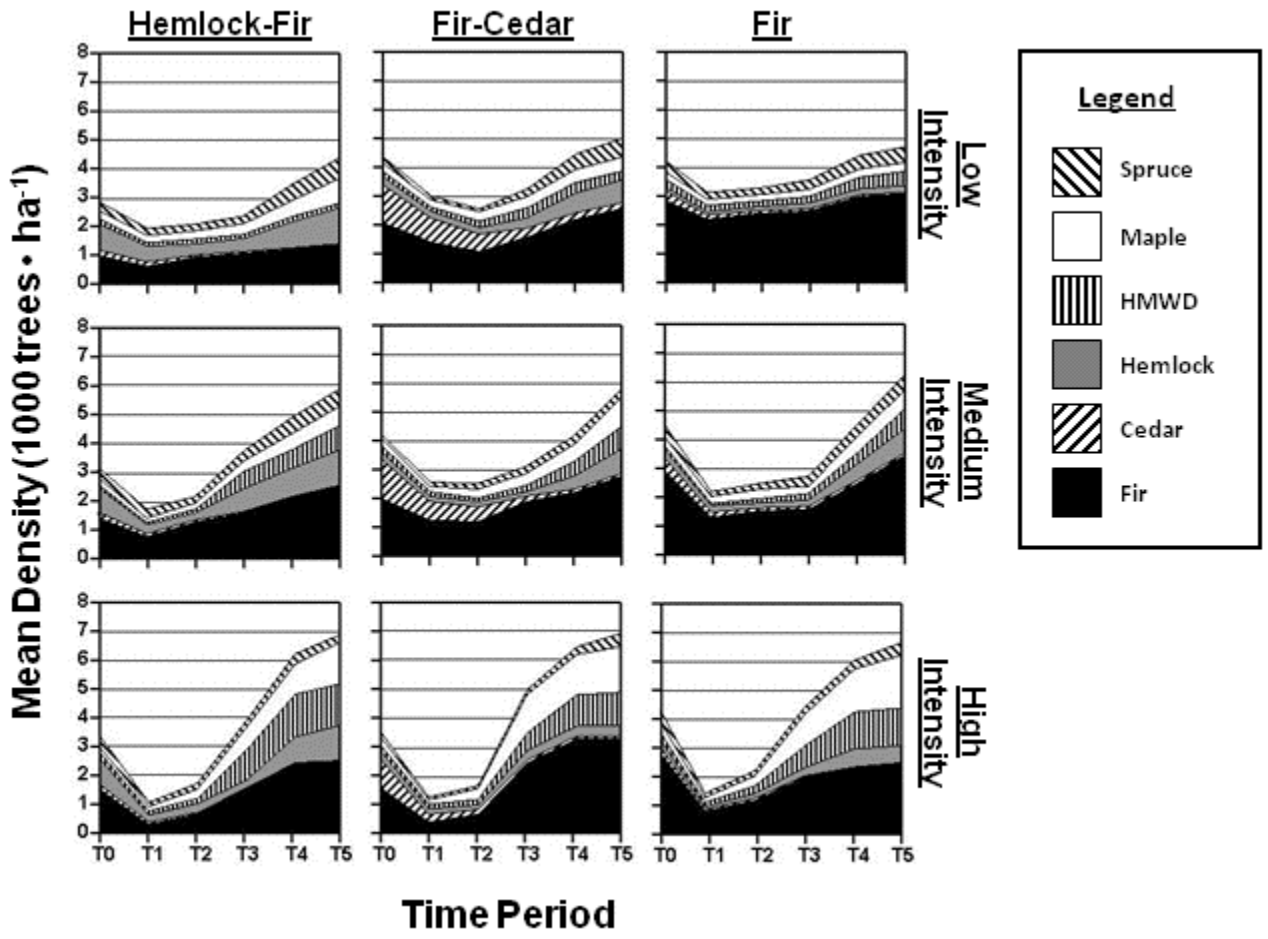

Figure 1.3. Changes in mean density by species for nine sub-clusters (three HDC by three PTC combinations) over six time periods from 1953-2005.

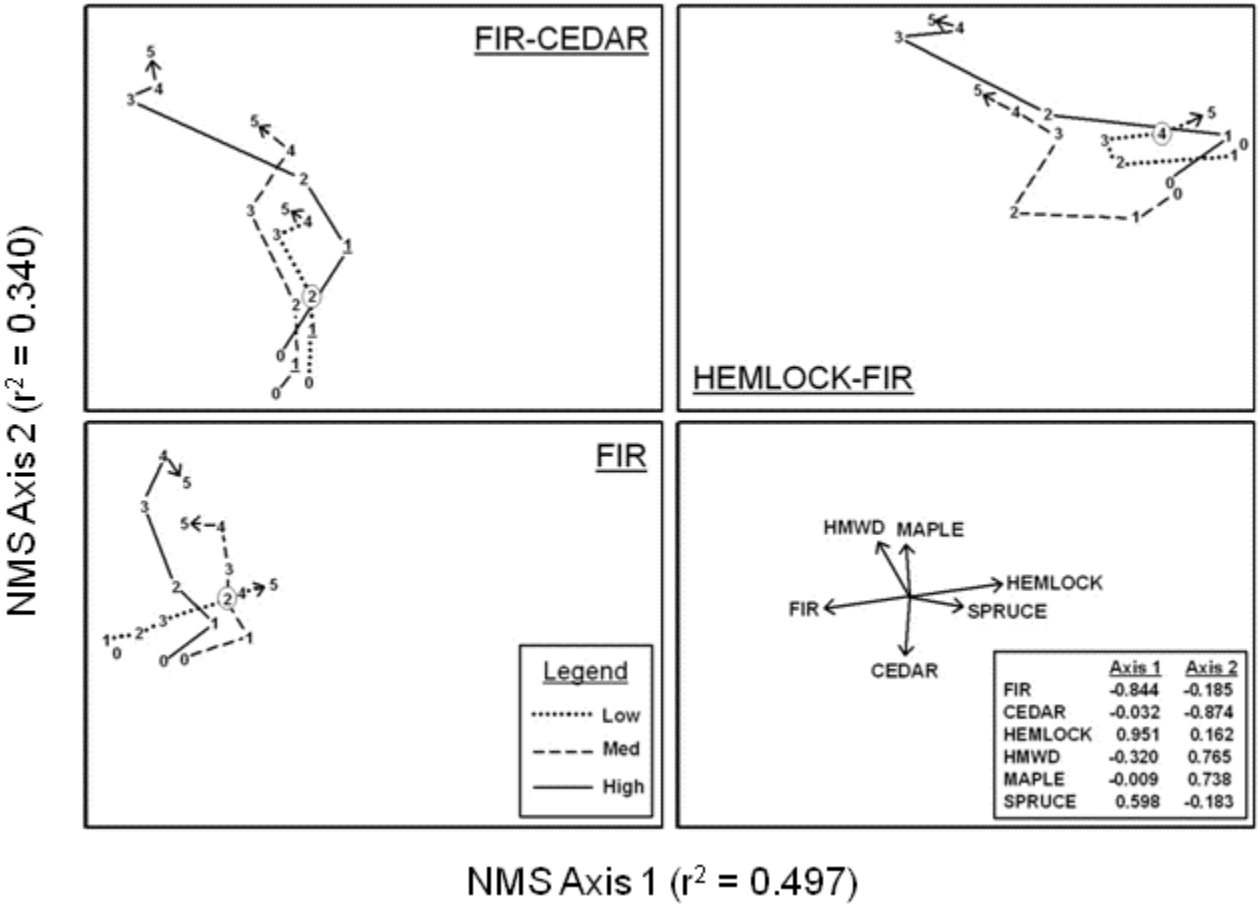

Figure 1.4. Developmental pathways of nine sub-clusters (three HDC by three PTC combinations) in NMS space over six time periods from 1953-2005. Bottom, right box contains NMS biplot and Pearson correlations between species importance values and NMS axes.

Prior to the start of the silvicultural treatments (denoted as 0; T0), harvest disturbance sub-clusters sorted out in ordination space by PTC particularly along axis 1 (Figure 1.4). After five decades, nearly all sub-clusters had migrated away from their starting positions in NMS space (denoted as 5; T5), with the exception of the low-intensity hemlock-fir cluster. The relative stability of the low-intensity hemlock-fir cluster also was reflected by minor changes in species importance over time (Figure 1.5), and absence of significant differences in species-level IV between T0 and T5 (Table 1.3). Despite changes observed in ordination space, fir importance did not change significantly in any sub-cluster over five decades.

Table 1.3. Results of Mann-Whitney tests comparing median importance values at T0 and T5 by species for each sub-cluster. Signs indicate significant differences ($\alpha = 0.05$) and whether importance value at T5 was greater (+) or less than (-) T0. L, M, and H are low, medium, and high intensity HDC types.

Species	**Hemlock-Fir**			**Fir-Cedar**			**Fir**		
	L	**M**	**H**	**L**	**M**	**H**	**L**	**M**	**H**
Fir									
Cedar		–	–	–	–	–	–	–	–
Hemlock		–	–				+		
HMWD		+	+		+	+		+	+
Maple			+		+	+		+	+
Spruce			–			–		–	–

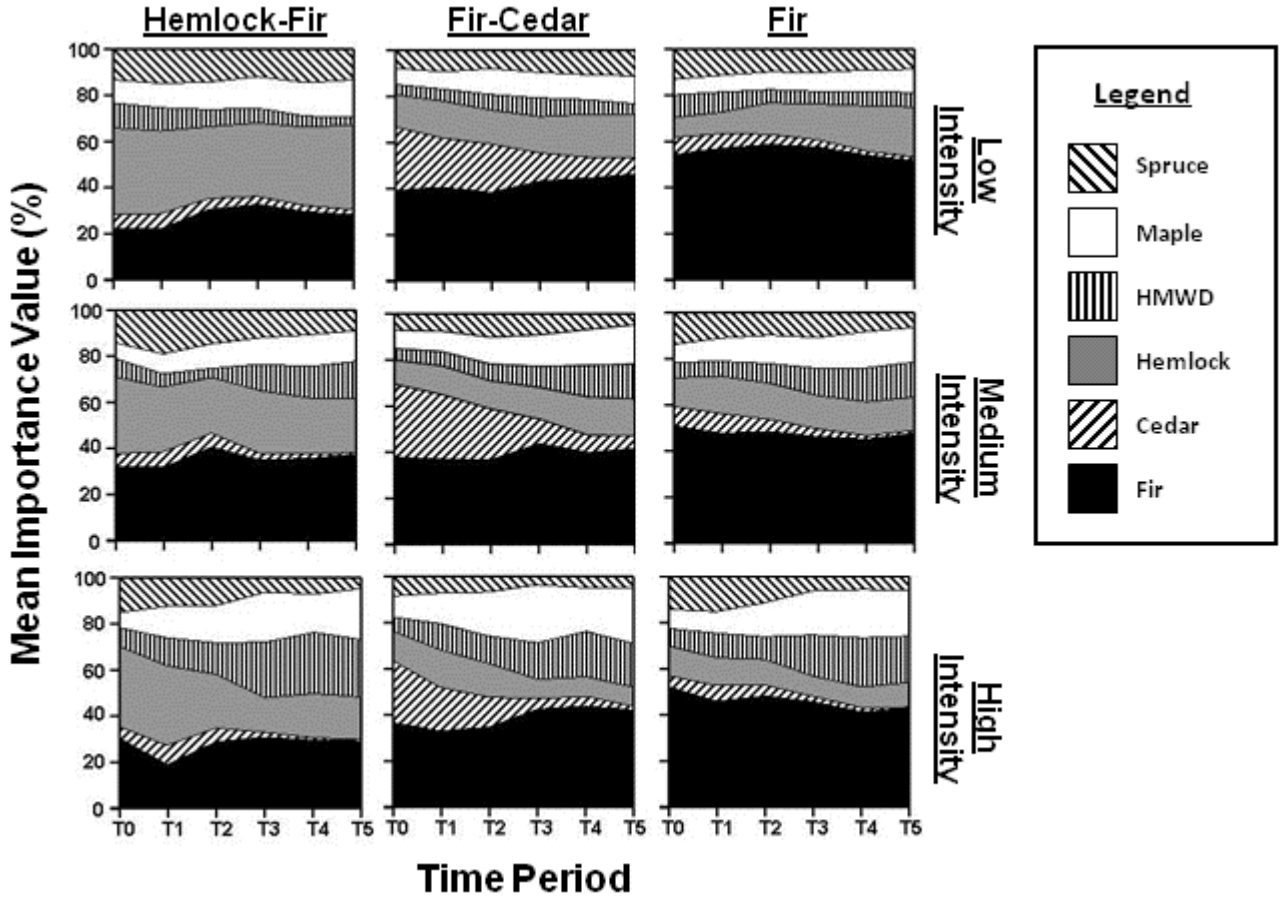

Figure 1.5. Changes in mean importance values by species for nine sub-clusters (three HDC by three PTC combinations) over six time periods from 1953-2005.

Harvest disturbance sub-clusters within PTC types were less clumped at T5 relative to T0, which indicated compositional divergence. Compositional divergence within PTC types was associated with increasing importance of HMWD, maple, or both as harvest intensity increased (Figure 1.5). Divergence within PTC types also was associated with significant decreases in spruce importance observed in all high-intensity sub-clusters. With the exception of maple importance in the medium-intensity hemlock-fir cluster, mean IV for HMWD and maple was significantly greater at T5 than T0 in medium and high-intensity sub-clusters (Table 1.3). The trend of increasing HMWD and maple importance was reflected by the vertical sorting of sub-clusters within PTC types at T5 (denoted as 5) along axis 2. Vertical sorting also was likely associated with declining cedar importance over the five-decade period, which was significant in nearly all sub-clusters except the low-intensity hemlock-fir cluster. In the case of hemlock-fir sub-clusters, compositional divergence also was linked with significant decreases in the importance of hemlock, and spruce in the high-intensity cluster, as harvest intensity increased; a pattern consistent with leftward shifts along axis 1 in NMS space.

By T5, patterns of species importance are similar within fir-cedar and fir sub-clusters (Figure 1.5), which was reflected by convergence in ordination space (Figure 1.4). Compositional convergence, in this case, was associated with significant decreases in cedar and spruce importance in fir-cedar and fir sub-clusters, respectively, similar trends in HMWD and maple importance over time, and maintenance of fir dominance.

1.5. DISCUSSION

From 1953-2005, varied silvicultural treatments generated an array of compositional outcomes at the sample plot level that were associated with differences in

initial tree species composition and subsequent harvest disturbances. Compositional outcomes ranged from no change from pre-treatment conditions (low-intensity, hemlock-fir sub-cluster) to significant changes after five decades for five out of six species in fir-cedar stands that were subjected to high-intensity harvesting. In ordination space, the developmental pathways of sub-clusters tended to migrate away from pre-treatment conditions and departures from pre-treatment conditions within PTCs increased with increasing harvest disturbance intensity. For example, developmental pathways of fir-cedar and fir sub-clusters converged in ordination space, which was likely due to large decreases in cedar importance in the fir-cedar sub-clusters and maintenance of fir dominance in both PTC sub-clusters.

A general trend that emerged from this investigation was that stand compositional changes were greatest in the high-intensity sub-clusters, intermediate in medium-intensity sub-clusters, and least in low-intensity sub-clusters. This trend across harvest intensities is consistent with an ecological release associated with increasing disturbance intensity. We use the term ecological release to describe a rapid increase in the importance of species that were previously subordinate. Disturbance can trigger an ecological release by increasing resource availability, which, in turn, creates recruitment opportunities for species with higher resource demands (Pickett and White 1985).

At the start of this long-term experiment, permanent sample plots were largely dominated by shade-tolerant, northern conifer species (fir, cedar, hemlock, and spruce), especially balsam fir (mean IV ranging from 29-53%) and a minor component (mean IV < 10%) of red maple and the hardwood-dominated mixedwood group (HMWD; a group comprised of shade-intolerant hardwoods and conifers). Apart from cedar (see Hofmeyer

et al. 2009), this group of northern conifer species is capable of regenerating prolifically in Acadian forest understories (Brissette 1996), which confers a level of compositional stability in the absence of major disturbances where these species are locally dominant (Frelich 2002). After five decades of various harvest treatments, the relative importance of red maple and the HMWD group increased mainly in plots that received medium- and high-intensity harvest disturbances. In the case of the high-intensity, hemlock-fir and fir-cedar sub-clusters, the importance of red maple, HMWD, or both surpassed that of a pre-treatment dominant species. Collectively, red maple and the species comprising the HMWD group are more light-demanding than the northern conifers and, in the case of the hardwoods, are vigorous sprouters capable of rapid re-growth enabling these species to behave opportunistically following disturbance. The ecological release observed in this study may be the product of prolific hardwood sprout development and increasing resource availability, particularly light, in plots where harvesting was historically more intense.

Over the study period, balsam fir maintained dominance across sub-clusters. Of the shade-tolerant, northern conifers found on the PEF, balsam fir is arguably the most opportunistic. Although considered short lived compared to other Acadian forest species, balsam fir compensates for its short longevity through the prolific development of advance regeneration (Seymour 1992). In a regeneration study using the same experiment, balsam fir regeneration was the most abundant among the silvicultural treatments (Brissette 1996). Not only does balsam fir gain a competitive advantage though prolific regeneration, but balsam fir height growth responds more quickly than co-occurring shade-tolerant conifers of similar size following overstory removal (Wetsveld

1931). In stands treated with single-tree selection, there appear to be no understory light conditions that favor either red spruce or hemlock over balsam fir of the same height (Moores et al. 2007). Based on the ecology and silviculture of balsam fir, the high relative density of balsam fir across sub-clusters observed in this study suggests that fir dominance was maintained by prolific regeneration.

Cedar, hemlock, and spruce were the only species that showed a significant decline in importance over the five-decade period. In the case of cedar, significant decline occurred in all but one sub-cluster and the largest decreases were observed in clusters where cedar was co-dominant (i.e., fir-cedar type). Cedar decline is pervasive throughout its geographic range, which has been linked to three main causes: regeneration failures, poor recruitment into sapling and pole size classes, and harvesting levels that exceed growth (Hofmeyer et al. 2009). With decreases in both basal area and density observed in all sub-clusters, cedar decline in this study may be linked to all three causes. Significant declines in spruce importance occurred in all three high-intensity sub-clusters, while hemlock importance decreased significantly in medium- and high-intensity hemlock-fir sub-clusters. For both spruce and hemlock, declines were associated with reductions in basal area since densities were higher at the end of the study period. This pattern was likely a result of removing overstory spruce and hemlock during harvesting, which simultaneously enhanced spruce and hemlock recruitment either through new seedling establishment or release of advance regeneration.

In this investigation, we were unable to assess the relative contribution of initial species composition and harvest disturbance intensity in driving neighborhood-scale (i.e., the scale of the permanent plot) pathways of compositional development. Using a similar

approach, Guild et al. (2005) found that initial species composition at the time of harvest was an important determinant of stand developmental pathways following clear-cutting in central hardwood forests of Pennsylvania and concluded that multiple pathways are likely to be observed in stands containing species with different modes of regeneration (e.g., sprouting, advance regeneration, post-disturbance establishment) and a range of shade tolerances. Collectively, the tree species in this study comprise an array of autecological and life history attributes, which likely contributed to variability in developmental pathways observed in this investigation. However, we were able to account for variation in both initial species composition and harvest disturbance history, which revealed that compositional development at the plot level was linked to variability in both factors. Based on our findings, pathways of neighborhood-scale compositional development were likely associated with variability in harvest disturbance history generated by the long-term silviculture experiment overlain on plot-to-plot variability in tree species composition at the time the experiment was established.

Long-term silviculture experiments provide opportunities to investigate the influences of forest management practices on regional forest dynamics across the range of temporal scales relevant to forest resource management. Our findings support the conventional wisdom about the effect of silviculture in Acadian mixedwoods dominated by northern conifers as outlined by Westveld (1953); specifically, heavier removal intensity increases hardwood dominance, while lighter harvesting (i.e., selection cutting) maintains the dominance of northern conifers. In a recent analysis of the same long-term experiment, Saunders and Wagner (2008) observed a dominance shift from northern conifers to hardwoods in both commercial clear-cut and fixed diameter-limit treatments,

while the 5-year selection treatment maintained a relatively stable species composition dominated by northern conifer species. Despite these compositional trends, we observed balsam fir dominance was maintained across a wide range of harvest intensities, suggesting that the development of balsam fir has contributed to the persistence of a northern-conifer component under higher harvest intensities. In contrast to balsam fir, the ubiquitous decline in cedar importance documented in this study suggests that additional treatments may be needed to enhance cedar recruitment in similar forest types where cedar regeneration is a management objective.

Chapter 2

EFFECT OF GAP HARVESTING ON UNDERSTORY SPATIAL PATTERN OF TREE REGENERATION AND CANOPY OPENNESS IN MIXEDWOOD STANDS OF CENTRAL MAINE

2.1. ABSTRACT

We used a long-term experiment to test the hypothesis that gap harvesting increased the grain of spatial patterning in understory tree regeneration and canopy openness in mixedwood stands of central Maine. Semivarigram estimation and spatial autocorrelation were used to describe spatial patterns in eight stands containing harvest gaps of various sizes (0.01 to 0.22 ha) and over a range of spatial scales from 3 to189 m. Multiple lines of evidence supported our hypothesis that gap harvesting created a coarse-grained spatial pattern in both understory tree density and canopy openness. Global pattern (i.e., pattern over the full spatial extent) interpreted from spatial correlograms suggested patchiness in understory regeneration and canopy openness among small (0.01-0.13 ha) and large (0.03-0.22 ha) harvest gap treatments, and an unharvested control. Although there were no differences in global spatial pattern among harvest gap treatments, the mean range of spatial dependence (both correlogram and semivariogram ranges) for tree regeneration and canopy openness increased along a gradient of harvest gap diameters that were created by the experiment, suggesting that gap harvesting influenced spatial patterning at smaller scales (< 50 m). Correlograms revealed a higher frequency of autocorrelation at larger scales (> 50 m) for stands with gap harvests compared to unharvested controls. Partial Mantel tests revealed positive associations

between tree regeneration density and canopy openness for the small-gap treatment only. Positive associations between tree regeneration and space in the large-gap treatment suggested that tree regeneration exhibited a spatial pattern that was independent of patterning in canopy openness. Results from partial Mantel tests and other analyses indicated that the spatial responses of tree regeneration and canopy openness among treatments was related to a pre-existing spatial pattern, as well as the creation of a new understory spatial pattern initiated by the harvest gap treatments. Using spatial analysis methods that can estimate patterns across multiple scales not only provided a better means to assess within-stand heterogeneity, but increased the range of spatial patterns that were detectable in this long-term experiment.

2.2. INTRODUCTION

Silvicultural regeneration methods inspired by patterns of natural disturbance are being applied with the intention of harvesting timber while keeping removals within the limits of natural variability in one or more disturbance attributes linked to overstory removal (Seymour et al. 2002). In the Acadian forest of the Northeastern United States, the dominant pattern of natural canopy disturbance and subsequent recovery is small-scale, gap dynamics (Lorimer 1977, Chokkalingam and White 2000, Fraver and White 2005); although outbreaks of the eastern spruce budworm (*Choristoneura. fumiferana*) occasionally cause canopy mortality severe enough to be considered stand-replacing (Seymour 1992). Ubiquitous and recurring gap disturbances in the Acadian forest in association with the cool and moist climate of the region has favored dominance by shade-tolerant tree species (Jacobson et al. 1987; Seymour et al. 2002), such as red spruce

(*Picea rubens*) and balsam fir (*Abies balsamea*) in the softwood-dominated forest and sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*) in the hardwood-dominated forest (Seymour 1995). Because of these patterns, gap-based methods of harvesting applied on an appropriate cutting cycle have been suggested as an improved silvicultural approach for regenerating and managing Acadian forest stands (Seymour et al. 2002, Seymour 2005).

Silvicultural regeneration methods are aimed at increasing the availability of resources that limit the process of regeneration through the deliberate removal of overstory trees (Smith et al. 1997). Since resource availability is spatially related to the pattern of tree removal, the spatial patterning of subsequent forest regeneration should mirror the spatial arrangement of harvested trees at sub-stand scales. For gap-based harvest methods, the spatial arrangement of gaps create patchiness in the availability of resources in the understory (Battaglia et al. 2002). Spatial patterning of understory resources is more heterogeneous over a range of scales in stands that are treated with gap-based methods compared to more uniformly applied regeneration methods such as clearcut and uniform shelterwood methods (Hansen et al. 1991, Kohn and Franklin 1997; Puettmann et al. 2009).

The degree to which understory spatial patterning in tree regeneration matches the size, spatial arrangement, and cumulative extent of harvest gaps is not well understood despite substantial research into examining the responses of forest regeneration to disturbance at both stand- and gap-scales (Puettmann et al. 2009). Whether or not the enhancement of sub-stand spatial heterogeneity is a silvicultural goal, quantifying multi-scale spatial heterogeneity can provide the level of spatial resolution needed to better

understand the effect of regeneration methods on ecological patterns and processes operating at sub-stand scales.

Although there has been substantial investment into research on the effects of gap-based regeneration methods on available resources and tree regeneration, most studies focus on either gap-scale or stand-average patterns (Puettmann et al. 2009). Few studies have made stand-scale, internal spatial heterogeneity a focus of investigation. The paucity of studies on within-stand spatial patterning is partly related to an emphasis on using sampling methods designed to capture mean responses at a specific scale, which limits their usefulness in spatial pattern analysis over multiple scales (Miller et al. 2002, Scheller and Mladenoff 2002, Puettmann et al. 2009). One approach to capturing spatial pattern across a range of spatial scales is to incorporate sampling and statistical methods designed to detect explicit spatial pattern (Rossi et al. 1992, Miller et al. 2002, Scheller and Mladenoff 2002, Fortin and Dale 2005). A commonly cited problem with spatial pattern analysis, particularly on unvariate patterning in space, is the difficulty of establishing causation between pattern and process in an ecological setting (Cale et al. 1989, Levin 1992, Legendre 1993). Although there are many pitfalls to inferring process from pattern, understanding the explicit spatial pattern of a phenomenon provides the investigator with a snapshot view of causal process reflected in the pattern itself.

Since spatial pattern analysis combines a suite of tools that can address issues in forest science that more traditional methods have been largely incapable of resolving, the analysis of spatial pattern has helped to advance forest ecology. One area in particular that has benefited substantially from the application of spatial statistics in forest ecology is the study of forest understories. In northern hardwood forests of the Lake States,

understory patch size of plants was negatively associated with sugar maple sapling density (Miller et al. 2002) and positively correlated with coarse woody debris and light heterogeneity (Scheller and Mladenoff 2002). Forest regeneration research has benefited from spatial analysis as well. McDonald et al. (2003) used point pattern analysis to assess oak regeneration failure in a Piedmont mixed-hardwood stand and concluded that oak regeneration was negatively associated with midstory red maple. In an old-growth hardwood stand in east-central Indiana, understory tree regeneration was aggregated at a scale substantially larger than the size of canopy gaps indicating that larger-scale factors have an effect on controlling regeneration distribution (Ward and Parker 1989). Houle (1994) described spatial and temporal pattern of forest regeneration and found environmental heterogeneity, in space and time, promotes competitive coexistence of northern hardwood species. In a related study, environmental variables exhibited less spatial heterogeneity than seedling emergence and survival suggesting that environmental variation alone does not explain the spatial pattern during the early stages of tree regeneration in northern hardwoods (Messaoud and Houle 2006).

Few studies have used explicit spatial pattern of understory attributes as response variables on the effects of forest management. Scheller and Mladenoff (2002) used geostatistical analysis of species patterns and plant diversity to compare old-growth and managed northern hardwoods. In a study on the effect of spatially-variable dispersion of overstory retention on understory light, Battaglia et al. (2002) found that spatial heterogeneity in understory light became increasingly coarse grained as overtory retention became more aggregated; and degree of overstory aggregation was found to increase biomass increment of planted longleaf pine seedlings (Palik et al. 2003).

Estimates of understory patch sizes of light and seedling abundance in a tropical wet forest were greatest in old-growth than selective harvested and unharvested second growth stands (Nicotra et al. 1999). Pardos et al. (2008) used point pattern analysis of natural regeneration to compare stands regenerated with uniform and group shelterwood methods and found conifer germinants were spatially clumped regardless of light conditions and the association between seedling distribution and light was stronger for the uniform shelterwood; which the authors concluded was more directly a response to greater harvest intensity and site preparation applied within the uniform system practiced traditionally in central Spain. Using spatial pattern of forest attributes that vary at sub-stand scales, like tree regeneration density and resource availability, as response variables provides another means of assessing the ecological effect of treatments in silviculture experiments.

The objective of this study was to investigate the effect of gap harvesting on spatial pattern of tree regeneration and canopy openness in the understory of mature mixedwood stands of the Acadian forest. Since the primary aim of gap-based, regeneration methods is to increase the availability of resources limiting tree regeneration, the spatial pattern of canopy openness was included as a second response variable to represent resource heterogeneity. We hypothesized that the grain (i.e., range of spatial dependence and the number of scales patchiness is observed) of understory spatial pattern in tree regeneration and canopy openness will increase in stands treated with gap harvesting and as the size of the harvest gaps increase. Specifically, we expected that: 1) the range of spatial dependence will increase with gap size, and 2) spatial pattern in unharvested stands would likely to be random at medium and larger

scales, while stands treated with gap harvesting would likely exhibit patchiness. Additionally, since we expected that both tree regeneration and resource availability would be spatially correlated, we also wanted to examine the association between regeneration and canopy openness, and if it existed, whether this association was affected by gap harvesting.

2.3. METHODS

2.3.1. Study Site

We used stands that are part of the long-term experiment of the Acadian Forest Ecosystem Research Program (AFERP) at the University of Maine. The AFERP study is located on the Penobscot Experimental Forest (PEF) in the towns of Bradley and Eddington, Maine (Figure 2.1). The PEF is a 1,680-ha predominantly forested tract owned by the University of Maine Foundation and is jointly managed by the University of Maine and the U.S. Forest Service, Northern Research Station. Climate in central Maine is cool and humid. February (avg. temp. -7.1º C) and July (avg. temp. 20.0º C) are the coldest and warmest months, respectively. Average precipitation is 107 cm. The growing season is approximately 160 days at the PEF. Soils at the PEF are complex resulting from glaciation. The dominant parent material is a fine-textured till derived from sedimentary bedrock which have developed into loams. Contained within the PEF is a broad gradient of soil drainage classes; however, a large portion of the forest grows on soils classified as somewhat poorly to poorly drained and moderate to low productivity (Sendak et al. 2003).

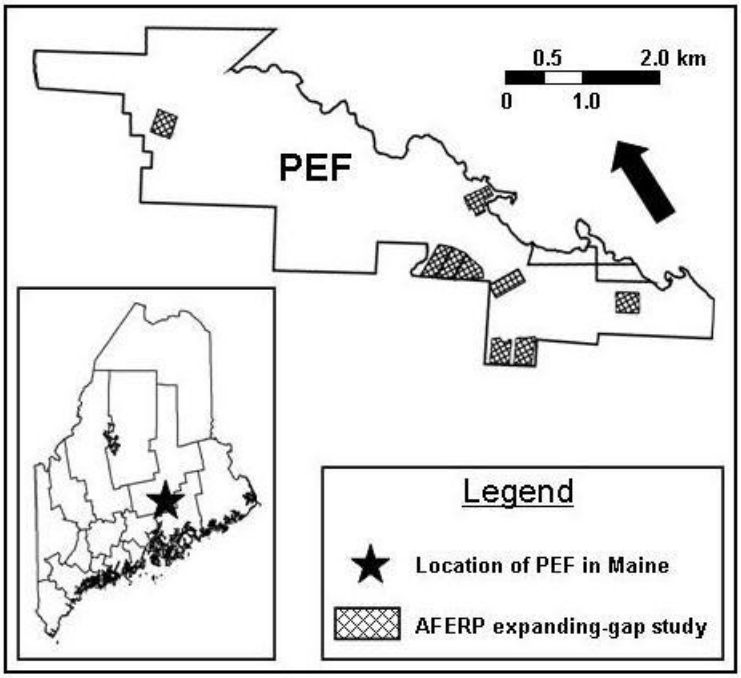

Figure 2.1. Location of AFERP study on the Penobscot Experimental Forest (PEF) with location of PEF in Maine (inset image, bottom left).

The PEF is within the Acadian Forest, a conifer-dominated mixedwood ecosystem covering much of Maine and the Canadian Maritimes (Halliday 1937). The dominant conifer species on the PEF include balsam fir, eastern hemlock (*Tsuga canadensis*), red spruce, white spruce (*Picea glauca*), northern white-cedar (*Thuja occidentalis*), and white pine (*Pinus strobus*), while the common hardwood species include red maple (*Acer rubrum*), paper birch (*Betula papyrifera*), gray birch (*B. populifolia*), trembling aspen (*Populus tremuloides*), and bigtooth aspen (*P. grandidentata*) (Sendak et al. 2003).

2.3.2. Experimental Design

AFERP is a long-term experiment established to investigate the ecological effects of expanding-gap silvicultural methods that were designed based on patterns of natural disturbance in the Acadian forest (Saunders and Wagner 2005, Seymour et al. 2006). The design of the expanding-gap experiment is a randomized complete block design with two levels of the expanding-gap harvesting and an unharvested control. Each treatment is replicated three times, yielding a total of nine experimental units. Each experimental unit or research area (RA) ranges in size from 9.0 to 11.5 ha. The two treatment levels of expanding-gap harvesting are based on total area harvested per entry and the regeneration period between gap expansions. In the small-gap treatment, 10 % of the RA is harvested during each entry with a 20-year regeneration period between expansions, meanwhile, in the large-gap treatment, 20 % of the RA is harvested per entry with a 10-year regeneration period between expansions. Target gap size was 0.1 and 0.2 ha, respectively, for small- and large-gap treatments. However, since the silvicultural prescription required working with the variation in stand conditions needing regeneration and understory release, gaps range in size from 0.01-0.13 ha for small-gap treatment and 0.03-0.22 ha for the large-gap treatment. See Saunders and Wagner (2005) for additional details about the AFERP experimental design.

For this study, we used two of three replicates, or six experimental units, of the AFERP study. Since the present investigation was initiated after the second harvest entry, or first round of gap expansions, into replicate block one (RAs 1-3), only replicate blocks two and three (RAs 4-6 and 7-9, respectively) were included in this investigation. Our intention was to investigate the effects of gap harvesting, not the expanding-gap

method, on understory spatial pattern. Additionally, we focused on spatial patterning of regeneration after the stands have had 10 years to recover from the first harvest gap entries. Data for this study was collected during the summers of 2006 (replicate block two) and 2007 (replicate block three).

Two of the six experimental units (RAs 4 and 7) are bisected into roughly equal sized stands by an unpaved, forest road (~ 6 m wide). From a spatial perspective, the road created a discontinuity in the understory pattern of both RAs. Therefore, we treated each half of the RAs 4 and 7 as separate stands or "pseudo" experimental units, providing a total of eight stands for this analysis. The divided stands are referred to here as RA4 west (RA4w) and east (RA4e), and RA7 north (RA7n) and south (RA7s).

2.3.3. Understory Vegetation and Light Sampling

Sampling was performed using unequally spaced plots placed along 100-m transects originally established for monitoring deadwood over the expanding-gap study. The center of each transect was positioned at the point of intersection on a 50 x 50 m grid that was located across each sample stand (RA). The orientation of each transect was determined by randomly assigning an azimuth to each transect. The number of transects varied from 30-36 depending on the size and number of grid points in each RA. The spatial arrangement of plots along each transect followed the repeating pattern, cyclical sampling design (RPCS) of Scheller and Mladenoff (2002). Conceptually, a cycle consists of a series of plots stacked end-to-end; a sub-set of which are sampled. Following Scheller and Mladenoff (2002), sampling was performed on 4 of 13 plots per cycle with sampling on the 1st, 2nd, 4th, and 10th plots in sequence (Figure 2.2). This

sampling method is designed to capture inter-plot variability across a range of spatial scales from shortest to furthest inter-plot distances, which can be used to estimate spatial pattern (Scheller and Mladenoff 2002). Two to four cycles were established per transect, which yielded a total 1,742 sample plots with the sample size ranging from 277 to 336 per RA.

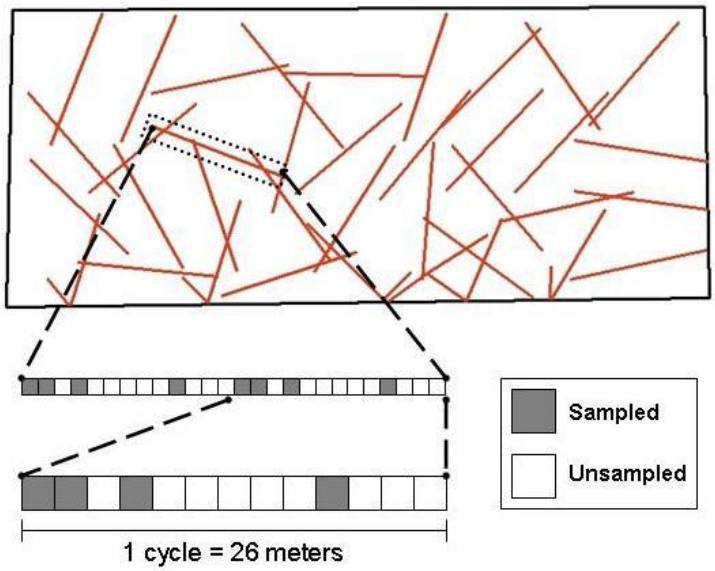

Figure 2.2. Depiction of randomly-oriented transects (solid red lines) within AFERP RA4 (control) and an illustration of the 4 of 13 RPCS design (image projected below).

Understory vegetation and light sampling were measured at each sample point along the transects in order to test for associations. Vegetation sampling included measuring all tree species in 4 m^2 (2 m x 2 m) plots and a single measurement of understory light was recorded above the center of each plot. Since the intent of this investigation was to evaluate the spatial response of established tree regeneration to gap harvesting, the lower size limit for regeneration sampling was set to 0.1 m tall to excluded recent germinants. Stem counts of all woody plants (≥ 0.1 m tall, but < 2 m tall) were tallied to species level and further stratified into three height classes: small (≥

0.1 m, but < 0.75 m), medium (≥ 0.75 m, but < 1.4 m), and large (≥ 1.4 m, but < 2 m) regeneration. During July of 2006 (RAs 4-6) and 2007 (RAs 7-9), the understory light environment was sampled using the LAI-2000 Plant Canopy Analyzer (LI-COR Inc. 1992). LAI-2000 was used to get an estimate of canopy openness measured directly as diffuse, non-interception (DIFN), which indicates that fraction of sky not being intercepted by foliage (LI-COR Inc. 1992). The value of DIFN was calculated based on the difference between readings taken from two sensors each with a 180° cover; one placed out in an open field greater than two canopy tree heights from the nearest forest edge (i.e., above canopy readings) and the other taken above the center of understory plots (i.e., below canopy readings). Readings from both sensors were recorded at a height of 1.4 m above the ground during the morning and evening to avoid exposure to direct sunlight.

2.3.4. Variables and Spatial Analysis

Analysis was performed using R statistical software, version 2.8.1 (R Development Core Team 2008). Stems counts of tree regeneration across the three height classes were combined by species to represent densities at the plot level. Diffuse, non-intercepted sunlight (DIFN) was used to represent canopy openness. DIFN was calculated using readings from the center three rings, or within 43° from vertical; a procedure recommended by Puettmann and Reich (1995) to restrict measures of light only to trees that directly influence the seedling light environment and to reduce the influence of sun flecks.

Exploratory data analysis (EDA) found all stand-level DIFN and species distributions were non-normal. Normality was achieved by taking the quarter-root of DIFN; however, normality could not be achieved with any simple transformation for species-level data. Normality was achieved by taking the square-root of total regeneration density for all tree species combined. The analysis that follows was performed on the square-root of plot-level densities of total tree regeneration (i.e., all species combined) and quarter-root of plot-level DIFN. Since only two replicates of the expanding-gap experiment were used for this study, traditional treatment comparisons (e.g., ANOVA) of stand-averaged responses were not included in this report due to insufficient replication. Instead, regeneration density and DIFN (henceforth referred to as canopy openness) were summarized as treatment-level means for both variables, percentage stocking for regeneration density, and coefficient of variation (CV) for canopy openness.

Analysis of spatial pattern often requires corroboration among two or more analytical methods to develop multiple lines of evidence for hypothesis testing (Legendre and Fortin 1989, Rossi et al. 1992, Fortin and Dale 2005). For this investigation, two methods of geostatistical analysis were used to determine spatial patterning: 1) semivariogram estimation and 2) spatial autocorrelation. Although both methods compare values of an attribute at different points in space as a function of their separation distance (i.e., spatial covariance), the two differ fundamentally in how points are compared. The semivariogram (henceforth called variogram) is based on the semivariance of the attribute in question and is a measure of dissimilarity; while spatial autocorrelation measures similarity between points for the variable of interest.

Variogram estimation, or variography (Cressie 1993), was used to describe small-scale spatial pattern of tree regeneration density and canopy openness, while spatial autocorrelation was used to describe and test the significance of spatial pattern across a range of spatial scales within each stand individually. Since the variogram emphasizes heterogeneity and spatial autocorrelation focuses on correlation, the combination of the two provides a more complete picture of spatial pattern and its significance (Scheller and Mladenoff 2002). Spatial patterns inferred from both methods were then used to determine the effects of gap harvesting on understory spatial pattern over a range of spatial scales. Assessing the effect, if any, of gap harvesting on small-scale patterning relied on comparisons based on both absolute differences among treatment means and overlap of the 95 % confidence intervals.

All pairs of plots were binned into distance classes of 6 m. Bin size was set to 6 m to capture at least 100 pairs in each bin; a binning procedure recommended by Fortin and Dale (2005) for estimating spatial structure. In four out of the eight stands included in this study (RAs 5, 6, 8, and 9), 32 distance classes per RA were selected to equal the number of 6 m distance classes necessary to represent approximately 1/2 the spatial extent of the area under investigation, a spatial extent of 192 m. Sixteen, 6-m distance classes were selected to represent the four split stands, an extent of 96 m. The two main effects of this difference in spatial extents on spatial analysis are that it limits both corroboration between replicates for control and small-gap treatments to scales below 96 m and the ability to make comparisons among treatments at spatial scales beyond 96 m (an impact for spatial autocorrelation analysis only). However, this difference will not affect corroboration between variogram and spatial autocorrelation analyses, which

applies primarily to small-scale pattern (< 50 m). Furthermore, we felt that the 6-m bin size and extent of at least 100 m was sufficient for capturing sub-stand spatial patterning of both response variables linked to canopy gaps, whether natural or harvest generated, in the stands included in this study.

After binning, a second round of EDA was conducted to test geostatistical assumptions of isotropy and stationarity for both tree regeneration and DIFN. Isotropy (i.e., distance dependence) was tested by examining variograms in four directions 90° apart starting from true north (or 0° azimuth). This procedure tests for anisoptropy or directional dependence (Cressie 1993). Stationarity was tested through trend surface analysis to detect larger scale trends, which checks for constancy in both mean and variance of response variables across the study area (Cressie 1993). EDA found no evidence of either anisotropy or non-stationarity. All analyses of spatial pattern assumed omni-directional spatial dependence.

The first step in variography was to create empirical variograms of both regeneration density and canopy openness for each stand. Empirical variograms were then checked for trends at larger scales, which served as a second check of the stationarity assumption. Next, three theoretical variogram models, 1) spherical, 2) exponential, and 3) Gaussian, were fit to empirical variograms using weighted least squares (WLS) estimation to determine which approach best approximated spatial dependence structure based on lowest minimized sums of squares. For WLS estimation, greater weight was given to shortest distance classes to better represent patterns at smaller scales; a procedure recommended by Cressie (1993). In all cases, either spherical or exponential models provided the best fit and, therefore, the Gaussian model was not

used (Appendix B). Estimates of the variogram range (separation distance where semivariance reaches a plateau), sill (value of semivariance where the plateau is reached), and nugget (semivariance when distance is zero) from best fit models were used to make comparisons among treatments (Appendix B). The variogram range was less than the spatial extent for all variograms indicating that all were bounded, which means spatial dependence was successfully captured (Cressie 1993). The variogram range was used as an estimate of patch diameter (Fortin and Dale 2005), while the nugget and sill were used to estimate structural strength, which indicates the strength of spatial dependence (Battaglia et al. 2002, Dent and Grimm 1999) captured by the variogram range. Structural strength was calculated by subtracting the nugget from sill, dividing the difference by the sill, and expressed as a percent by multiplying the quotient by 100 (Battaglia et al. 2002, Dent and Grimm 1999). Kriged maps were generated for regeneration density only using the best fit theoretical variogram models, including estimates of the range, sill and nugget, in ArcGIS version 9.0.

Spatial autocorrelation was estimated using Moran's Index (Moran's I). Moran's I is commonly used to represent spatial autocorrelation in ecological data (Fortin and Dale 2005). Moran's I indicates the direction (positive or negative) and magnitude of spatial autocorrelation between two or more pairs of observations a set distance apart (Fortin and Dale 2005). Significance testing of Moran's I values across a range of spatial scales were performed using Moran's tests. Since multiple tests of autocorrelation were performed on each dataset (either 16 or 32 Moran's tests), the p-value was adjusted using the Bonferroni correction. The *a priori* alpha for this investigation was set at 0.05;

therefore, significance was assessed according to a corrected p-value of 0.0031 and 0.0016 for 16 and 32 tests, respectively.

In order to facilitate interpretations of spatial patterns from autocorrelation analysis, the value and significance of Moran's I values were graphically displayed as spatial correlograms using their respective distance class midpoints for spatial reference. The spatial correlogram is an efficient means of determining spatial pattern and its statistical significance, which can reveal spatial patterns of ecological importance, such as patchiness and gradients (Sokal 1979, Legendre and Fortin 1989, Legendre and Legendre 1998, Fortin and Dale 2005). The procedures used for interpreting pattern from spatial correlograms follow the recommendations of Fortin and Dale (2005). Spatial pattern inferred from correlograms can be divided into two categories: 1) *global*, or pattern over the full spatial extent (e.g., randomness, gradients, and patchiness), and 2) *local*, or pattern at scales smaller than the full extent (e.g., patch size and distance between adjacent patches). However, global significance of correlograms should be determined before interpreting their pattern. For this investigation, Moran's test needed to be significant at the appropriate Bonferroni-corrected critical value for at least one distance class. Failure to satisfy global significance indicates a globally-random spatial distribution, which may reflect complete spatial randomness or fine-grain pattern below the scale of spatial resolution over the analysis extent (Sokal 1979, Legendre and Fortin 1989, Legendre and Legendre 1998, Fortin and Dale 2005). Spatial correlograms were compared with theoretical correlogram structures published in Fortin and Dale (2005) in order to determine global patterns. Range of autocorrelation, measured as the mid-point of the distance class farthest from zero along a consecutive series of significant and

positive Moran's I values, was used as a second estimate of patch size for both tree regeneration density and canopy openness along with variogram ranges. For significant Moran's I values beyond the correlogram range, positive values (i.e., peaks) were interpreted as scales where patches coexist, while negative values indicated scales in which areas of high and low (i.e., troughs) density coexist. Peaks and troughs on correlograms were interpreted as indicators of pattern at larger spatial scales, which was used to compare treatments.

Additional tests were performed to examine the relation between regeneration density and canopy openness. Since timber harvesting indirectly affects regeneration by altering understory conditions, we expected tree regeneration density and canopy openness to be positively correlated. Because the sampling method used in this study violated the assumption of independence, testing this hypothesis required a method that could account for spatial autocorrelation intrinsic to both factors in order to obtain an estimate of their pure association (Legendre and Legendre 1998). The partial Mantel's test can be used to partial out the effect of space when testing for associations among factors (Leduc et al. 1992, Legendre and Legendre 1998). Mantel's test is a correlation performed on distance matrices (either similarity or dissimilarity) representing each factor (see Legendre and Legendre 1998 for more details on the Mantel's Test). For this analysis, dissimilarity matrices were created using Euclidean distance to represent three factors: tree regeneration density, canopy openness, and two-dimensional coordinates (i.e., space). Partial Mantel's tests were performed using R to identify potential associations among these factors for each stand individually and significance was assessed at $p < 0.017$ (Bonferroni-correction for three tests at $\alpha = 0.05$).

2.4. RESULTS

2.4.1. Non-Spatial Pattern

Treatment-averaged regeneration density ranged from 18,986 stems ha^{-1} in the control treatment to 25,410 stems ha^{-1} in the large-gap treatment, while mean stocking ranged from 91.2% to 98.6% for control and large-gap treatments, respectively (Table 2.1). The trend in treatment-averaged densities, stocking, and canopy openness among the three treatments followed the pattern: large-gap > small-gap > control. The coefficient of variation (CV) for canopy openness was included as stand-level index of heterogeneity (i.e., higher values indicate greater inter-plot variability). The trend in treatment-averaged coefficient of variation for canopy openness across treatments followed the trend: small-gap > large-gap > control. However, the difference between mean CVs for the control and large-gap treatments was minor compared to differences with the mean for the small-gap treatment.

Table 2.1. Treatment comparison of mean density and stocking of regeneration and mean and coefficient of variation of canopy openness for AFERP study sites.

Treatment	**Regeneration Density**		**Canopy Openness**	
	Mean (stems per ha)	**Stocking (%)**	**Mean (%)**	**C.V.* (%)**
Control	18,986	91.2	4.6	79.6
Small-gap	22,443	92.6	7.9	98.1
Large-gap	25,410	97.5	10.8	79.2

* Coefficient of variation

2.4.2. Univariate Spatial Pattern

Global significance was satisfied for all spatial correlograms of both tree regeneration density and canopy openness; therefore, none were globally random (Figure 2.3). Compared to correlograms published by Fortin and Dale (2005), correlograms from this analysis most closely resembled structures indicating patchiness; suggesting that the distribution of understory tree regeneration and canopy openness were patchy in all stands. For tree regeneration, this patchy sub-stand pattern was seen in kriged maps, which was represented by relatively discrete hotspots in regeneration density (Figure 2.4). Because global patterns were relatively similar across treatments, spatial correlograms for RAs 5, 6, 8, and 9 were truncated to 93 m for graphical convenience.

Moran's I was positive and significant ($P < 0.0031$ and 0.0016 for 16 and 32 tests, respectively) at the first distance class (0-6 m class) for all correlograms. With the exception of four correlograms for the control treatment, (two for regeneration [RA4w and RA4e] and two for canopy openness [RA4w and RA8]), correlogram ranges extended beyond the first distance class. Although not shown on several of the truncated correlograms, 13 of 16 correlograms for both variables combined had significant spatial autocorrelation beyond the correlogram range. The only correlograms without significant autocorrelation detected beyond the range were regeneration density for RA7n and both regeneration density and canopy openness for RA4e; small-gap and control treatments, respectively. Moran's test detected significant spatial autocorrelation at a greater number of scales beyond the correlogram range for the gap-harvested treatments compared to the unharvested controls. In most cases, significant autocorrelation beyond the correlogram range was positive.

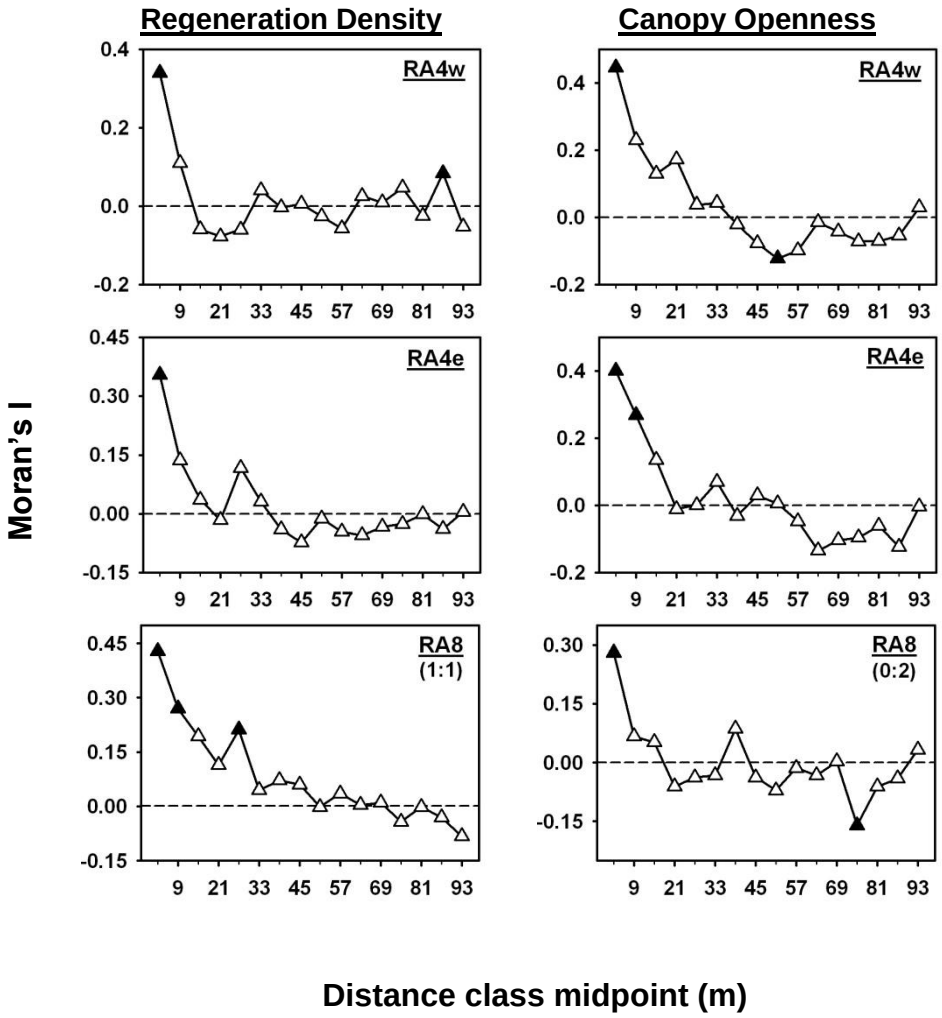

Figure 2.3. Spatial correlograms of regeneration density (left column) and canopy openness (right column) over a 93-meter extent in the understory of controls (RAs 4w, 4e, and 8), small-gap (RAs 5, 7n, and 7s), and Large-gap (RAs 6 and 9) treatments of the AFERP study. Solid triangles represent significant spatial autocorrelation at Bonferroni-corrected $\alpha = 0.05$. Ratio in Parentheses is the proportion of significant testst hat are positive from 99 to 189 meters for truncated correlograms (RAs 5, 6, 8, and 9).

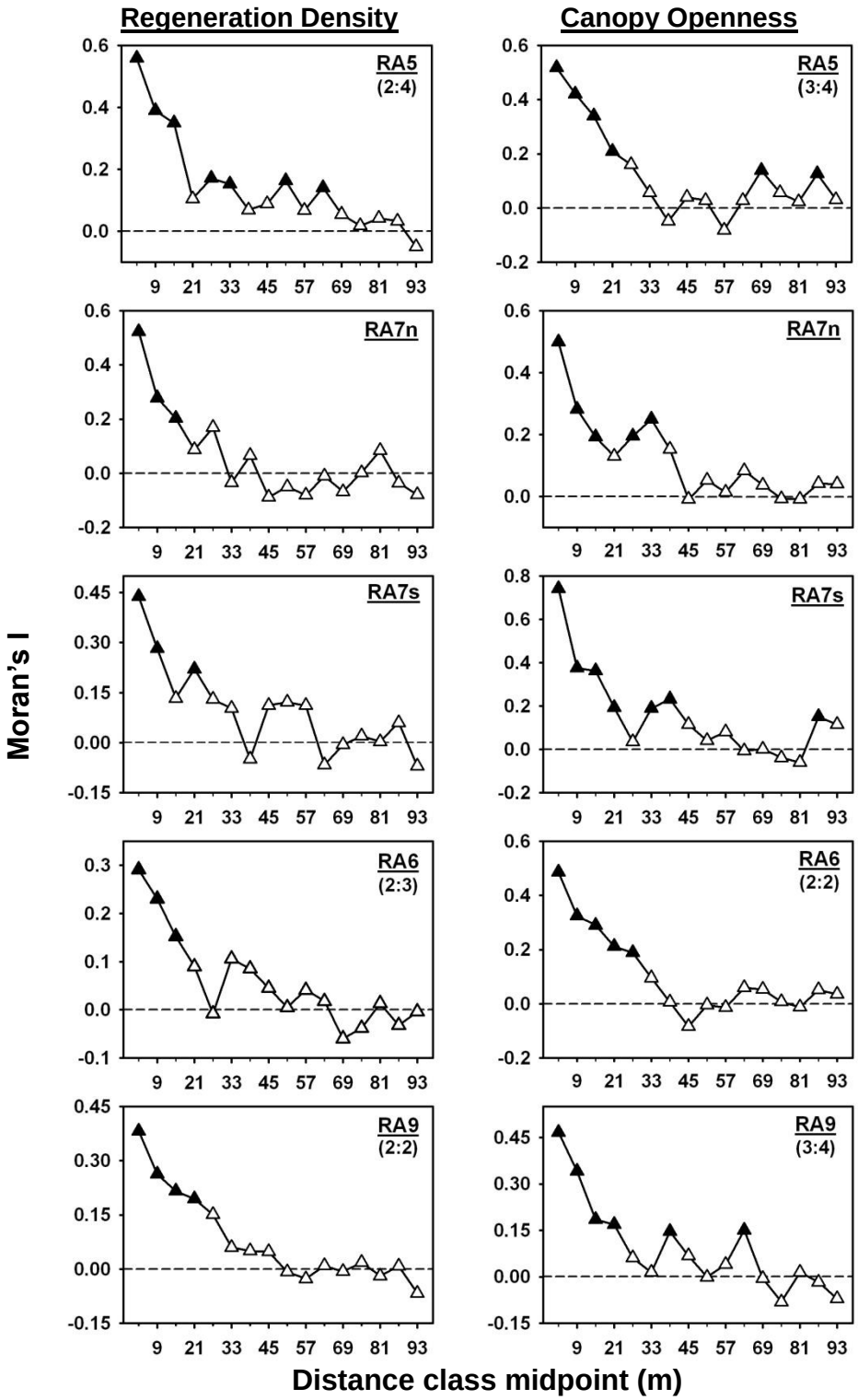

Figure 2.3. Continued.

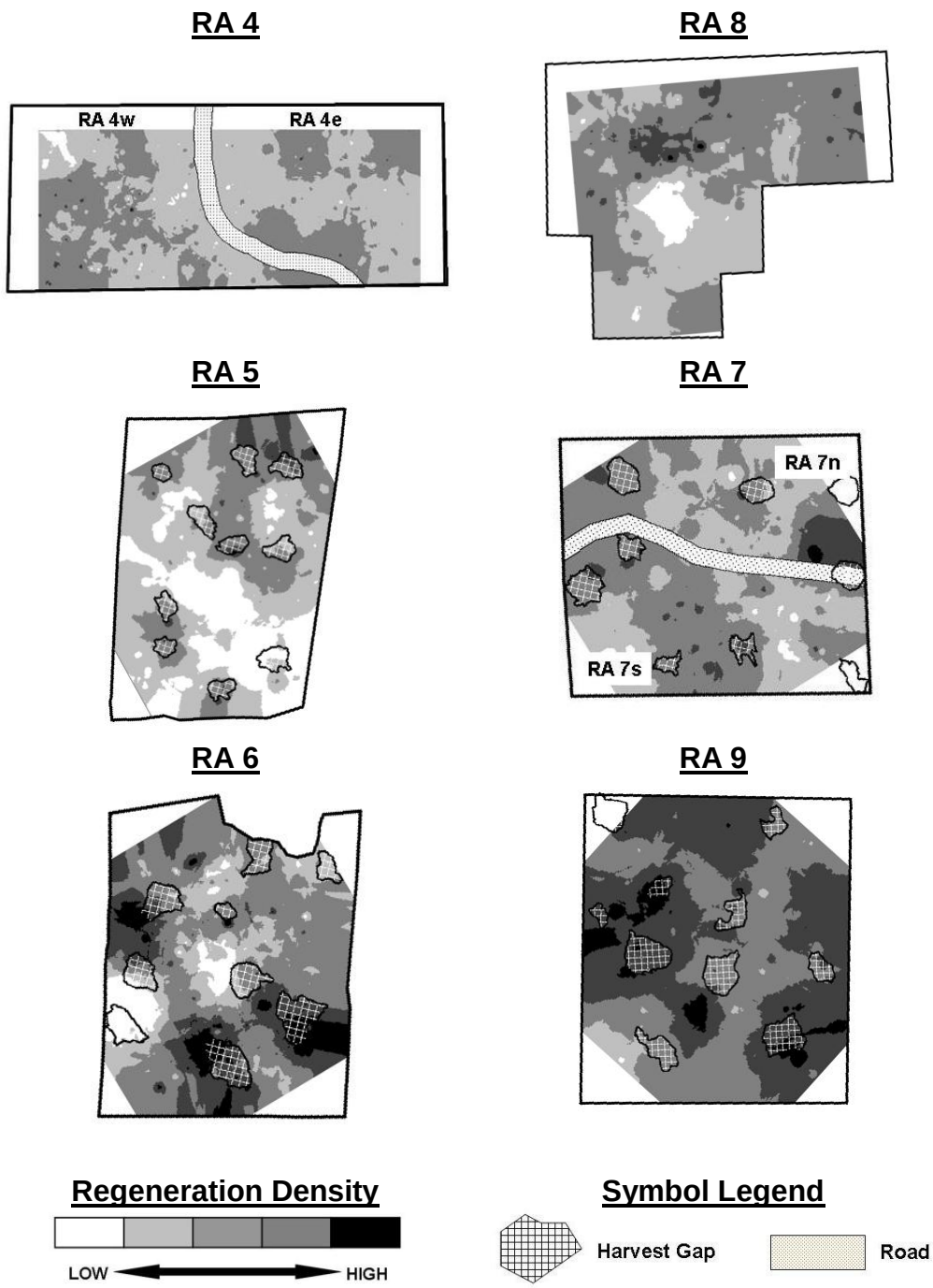

Figure 2.4. Interpolated maps of regeneration density generated by Ordinary Kriging for control (top row), small-gap (middle row), and large-gap (bottom row) treatments of the AFERP study. White zones on the edges of maps represent unsampled areas.

Mean structural strength (SS) for both regeneration density and canopy openness across treatments are presented in Table 2.2. The trend in SS over treatments followed the pattern: small-gap > control > large-gap; which indicates that SS for both factors did not follow the trend in gap size.

Table 2.2. Treatment comparisons of mean structural strength of theoretical variogram models of regeneration density and canopy openness for the AFERP study. Values in parentheses are two standard deviations.

Treatment	**Structural Strength (%)**	
	Regeneration Density	**Canopy Openness**
Control	71.1 (9.7)	73.9 (30.6)
Small-gap	76.7 (13.9)	78.2 (27.8)
Large-gap	53.7 (13.5)	61.8 (13.8)

Mean variogram and correlogram ranges for tree regeneration density and canopy openness increased with increasing mean canopy gap diameter among treatments (Figure 2.5). Both mean range values were notably less than the mean gap diameter in harvest units and more equable in controls. Comparing both variables within treatments, the mean variogram range was consistently larger than the mean correlogram range, yet their 95% confidence intervals overlapped.

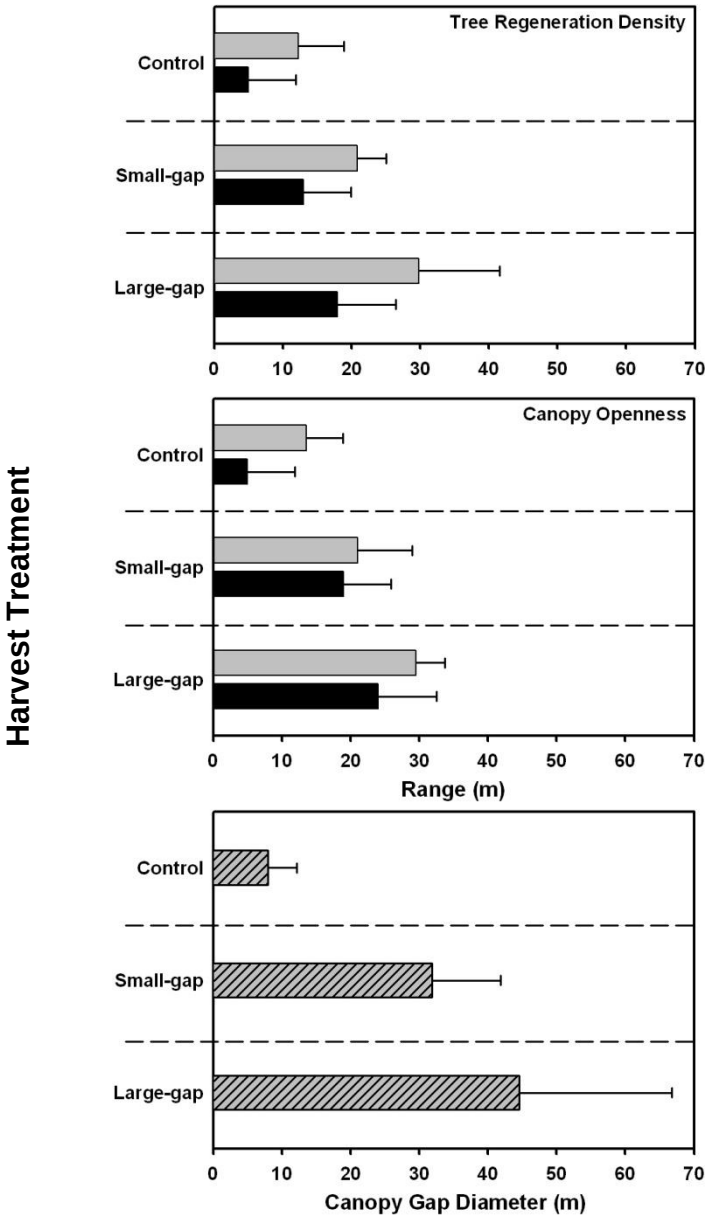

Figure 2.5. Mean ranges of spatial autocorrelation (black bars) and semivariance (gray bars) for regeneration density (top figure) and canopy openness (middle figure) along with mean canopy gap diameter (bottom figure; hatched bars). Error bars are two standard deviations. Note: mean gap size for the control treatment comes from Schofield (2003).

2.4.3. Association Between Regeneration Density and Canopy Openness

Although Mantel correlation coefficients were small (Table 2.3), other studies using partial Mantel tests have noted relatively small magnitudes even when highly significant statistically (Leduc et al. 1992, Urban 2003). Partial Mantel's tests detected at least one significant correlation ($p < 0.017$) in all stands except RA4e. A significant correlation was detected between regeneration density and canopy openness only for stands treated with small-gap harvesting. The correlation between canopy openness and space was positive and significant in seven of eight stands, while a significant positive correlation between regeneration density and space was detected in all treatments for at least one stand and in both stands treated with large-gap harvesting. With the partial Mantel's test, a correlation with space represents spatial pattern in one factor that is unique from pattern in other factors included in the analysis. Therefore, at least part of the spatial pattern exhibited by canopy openness was not associated with tree regeneration in nearly all stands, while regeneration exhibited spatial pattern not associated with canopy openness in at least one stand for all treatments. This interpretation is further evidenced by variability in the correspondence between locations of harvest gaps and regeneration density hotspots seen on Kriged maps (Figure 2.4).

Table 2.3. Partial Mantel correlations and test results by stand and treatment. Columns are combinations of partial Mantel tests among regeneration density (Regen), canopy openness (Canopy), and spatial location of plots (Space).

		Partial Mantel Test Results		
Treatment	**Stand**	****Regen by Canopy / Space**	**Regen by Space /Canopy**	**Canopy by Space**
Control	RA4w	0.016	0.064	0.142*
	RA4e	-0.002	0.038	0.049
	RA8	-0.013	0.081*	0.134*
Small-gap	RA5	0.238*	0.058	0.065*
	RA7n	0.084*	0.072*	0.163*
	RA7s	0.066*	0.038	0.093*
Large-gap	RA6	0.036	0.071*	0.074*
	RA9	0.027	0.084*	0.171*

* Statistically significant at $p < 0.017$
** In column titles: the first and second names represent association components, while the third name is the factor accounted for in partial correlation.

2.5. DISCUSSION

2.5.1. Non-Spatial Pattern

The high regenerative capacity often attributed to the Acadian forest (Davis 1991, Smith 1991, Seymour 1992) was reflected in the high density and stocking of tree regeneration (≥ 0.1 m tall, but < 2 m tall) across treatments (> 19,000 stems ha^{-1} and 90%, respectively). A previous study on regeneration density, stocking, and composition from a long-term silviculture experiment near this study found natural regeneration to be abundant and well-distributed among a wide range of harvest treatments (Brissette 1996).

Trends in stand-level tree regeneration density and stocking and canopy openness across treatments in this experiment suggested these variables responded positively to

increasing gap size. The increase in stand-level tree regeneration density and canopy openness observed across treatments was consistent with what is generally accepted about the effects of overstory removal intensity on both factors (Smith et al 1997, Barnes et al. 1998). Past studies have found a higher density of tree regeneration within canopy gaps than non-gap areas (Runkle and Yetter 1987, McClure and Lee 1993, Schofield 2003); a condition for which the gap phase of Watt (1947) largely hinges upon. At the stand-level, one would expect stocking of tree regeneration to increase along with gap size, since larger gaps translates to greater canopy openness and, therefore, greater site occupancy by regeneration over the stand area.

2.5.2. Treatment Effects on Univariate Spatial Patterns

Results from our geostatistical analysis indicated that gap harvesting influenced the spatial patterning of both regeneration density and canopy openness at small spatial scales. Specifically, the effect of gap harvesting on small-scale patterning was indicated by the increase in both variogram and correlogram ranges of spatial dependence along the gap-size gradient produced in this experiment. These results support the hypothesis that the ranges of spatial dependence of both tree regeneration and canopy openness responded positively to gap size. Since the range of spatial dependence can be used as an estimate of patch size, one ecological interpretation is that understory patch diameters for tree regeneration and canopy openness increased with increasing gap size. Although this result may seem intuitive, few studies have documented a relationship between gap size and the explicit spatial patterning of understory attributes in stands treated with gap harvesting. Results from our analysis of canopy openness corroborate those from a study

on explicit spatial pattern in a longleaf pine forest by Battaglia et al. (2002) who found the understory light environment became increasingly coarse textured as gap size increased. To the best of our knowledge, our study is the first to document a positive response between the size of harvest gaps and the patch diameter understory regeneration.

In contrast to Battaglia et al. (2002), however, who found that the structural strength of spatial dependence (SS) for understory sunlight increased with gap size, results from our study found SS for canopy openness was highest for the small-gap treatment, lowest for the large-gap treatment, and intermediate for the unharvested control; a pattern also reflected by the trend in SS for regeneration density across treatments. In terms of ecological patterning, Battaglia et al. (2002) interpreted SS as an indication of how strong the variability in a factor was spatially defined (i.e., discrete boundaries defining patches in space). Borrowing from this interpretation, our results indicated that understory patchiness in both variables was most spatially discrete in the small-gap treatment and more spatially discrete in the unharvested control than in the large-gap treatment. Compared to the open-canopied, longleaf pine forest studied by Battaglia et al. (2002), mature, Acadian forest stands tend to be structurally heterogeneous and support a diverse mixture of tree species (Seymour 1992), which may partly account for the differing trends in SS observed here. Furthermore, treatment levels in Battaglia et al. (2002) were set to create different gap sizes holding basal area reductions equal among treatments, while treatment levels in our experiment were set according to both gap size and total harvest area, roughly a doubling for both in the large-gap treatment compared to the small-gap treatment; which could also account for

differences in SS for estimates of sunlight between studies. Finally, the present study captured spatial patterning ten years after treatments were applied, while Battaglia et al. (2002) analyzed patterning in understory sunlight shortly after harvest. Collectively, differences in ecological setting, experimental treatments, and time lag between treatment application and spatial sampling between these two studies likely contributed to differences in observed spatial patterning.

Although analysis of spatial autocorrelation was effective in addressing spatial pattern over multiple scales in this investigation, no studies to our knowledge have used correlograms to examine the effect of gap harvesting on spatial patterning of understory vegetation attributes across a range of sub-stand scales. Both regeneration density and canopy openness exhibited some degree of patchiness over the extents analyzed regardless of treatment, particularly at small spatial scales. Interpretations of correlograms for both variables at least partially support the hypothesis that spatial pattern in unharvested stands were more likely to exhibit randomness at larger scales, while stands treated with gap-harvesting were more likely to exhibit patchiness. Although all treatments had at least one correlogram with significant autocorrelation detected the beyond the correlogram range, support comes from the higher frequency of significant autocorrelation at larger spatial scales for stands treated with gap harvesting compared to the unharvested controls over the range of scales analyzed. Significant autocorrelation at larger scales beyond the correlogram range were predominantly positive, which can be interpreted as spatial scales where patches coexist (Fortin and Dale 2005). Therefore, the non-random patterns at larger scales detected from correlograms may have reflected the scales where patches co-occurred. Greater patchiness at larger

spatial scales in harvested stands compared to unharvested control stands appears to be linked to the creation of multiple harvest gaps dispersed over the stands treated with gap harvesting, which in turn, generated internal heterogeneity in canopy openness and tree regeneration at the stand level. This interpretation is partially corroborated by the, albeit weak, spatial congruity between harvest gaps and regeneration patches on interpolated maps. Accepting this interpretation, however, requires caution as we were unable to use correlograms for corroborating spatial patterning beyond a scale of 93 m for control and small-gap treatments due to the effect of dividing the RAs.

2.5.3. Regeneration Density and Canopy Openness Associations

The above discussion has focused on the effects of gap harvesting on univariate spatial patterns of tree regeneration and canopy openness individually, and assumed there was no relation between the two. Implicit in theories of forest stand dynamics, such as the gap-phase concept (Watt 1947), and gap-based silviculture regeneration methods, such as single-tree and group selection (Smith et al. 1997), is that canopy gaps facilitate tree regeneration and subsequent recruitment into the canopy. The role of gaps in driving forest regeneration, however, should be more direct because increased seedling survival and growth in canopy openings generally occurs in response to the increased availability of above- and below-ground resources previously held by gap makers (Watt 1947, Runkle 1998).

Since canopy openness can be considered as a surrogate for resource availability, we had predicted that tree regeneration and canopy openness would be positively correlated. Although concurrent increases in range estimates for regeneration density and

canopy openness across treatments suggested a positive association, results from partial Mantel tests indicated that tree regeneration density was positively associated with canopy openness in all three stands for the small-gap treatment, but not the other treatments. Nicotra et al. (1999) also observed similar scales of autocorrelation for both understory light and tree regeneration, but weak associations between the two according to partial Mantel tests. We do not interpret these results to indicate that tree regeneration and canopy openness were not associated in control and large-gap treatments, but that the strength of this association varied by treatment. In other words, these results suggested that the understory distribution of tree regeneration in the small-gap treatment was more tightly coupled with the spatial pattern of canopy openness, while the spatial covariance between regeneration and canopy openness in control and large-gap treatments was similarly weak. A similar conclusion was reached by Pardos et al. (2008) who found the spatial association between scots pine regeneration and understory sunlight was attenuated by two spatially-distinct forms of the shelterwood method (i.e., uniform vs. group shelterwood methods).

Two additonal lines of evidence, 1) the coefficient of variation (CV) for canopy openness and 2) structural strength of spatial dependence (SS) for both variables, corroborated differences in the association between regeneration density and canopy openness among treatments. When interpreting the CV as an index of heterogeneity, the pattern across treatments suggested that canopy openness was most heterogeneous in the small-gap treatment and that heterogeneity was similar for control and large-gap treatments. Factoring in treatment-associated differences in mean canopy openness along with CVs suggested that canopy openness was uniformly high in the large-gap treatment,

uniformly poor in controls, and moderate but more spatially variable in the small-gap treatment. Since resource heterogeneity, particularly sunlight, is considered a key determinant in structuring understory spatial patterning of tree regeneration (Canham et al. 1994, Pacala et al. 1994), the heterogeneity of understory regeneration density may have been reflected to some degree the spatial variability in resources. This position is supported by the covariance between SS values for both variables across treatments. For the small-gap treatment, stronger small-scale spatial dependence in canopy openness appeared to have structured stronger patchiness in regeneration density; meanwhile, weaker spatial dependence for both variables in large-gap and control treatments suggested weaker associations. Together with the partial Mantel results, our findings suggested that regeneration density was most strongly linked to the spatial distribution of canopy openness in stands treated with small-gap harvesting, while the association was weaker for the unharvested controls and large-gap treatments.

2.5.4. Understory Dynamics Inferred from Spatial Pattern

Assuming similar starting conditions for all stands in this study, the fine-grained patchiness observed in the unharvested controls existed in all stands before harvesting. Harvest gaps were deliberately placed to release advance regeneration where possible (Bob Seymour, personal comm.). Applying stand dynamics principles, most of the areas treated with gap harvesting were in the understory reinitiation stage of development (Oliver and Larson 1996). Since a stand's progress into this stage is usually caused by canopy mortality and the formation of persistent canopy openings, tree regeneration and canopy openness are expected to be spatial linked during the understory reinitiation stage.

This assumption suggests that the effect of gap harvesting is to shift the grain of understory spatial pattern from fine to more coarse, which, for tree regeneration, was produced largely through the release of advance regeneration and establishment of new stems from seed or sprouting and, for canopy openness, was produced largely by removing residuals from the overstory. In other words, gap harvesting reinforced an existing understory spatial pattern.

Because of the high relative importance of advance regeneration to the dynamics of the Acadian forest (Westveld 1931, Seymour 1992), pattern reinforcement in response to multiple gaps formed over time seems likely in Acadian forest understories. Pardos et al. (2008) observed pattern reinforcement after shelterwood cutting, which they linked to concurrent changes in understory sunlight and seedbed conditions. Additional evidence for this interpretation comes from Schofield (2003) researching the same stands used in this study who found regeneration was most abundant in harvest gaps, but regeneration was not correlated with gap origin (i.e., regenerated before gap or after gap formation).

Evidence from this study suggested that the effect of gap harvesting on understory spatial pattern was more than a simple reinforcement of an existing understory pattern across treatments. Compared to the control treatment, the increase in understory patch size and SS, prevalence of non-random patterns at larger scales, and positive association between regeneration density and canopy openness for the small-gap treatment appear to represent stronger reinforcement of an existing spatial pattern. The effect of the large-gap treatment appears to be both a reinforcement of pre-treatment pattern as well as the creation of new understory patterning in both canopy openness and regeneration density. Larger understory patch sizes in the large-gap treatment supports reinforcement, while

low values of structural strength and non-significant associations between regeneration and canopy openness seem to indicate a degree of spatial pattern divergence when comparing small-gap and large-gap treatments and a level of convergence for large-gap and control treatments. As previously stated, the two levels of gap harvesting differed according to gap-size and total area harvested, which translated into roughly a doubling of total stem volume harvested, abundance and distribution of skid trails and, therefore, stand-level harvest impact. Additionally, the large-gap treatments also received light partial cutting in the matrix between harvest gaps to remove short-lived species across canopy strata (Seymour 2005). Doubling the area treated by gap harvesting in conjunction with dispersed cutting in the matrix likely produced a more homogeneous understory environment in the large-gap treatment compared to the small-gap treatment, particularly in the matrix, which was reflected by the lower CV in canopy openness for the large-gap treatment relative to the small-gap treatment. Meanwhile, the large-gap treatment appeared to have generated a degree of within-stand heterogeneity more comparable to the unharvested controls, which was reflected by low SS values for both variables and, especially, similar CVs for canopy openness. In both stands treated with large-gap harvesting, regeneration was positively correlated with space while accounting for the association with canopy openness suggesting that regeneration density in the large-gap treatment consistently exhibited patterning in space not association with variability in canopy openness. Perhaps, this significant association represented tighter connections with other factors affecting regeneration as harvest intensity increases, including damage to advance regeneration (Smith 1991), scarification and soil disturbance (McInnis and Roberts 1991), and greater light penetration into the matrix

from gap and trail edges; all of which may have weakened spatial congruity between canopy openness and tree regeneration and increased homogeneity in the understory environment.

2.5.5. Overlay of Factors Structuring Spatial Pattern of Regeneration

In mature forests, the understory spatial pattern of tree regeneration reflects more than the size, shape, and dispersion of canopy gaps. Conceptually, total variation in understory spatial pattern in tree regeneration can be thought of as the effect of spatial and temporal variability in factors, deterministic and stochastic, influencing regeneration overlain on an area producing pattern across scales (Clark et al. 1998, Beckage and Clark 2003, Dovčiak 2003, McEuen et al. 2004, Messaoud and Houle 2006). Large scale factors like depth to water table can structure gradients in forest understories, while fine-grained factors like resource heterogeneity typically produce small-scale patchiness (Ehrenfeld et al. 1997). Nicotra et al. (1999) concluded that the spatial pattern of tree regeneration they observed in the understory of tropical wet forests was partly the legacy of past understory light environments. For all eight stands included in this study, spatial pattern analysis of tree regeneration density and canopy openness found that distributions were patchy at the sub-stand scale, that understory patch size increased with gap size, and that none of the global patterns exhibited a gradient in density. Collectively, these findings suggested that small-scale heterogeneity in the understory environment, perhaps linked to local variability in canopy structure and composition (i.e., neighborhood scale), may be more important in structuring understory spatial pattern over the range of spatial scales included in this investigation than factors operating at larger scales.

In the Acadian forest, regeneration can accumulate under a wide range of canopy conditions; an emergent property of the region partially afforded by an ecologically-diverse composition of tree species (Davis 1991, Smith 1991, Seymour 1992). In this study, regeneration was dominated by shade-tolerant northern conifers that are capable of surviving in the understory for decades after germination, primarily balsam fir, hemlock, and the spruces (red and white). Although understory composition was dominated by shade tolerant conifers, these stands also supported populations of relatively shade intolerant species that depend more on canopy disturbances for regeneration, primarily white pine, red oak (*Quercus rubra*), red maple, and the aspens (bigtooth and trembling) and birches (paper, gray, and yellow (*B. alleghaniensis*)). Since analysis in this study was performed on total regeneration, which included all species tallied during regeneration sampling, the spatial patterns detected were the combination of species-specific regeneration responses to factors structuring understory distribution. Furthermore, the estimate of regeneration density in this study captured trees ranging in size from 0.1 to 2 m tall, which likely translates to a wide range of tree ages, particularly in the northern conifers, and therefore, spatio-temporal variability in the timing of regeneration. As a result of both wide ecological amplitude among tree species and range of sizes included in the analysis, the spatial patterns detected likely reflect spatio-temporal variation in individualistic responses at the species-level that span several decades.

Not only is ecological pattern scale dependent, but a general conclusion on the nature of pattern and scale is that stochasticity of a pattern is inversely proportional to the scale of observation (Levin 1992). Compared to large-scale factors such as climate,

factors linked to uniqueness in the local environment (e.g., disturbance history, chance dispersal and colonization, herbivory and seed predation) become increasingly influential in structuring ecological pattern as the observational scale decreases (Levin 1992). The range of scales covered in this investigation, from 3 m to either 93 or 189 m, is relatively small when considering factors influencing spatial pattern in forest ecosystems, which vary at sub-continental scales. The degree of stand-level uniqueness seen in spatial analysis results suggested that local uniqueness, perhaps linked to stand history, has also played an important role in structuring regeneration patterns observed in this study.

2.5.6. Spatial Pattern Analysis in Silviculture Research

While many silviculture experiments have used non-spatial response variables to represent understory attributes in mature stands, few have examined spatial pattern as a response variable. Since the process of tree regeneration operates at the sub-stand level, traditional experiments that simply characterize stand-averaged, understory regeneration are likely to miss small-scale heterogeneity in understory spatial pattern associated with regeneration (Puettmann et al 2009). For example, a typical measure of site occupancy used in silviculture is percent stocking. In this study, treatment-averaged percent stocking was high regardless of treatment (> 90 %), which did not indicate how the regeneration was spatially distributed within these stands. Spatial pattern analysis indicated that mean understory patch size of tree regeneration increased along the gap-size gradient produced by the experiment, which more adequately described within-stand spatial heterogeneity in tree regeneration.

This is not to say that stand-averaged estimates of tree regeneration density do not carry important information for forestry research. On the contrary, we feel this investigation showed that analysis of understory spatial pattern provided additional information for interpreting the ecological effects of gap harvesting on forest regeneration, which complemented the analysis on non-spatial pattern and provided greater coverage in the range of responses detectable in silvicultural research. Furthermore, we concur with the conclusions of Nicotra et al. (1999), Miller et al. (2002), Scheller and Mladenoff (2002), and Puettmann et al. (2009) that incorporating spatial pattern analysis in long-term forest research can provide observational scales necessary to capture patterns that are linked to processes operating over a range of sub-stand scales, such as forest regeneration.

2.6. CONCLUSIONS

We accepted our general hypothesis that gap harvesting fosters development of a coarse-grained understory spatial pattern in both tree regeneration and canopy openness. This conclusion was supported by increasing patch sizes with increasing gap size, and the prevalence of peaks on correlograms at medium and large spatial scales in stands treated with gap harvesting; which to the best of our knowledge has not been observed in any previous research. Further evidence suggested that gap harvesting increased the grain of understory spatial pattern not only by favoring the establishment of new regeneration (i.e., seed origin and sprout origin), but also accentuated a pre-existing spatial pattern through the release of advance regeneration; a pattern related to numerous factors varying over both space and time (Clark et al. 1998, Beckage and Clark 2003, Messaoud and

Houle 2006). As reported by Pardos et al. (2008), we also found that the association between the abundance of tree regeneration and canopy openness, which we used as a surrogate for resource availability, was affected by gap harvesting. A further conclusion of this study was that spatial pattern analysis has both immediate and long-term value in understanding forest regeneration in silvicultural experimentation.

Chapter 3

FACTORS AFFECTING SPECIES RICHNESS OF TREE REGENERATION IN MATURE, ACADIAN MIXEDWOODS OF CENTRAL MAINE

3.1. ABSTRACT

Since tree regeneration in forest understories is largely a neighborhood-scale process, modeling approaches that utilize within-stand heterogeneity of both local environmental conditions and understory vegetation have been used to describe patterns of tree regeneration. The objective of this research was to identify small-scale, ecological correlates affecting the species richness of tree regeneration in Acadian mixedwoods of central Maine. A cyclic sampling method was used in a long-term gap-harvesting experiment to capture the explicit spatial pattern of understory tree regeneration concurrently with local environmental conditions (i.e, overstory basal area, canopy openness, and depth-to-water-table). Since balsam fir regeneration was prolific under a wide range of local environmental conditions, we hypothesized that the species richness of tree regeneration was negatively associated with balsam fir regeneration density; therefore, balsam fir density was included as an explanatory variable. Ecological correlates were identified for small (height ≥ 0.1 m, but < 0.75 m) and large (height ≥ 0.75 m, but < 1.4 m) regeneration richness using maximum likelihood methods. Estimates of total regeneration richness (small and large combined) were relatively uniform across stands and treatments, yet was highest in a replicate of the large-gap treatment. Despite a high amount of variation that remained unaccounted for by the models, consistent associations among factors were found across models. Negative

associations between the species richness of small regeneration and balsam fir regeneration density were found, while there was a positive association between the species richness of small and large regeneration and canopy transmittance. Based on these findings, the persistence of some tree species may not be sustainable through gap-harvesting alone if the density of understory fir is not reduced.

3.2. INTRODUCTION

Spatio-temporal patterning of tree regeneration in forest understories is structured by one or more factors operating largely at small-spatial scales, including: seed rain (Clark et al. 1998, Dovciak et al 2003), safe sites (Harper et al. 1961, Dovciak et al 2003), resource availability (Houle 1994, Pacala et al 1994, Beckage et al. 2000, Dovciak et al 2003, Messaoud and Houle 2006), and natural enemies (Wills et al. 1997, Schnurr et al 2004). Previous research on ecological patterning in forest understories has established spatial associations between the distributions of tree regeneration and local canopy structure and species composition (e.g., Pacala et al 1994, Frelich et al 1998). The spatial extent of this local canopy influence is commonly referred to as the neighborhood (Frelich et al 1998, Canham and Uriarte 2006), which is the fundamental scale of interaction in the emerging theory of neighborhood dynamics (see review by Canham and Uriarte 2006). When scaling up from neighborhood to stand scales, spatial patterning of tree regeneration forms within-stand heterogeneity reflecting contemporary and historic variability in the process of forest regeneration (Nicotra et al 1999).

Forest regeneration is widely regarded as a complex process (Peterson and Pickett 1995, Clark et al. 1998, Rees et al. 2001, Gratzer et al. 2004, Miina and Heinonen 2008).

When the objective of research is to increase understanding of complex ecological processes, maximum likelihood approaches based on model selection and parameter estimation are good alternatives to more traditional methods of hypothesis testing (Burnham and Anderson 1998, Johnson and Omland 2004, Canham and Uriarte 2006). Past studies that use modeling approaches to gain deeper understanding of neighborhood-scale patterns of forest regeneration have typically focused on regeneration abundance as a response variable (e.g., Lichstein et al. 1994, Messaoud and Houle 2006). However, relatively few studies have used tree species diversity in modeling forest regeneration (e.g., Nicotra et al. 1999). This observation is surprising when one considers the large body of work in ecology dedicated to explaining patterns of biological diversity across spatial, temporal, and organizational scales.

The main objective of this investigation was to identify local environmental variables that were correlated with species richness of tree regeneration in mature, Acadian mixedwoods of central Maine (henceforth referred to as ecological correlates). Determination of ecological correlates was accomplished through developing empirical models relating species richness of tree regeneration (henceforth referred to as tree regeneration richness) with measurements of the local environment of understory vegetation plots. Specifically, we examined the association between balsam fir regeneration density and tree regeneration richness. We hypothesized that tree regeneration richness would be negatively associated with balsam fir density.

3.3. METHODS

3.3.1. Study Site

We used stands that are part of the long-term experiment of the Acadian Forest Ecosystem Research Program (AFERP) at the University of Maine. The AFERP study is located on the Penobscot Experimental Forest (PEF) in the towns of Bradley and Eddington, Maine (Figure 3.1). The PEF is a 1,680-ha predominantly forested tract owned by the University of Maine Foundation and is jointly managed by the University of Maine and the U.S. Forest Service, Northern Research Station. Climate in central Maine is cool and humid. February (avg. temp. -7.1º C) and July (avg. temp. 20.0º C) are the coldest and warmest months, respectively. Average precipitation is 107 cm. The growing season is approximately 160 days at the PEF. Soils at the PEF are complex resulting from glaciation. The dominant parent material is a fine-textured till derived from sedimentary bedrock which have developed into loams. Contained within the PEF is a broad gradient of soil drainage classes; however, a large portion of the forest grows on soils classified as somewhat poorly to poorly drained and moderate to low productivity (Sendak et al. 2003).

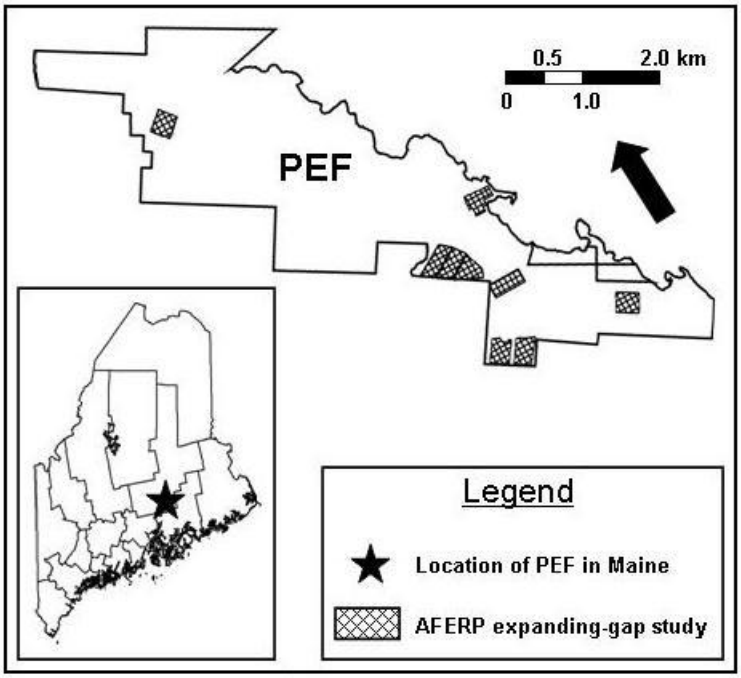

Figure 3.1. Location of AFERP study on the Penobscot Experimental Forest (PEF) with location of PEF in Maine (inset image, bottom left).

The PEF is located within the Acadian Forest, a conifer-dominated ecosystem covering most of Maine and the Canadian Maritimes (Halliday 1937). Dominant conifer species include balsam fir (*Abies balsamea*), eastern hemlock (*Tsuga canadensis*), red spruce, white spruce (*Picea glauca*), northern white-cedar (*Thuja occidentalis*), and white pine (*Pinus strobus*), while the common hardwood species include red maple (*Acer rubrum*), paper birch (*Betula papyrifera*), gray birch (*B. populifolia*), trembling aspen (*Populus tremuloides*), and bigtooth aspen (*P. grandidentata*) (Sendak et al. 2003).

3.3.2. Experimental Design

AFERP is a long-term experiment established to investigate the ecological effects of expanding-gap silvicultural methods that were designed based on patterns of natural disturbance in the Acadian forest (Saunders and Wagner 2005, Seymour et al. 2006). The experiment is a randomized complete block design with two levels of the expanding-gap harvest and an unharvested control that are each replicated three times, yielding a total of nine experimental units. Each experimental unit or research area (RA) ranges in size from 9.0 to 11.5 ha. The two treatment levels of expanding-gap harvesting are based on total area harvested per entry and the regeneration period between gap expansions. In the small-gap treatment, 10 % of the RA is harvested during each entry with a 20-year regeneration period between expansions, meanwhile, in the large-gap treatment, 20 % of the RA is harvested per entry with a 10-year regeneration period between expansions. Target gap size was 0.1 and 0.2 ha, respectively, for small- and large-gap treatments (see Saunders and Wagner (2005) for additional details about the AFERP experimental design). However, since the silvicultural prescription required working with the variation in stand conditions needing regeneration and understory release, gaps range in size from 0.01-0.13 ha for small-gap treatment and 0.03-0.22 ha for the large-gap treatment. The high within-stand environment heterogeneity in the understory produced by gap harvesting makes this experiment ideal for identifying the small-scale (i.e., plot-level neighborhood) associations with tree regeneration richness.

Data for this study were collected on two of three replicates, or six experimental units, of the expanding-gap experiment. This investigation was initiated after the second harvest entry, or first round of gap expansions, into replicate block one (RAs 1-3). Since

our intention was to investigate the effects of gap harvesting, rather than the expanding-gap method, on understory regeneration pattern, only replicate blocks two and three (RAs 4-6 and 7-9, respectively) were included in this investigation. Additionally, we were interested primarily in spatial patterning after stands have had time to recover from harvesting rather than the immediate post-harvest environment. Data for this study were collected during the summers of 2006 (RAs 4-6) and 2007 (RAs 7-9).

Two of six experimental units (RAs 4 and 7) are bisected into two, roughly equal sized stands by an unpaved, forest road (~ 6 m wide). The road represents a spatially discontinuity in the understory pattern of both RAs. Therefore, part of this analysis was performed on each of the stands, making a total of eight stands. ArcGIS was used to remove plots from the analysis that fell within within an 8-m buffer on each side of the forest roads (diameter = 16 m) to minimize road effects (e.g., edge effects, road erosion). The split stands will be referred to herein as RA4 west (RA4w) and east (RA4e) and RA7 north (RA7n) and south (RA7s) for the rest of this report.

3.3.3. Vegetation and Light Sampling

Sampling was performed using unequally spaced plots placed along 100-m transects originally established for monitoring deadwood over the expanding-gap study. The center of each transect was positioned at the point of intersection on a 50 x 50 m grid that was located across each sample stand (RA). The orientation of each transect was determined by randomly assigning an azimuth to each transect. The number of transects varied from 30-36 depending on the size and number of grid points in each RA. The spatial arrangement of plots along each transect followed the repeating pattern, cyclical

sampling design (RPCS) of Scheller and Mladenoff (2002). Conceptually, a cycle consists of a series of plots stacked end-to-end; a sub-set of which are sampled. Following Scheller and Mladenoff (2002), sampling was performed on four of 13 plots per cycle with sampling on the 1st, 2nd, 4th, and 10th plots in sequence (Figure 3.2). This sampling method is designed to capture inter-plot variability across a range of spatial scales from shortest to furthest inter-plot distances, which can be used to estimate spatial pattern (Scheller and Mladenoff 2002). Two to four cycles were established per transect, which yielded a total 1,742 sample plots with the sample size ranging from 277 to 336 per RA.

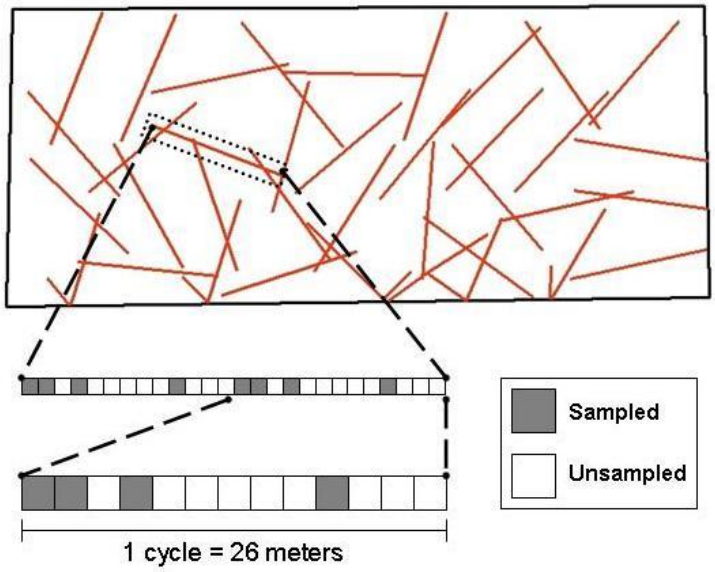

Figure 3.2. Depiction of randomly-oriented transects (solid red lines) within AFERP RA4 (control) and an illustration of the 4 of 13 RPCS design (image projected below).

Understory vegetation, overstory composition, and available light measurements were spatially linked to test for associations. Understory vegetation sampling included woody plants (height ≥ 0.1 m, but < 2 m) rooted within 4 m^2 (2 x 2 m) plots. Since the intent of this investigation was to evaluate the spatial response of established tree regeneration to gap harvesting, the lower size limit for regeneration sampling was set to 0.1 m tall to excluded recent germinants. Woody plants encountered within plots were identified to lowest taxonomic level. Stem counts of woody plants tall were stratified into three height classes: small (≥ 0.1 m, but < 0.75 m), medium (≥ 0.75 m, but < 1.4 m), and large (≥ 1.4 m, but < 2 m) regeneration. The local overstory (stems ≥ 9.5 cm dbh) was sampled for composition by tallying the number of species captured with a 4-m^2 prism point above the center of each understory plot. During July of 2006 (RAs 4-6) and 2007 (RAs 7-9), the understory light environment was sampled above plot center using the LAI-2000 Plant Canopy Analyzer (LI-COR Inc. 1992). The LAI-2000 was used to estimate diffuse, non-interceptance (DIFN), which is a measure of the fraction of sky not blocked by foliage (LI-COR Inc. 1992). The value of DIFN was calculated as the ratio between readings taken from two sensors; one taken above the center of understory plots (i.e., below canopy readings) and the other placed out in an open field greater than two canopy tree heights from the nearest forest edge (i.e., above canopy readings). DIFN was converted to percent canopy transmittance (referred to as %T in this report) above each understory plot simply by multiplying by 100. Readings from both sensors were recorded at a height of 1.4 m above the ground during the morning and evening to avoid exposure to direct sunlight.

3.3.4. Statistical Analysis

The data used in this study came from a sampling method designed to capture within stand variability both in response variables and potential explanatory variables rather than data from a controlled experiment where potential explanatory variables were set according to a range of fixed levels. In this case, species richness of tree regeneration in each 4-m^2 plot was the response variable of interest. Species richness was used since it is the simplest indicator of biological diversity, yet forms the basis of many ecological theories (Gotelli and Colwell 2001). Thus, statistical modeling in this study was applied more as an approach to gain better understanding of the factors associated with tree regeneration richness in forest understories based on interpretation of model parameters than the development of predictive models to project changes under alternate future scenarios. Modeling results were compiled as multiple lines of evidence in order to make inferences on ecological correlates of tree regeneration richness in Acadian mixedwood stands. All analysis was performed using R 2.8.1 (R Development Core Team 2008).

Exploratory data analysis (EDA) was performed to reduce dataset size by either removing or pooling data to identify appropriate data transformations when needed. The first round of EDA identified two tree regeneration richness response variables and three potential explanatory variables (OBA, DWT, and %T) for model building (see Table 3.1 for description). A second round of EDA revealed balsam fir regeneration (BFr; Table 3.1) density was abundant and well distributed over wide ranges of OBA, DWT, and %T (Appendix C); while pair-wise correlations between BFr and the 3 variables above were weak ($|r| < 0.333$); specifically, Pearson correlation coefficients of -0.225, 0.018, and 0.021 for OBA, DWT, and %T, respectively. Based on these results and an

understanding of the autecology of balsam fir (see Frank 1990), we postulated that increasing fir regeneration density had reduced tree regeneration richness; perhaps related to unfavorable conditions beneath understory balsam fir limiting regeneration and, therefore, reducing tree regeneration richness locally when fir regeneration was abundant. Thus, balsam fir regeneration density was included as a fourth explanatory variable. When transformation was warranted, variables were either square-root or quarter-root transformed to achieve normality (Shapiro-Wilk W ≥ 0.9).

Table 3.1. Description of response and explanatory variables.

Variable	**Description**
Response	
Small regeneration richness	Number of tree species ≥ 0.1 m tall, but < 0.75 m tall tallied within each 2 m x 2 m plot minus balsam fir.
Large regeneration richness	Number of tree species ≥ 0.75 m tall, but < 1.4 m tall tallied within each 2 m x 2 m plot minus balsam fir.
Explanatory	
Depth to water table	DWT; Distance (m) from soil surface to normal high level of the water table extracted from a raster model (Castonguay 2006).
% Canopy transmittance	%T; Fraction of sky not blocked by foliage; measured using the LAI-2000.
Fir regeneration density	BFr; Number of fir stems ≥ 0.1 m tall, but < 2 m tall tallied within each 2 m x 2 m plot.
Overstory basal Area	OBA; Number of trees ≥ 9.5 cm dbh captured with a 4-m^2 prism multiplied by 4.

All possible combinations of the four explanatory variables were considered when modeling small and large regeneration richness separately using non-spatial modeling procedures. First, linear mixed-effects (LME) models were developed to select an appropriate model to represent the entire dataset (n = 1307; Appendix D). Gap treatment was included as a random effect along with replicate block to correct for low treatment replication (only two replicates per treatment). The effect of treatment was assessed by examining nested, stand-level model parameters (slopes and intercepts) for trends across treatments. Next, ordinary least-squares (OLS) regression was used to select the appropriate model for each stand individually (Appendix D). Since our sampling design (RPCS) was used not only to sample forest vegetation and understory light but space itself, the OLS residuals from models selected for each stand were checked for spatial autocorrelation using Moran's tests. In the presence of spatial dependence in the response variable (endogenous autocorrelation), not accounting for endogenous autocorrelation can affect parameter estimates and obfuscate interpretation of relationships between the response and explanatory variables (Legendre 1993, Lichstein et al 2002, Bahn et al. 2006). OLS residuals were checked for autocorrelation at 16 distance classes of 6 m (spatial extent = 96 m). Distance classes were set to 6 m to capture at least 30 distance pairs within each class; a procedure recommended by Fortin and Dale (2005) for detecting spatial dependence. OLS regression coefficients were interpreted as ecological correlates only when spatial autocorrelation was not detected in residual variance. The *a priori* alpha for this investigation was set at 0.10 and spatial autocorrelation was significant at $p < 0.0063$ (Bonferroni-corrected for 16 simultaneous tests).

When significant spatial autocorrelation was detected, models were rebuilt for each stand using a conditional autoregressive (CAR) modeling procedure. CAR models are used to simultaneously account for spatial dependence in the response variable along with potential explanatory variables (endogenous and exogenous factors, respectively). Simultaneously accounting for endogenous autocorrelation not only helps isolate the effect of explanatory variables on the response, but contains information regarding univariate spatial patterning in the response (Legendre 1993). The CAR procedure requires a neighborhood-size term from the user in order to estimate spatial dependence in the response variable. Neighborhood size was estimated from the pattern and significance of small-scale spatial autocorrelation in OLS residuals. Actual neighborhoods were formed using a binary coding to establish neighboring plots, which only considered plots falling within a pre-specified radius around plots as neighbors. When spatial autocorrelation in model residuals is detected at larger-scales, trend surface terms based on plot coordinates can be added to estimate large-scale spatial dependence in the response (e.g., Lichstein et al 2002, Bahn et al. 2006). Variance explained (R^2) by CAR models was calculated from log-likelihoods using the formula given by Nagelkerke (1991).

3.4. RESULTS

3.4.1. Tree Regeneration Richness across Stands and Treatments

Twenty-four tree species were identified in the understory sample (Appendix E). Of the 24 species, thirteen were encountered in every stand and 18 were regenerating in all treatments. Along with shade-tolerant conifers (balsam fir, red and white spruce,

eastern hemlock, and northern white-cedar), other common tree species included: red maple, paper birch, American beech, white ash, white pine, the aspens, and northern red oak. Grey birch, red pine, and red cherry, all considered shade intolerant, were found regenerating in harvest treatments only. Several hardwood species typical of high-quality sites, such as sugar maple, yellow birch, eastern hophornbeam, and American basswood, were tallied during 2006 and 2007.

Total tree regeneration richness (small and large combined plus balsam fir) estimated using both sample-based rarefaction from a minimum of 240 plots (EstimateS software; Colwell 2000) and actual number of species from all plots are presented in Figure 3.3. Estimates of tree regeneration richness were relatively uniform across stands and treatments; particularly replicates of control and small-gap treatments. Stand RA6, replicate of the large-gap treatment, had the highest rarefaction and actual number of tree species observed in the understory; conversely, RA7, a small-gap treatment, had the lowest estimates of understory tree richness. Both rarefaction and actual richness were consistently higher across treatments in replicate block 2 (RAs 4, 5, & 6).

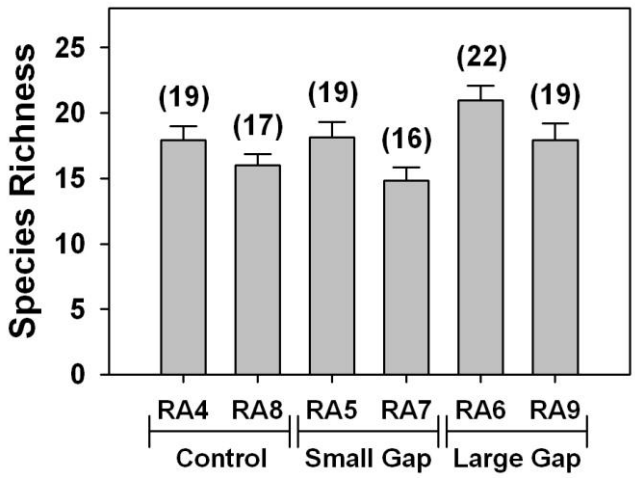

Figure 3.3. Treatment comparison of tree regeneration richness estimated by sample-based rarefaction. Error bars equal two standard deviations. Number in parentheses is actual richness based on all plots.

3.4.2. Models of Plot-Level Richness

Model selection identified the same 3-variable LME model for small and large regeneration richness composed of BFr, OBA, and %T (Table 3.2). Both LME models are significant ($p < 0.01$), yet adjusted-R^2 values are low (0.12 and 0.20 for small and large regeneration richness, respectively). Positive slopes of %T were estimated for both tree regeneration richness models. In contrast, slopes for BFr and OBA were quite different between regeneration richness LME models. BFr slopes for small and large regeneration richness are negative and positive, respectively, while slopes for OBA are positive and negative. Slope p-values indicated small regeneration richness was most strongly associated with BFr and least with OBA; meanwhile large regeneration richness was most closely linked to %T and least with BFr. An effect of gap-harvest treatment on nested, stand-level model parameters was not evident (Append F).

Table 3.2. Model details and slope estimates for closest approximating models of (A) small and (B) large regeneration richness. Value in parentheses is slope p-value.

	Control			Small Gap			Large Gap		
Model	**RA4w**	**RA4e**	**RA8**	**RA5**	**RA7n**	**RA7s**	**RA6**	**RA9**	**All *** RAs**
A) Small regeneration richness									
Detail									
nb *	NI **	NI	6	8	NI	NI	14	12	NI
adj-R^2	0.09	0.07	0.09	0.22	0.06	0.10	0.19	0.16	0.12
p-value	<0.01	0.01	<0.01	<0.01	0.05	0.01	<0.01	<0.01	<0.01
	Slope (p-value)								
Correlate									
BFr	-0.20 (<0.01)	-	-0.19 (<0.01)	-0.19 (<0.01)	-	-0.11 (0.04)	-0.30 (<0.01)	-0.22 (<0.01)	-0.18 (<0.01)
OBA	-	-	-	0.02 (0.01)	-	-	0.02 (0.01)	0.02 (<0.01)	0.01 (0.07)
DWT	-	-	-	0.13 (0.09)	-0.04 (0.10)	-	-	-	-
%T	0.12 (0.04)	0.27 (0.03)	0.18 (0.10)	0.26 (<0.01)	-	-	-	-	0.35 (0.01)
B) Large regeneration richness									
Detail									
nb	8	6	12	10	NI	9	25	21	NI
adj-R^2	0.16	0.14	0.18	0.32	0.12	0.21	0.28	0.27	0.20
p-value	<0.01	<0.01	<0.01	<0.01	0.01	<0.01	<0.01	<0.01	<0.01
	Slope (p-value)								
Correlate									
BFr	-	-	-	0.08 (<0.01)	-	-	0.13 (0.12)	-	0.08 (0.06)
OBA	-0.01 (0.05)	-0.01 (0.08)	-	-	-	-0.02 (0.02)	-	-0.02 (0.04)	-0.02 (0.02)
DWT	-	-	-0.74 (0.01)	-0.11 (0.05)	-	0.25 (0.03)	0.16 (0.06)	-	-
%T	0.47 (<0.01)	0.34 (<0.01)	0.56 (<0.01)	0.31 (<0.01)	0.38 (<0.01)	0.35 (<0.01)	0.44 (<0.01)	0.57 (<0.01)	0.51 (<0.01)

***** Neighborhood; Represents neighborhood diameter in meters used in CAR models.
** Not included; No spatial autocorrelation detected in OLS residuals so neighborhood factor was not included; Model fit and slope coefficients from OLS regression.
******* Results in column are for LME models using all RAs simultaneously.

Due to the presence of spatial autocorrelation in several stand-level models, tree regeneration richness models were built using a combination of OLS and CAR. Spatial autocorrelation ($p < 0.0063$; Bonferroni-corrected) was detected in OLS residuals for four small and seven large regeneration richness models (Table 3.2), which were re-run as CAR models. Since autocorrelation was observed primarily at small spatial scales, no trend surface terms were included. For both tree regeneration richness models, only OLS models for split-stands (stands labeled with a lower case letter) lacked significant spatial autocorrelation, and therefore, OLS results were interpreted only for split-stands. This trend could have been related to smaller sample sizes of split-stands. However, the difference in endogenous autocorrelation between height-stratified richness classes suggested large regeneration richness exhibited autocorrelation in more of the stands analyzed; a pattern perhaps linked to processes of species recruitment between the height classes. Estimated neighborhood sizes were consistently larger for large regeneration richness and in the large-gap treatment for both richness classes; meanwhile, neighborhoods were more equable in size between small-gap and control treatments. Neighborhoods representing large regeneration richness in large-gap treatments were nearly two-times larger than other treatments.

Similar to LME results, model fits were low for all stand-level models with adjusted-R^2 ranging from 0.07-0.22 and 0.12-0.32 for small and large regeneration richness, respectively (Table 3.2). For both forms of tree regeneration richness, RA5 CAR models had the highest adjusted-R^2 values (0.22 and 0.32 for small and large classes, respectively), while RA7n OLS models had the lowest (0.06 and 0.12, respectively). Model fits were consistently stronger within stands for large regeneration

richness, which was reflected in correspondingly higher model p-values (all have $p \leq 0.01$). CAR models tended to have higher adjusted R^2 values than OLS models, which appeared to be at least partly related to inclusion of an additional parameter to account for spatial dependence in the response.

According to model selection, there was little consistency among stand-level models based on composition of explanatory variables (Table 3.2); a trend associated with the relatively weak models evidenced by low adjusted-R^2 values. However, there was partial agreement within some treatments. For example, small regeneration richness models in control and large-gap treatments and models of large regeneration richness in controls only. Also, slope signs of individual explanatory variables with the exception of DWT were consistent across models (including LME models) within both tree regeneration richness classes, while actual slope values showed less consistency. Some variables were selected more often than others across models. For small regeneration richness, slopes for BFr were included in six of eight stand-level models, all of which are negative and relatively similar in value. All stand-level models selected for large regeneration richness contained a slope of %T, all of which were positive and relatively similar in value. Consistent with LME results, BFr and %T were often the most important explanatory variable in models of small and large regeneration richness, respectively, according to slope p-values. Not only was DWT omitted from LME models, but slopes of DWT in stand-level models were inconsistent in for both small and large regeneration richness models.

3.5. DISCUSSION

3.5.1. Experiment-Wide Patterns of Tree Regeneration Richness

The Acadian forest is an ecotone between eastern broadleaf and boreal forest biomes (Rowe 1972). Because of this unique geography, the Acadian forest is relatively rich in tree species despite its cool climate and northern latitude. We found forest understories of the expanding-gap experiment supported a species-rich and ecologically-diverse assemblage of tree regeneration. The 24 species of tree regeneration observed in this study were similar to findings by Schofield (2003) who identified approximately 25 tree species across all nine RAs in this same experiment. Our results also showed that the gap-harvest treatments were regenerating intolerant and mid-tolerant tree species in mixture with shade tolerant species and, therefore, regenerating a mid-successional composition; a silvicultural objective of the large-gap treatment.

Comparisons based on rarefaction and actual species counts indicated no substantial differences in tree regeneration richness among treatments, which likely reflected substantial within-stand heterogeneity captured by the relatively large size of experimental units (average stand area ~ 10 ha or 25 ac), ecologically-diverse tree assemblage, and the prevalence of shade-tolerant tree species that readily establish advance regeneration. Brissette (1996) found high abundance and stocking of regeneration composed of an array of species across partially-cut stands on the PEF, a pattern he linked to within-stand environmental heterogeneity and wide adaptability of native tree species. Therefore, one plausible explanation for equability in richness across this experiment is that high sub-stand heterogeneity in the understory environment

interacts with the autecological characteristics of an ecologically-diverse tree species assemblage to promote tree regeneration richness across treatments (Brissette 1996).

There are many possible causes of environmental heterogeneity, including variation attributable to heterogeneity in soils and canopy structure and composition. Mixedwood canopies on the PEF commonly support tree species that span a wide spectrum of shade tolerance ratings (e.g., trembling aspen < white pine < red maple < balsam fir < eastern hemlock), which can engender sub-canopy light heterogeneity commensurate with inter-specific variation in canopy light penetration (Canham et al 1994). Although the soils of the PEF are largely considered low to mid-quality (Sendak et al 2003), the presence of hardwood species associated with high-quality sites, including sugar maple, yellow birch, and American basswood, suggests suitable habitat exists for site-demanding species to persist alongside site-insensitive tree species, such as red maple, white pine, and balsam fir; further supporting the hypothesis that similarity in richness is at least partly linked to environmental heterogeneity not associated with gap-harvest treatments. Dominant tree species of the PEF, such as balsam fir, eastern hemlock, the spruces, and red maple, are considered tolerant to very tolerant of shade and, therefore, capable of persistence in the understory as advance regeneration. Thus, similarity in tree regeneration richness among stands and treatments is at least partially linked to abundant and well-distributed seedling banks of shade tolerant tree species across the expanding-gap experiment.

3.5.2. Ecological Correlates of Tree Regeneration Richness

Considering the complex nature of forest regeneration, it was not surprising that closest approximating models had high unexplained variance and model selections were inconsistent across models. Past research on forest regeneration has also reported high unexplained variance (e.g., Nicotra et al. 1999, Dovciak et al 2003, Lichstein et al 2004) and inconsistency in model selection (e.g., Nicotra et al 1999). Poor model fits in this analysis are likely associated with too few explanatory variables leading to unmodeled trends falling into the error term, local uniqueness of individual stands, and the importance of stochasticity in the regeneration process. A major obstacle for forest regeneration research is the low signal-to-noise ratio inherent to process of regeneration (Miina and Heinonen 2008), which is largely attributable to stochastic elements in the regeneration process that are particularly important at small scales, such as chance dispersal and colonization (Houle 1994, Peterson and Pickett 1995, Clark et al. 1998, Dovciak et al 2003, Messaoud and Houle 2006).

Although the composition of models varied considerably, we found some corroborative support for individual explanatory variables across models. Small and large regeneration species richness was not strongly associated with estimated depth-to-water-table (DWT). The actual estimate of DWT was extracted from a 10-m raster of predicted DWT, an area 25-times larger than regeneration plots. So it is plausible that spatial grain of DWT was too coarse to effectively account for the plot-to-plot variability typically associated with tree regeneration patterns. For both small and large regeneration richness combined, most models indicated that species richness was enhanced by increasing canopy transmittance (%T); especially large regeneration

richness. Previous studies on understory vegetation pattern cite the availability of limiting resources as important determinants of pattern in forest understories; especially heterogeneity of sunlight (Canham et al. 1994, Pacala et al 1994). Since %T can be used as a surrogate for canopy openness and, therefore, resource availability, positive associations between canopy openness and regeneration richness observed in this study indicated a degree of spatial correspondence between richness hotspots and resource patchiness. In contrast to our results, Nicotra et al. (1999) found associations between tree regeneration richness and estimates of light availability (including %T) were quite variable in species-rich tropical forests. These authors, however, concluded that contemporary patterning of tree density and species richness is partly a legacy of historical light conditions and, therefore, reflected regeneration responses to the understory light environment more as a shifting mosaic over time rather than a static template.

Grime (1998) theorized that subordinate species can indirectly influence ecosystem structure and function through effects on the regeneration of dominant species, which he termed filter effects. There is a growing body of literature on the filter effects of understory vegetation on tree regeneration worldwide (see the review by Royo and Carson 2006). Based on an initial exploratory analysis and understanding of its autecology and life history (Frank 1990), we hypothesized that balsam fir fits Grime's (1998) description of a subordinate species in mature Acadian mixedwoods. Specifically, we expected balsam fir regeneration density (BFr) would deplete tree regeneration richness locally where BFr is high. Our results confirmed this expectation for small regeneration richness. Similar patterns were observed by Wyckoff and Webb (1996) who

found lower plant species richness beneath the shade-tolerant, invasive Norway maple in eastern North America. In contrast, associations between large regeneration richness and BFr were positive and relatively weak. We do not interpret this association as facilitation, but as a shared response among all regenerating species to available resources in an otherwise resource-limited understory environment.

Overstory-understory interactions are inherently asymmetrical favoring dominant overstory vegetation, which often assumes a negative, one-way interaction (Frelich 2002). This pattern of interaction was partly upheld by the negative associations between overstory basal area (OBA) and large regeneration richness inferred from negative slopes. However, the association with OBA was reversed for small regeneration richness. Interestingly, OBA was included in stand-level models of small regeneration richness in harvest treatments only, including both replicates of the large-gap treatment. The large-gap treatment was also treated with light, partial cutting in the matrix between gaps to remove short-lived species (Robert Seymour, personal comm.) which may have improved understory conditions for regeneration while having little effect on OBA in the matrix.

The focus thus far has been on pair-wise associations. However, it is plausible that patterns of tree regeneration richness involve the interplay of ecological correlates. Although interactions among correlates were not tested statistically, potential trends emerged when examining associations across stands. For example, positive associations between small regeneration richness and OBA in harvest treatments might also reflect the exclusion of small regeneration from harvest-gap areas as advance regeneration, composed predominately of balsam fir, was released following the harvest, while scarification from harvesting and light penetration along gap edges into the matrix could

explain the positive associations with both %T and OBA. For large regeneration richness, the positive association with BFr appears to indicate the importance of increasing %T or canopy openness to the process of sapling recruitment for all tree species; an ecological pattern observed in many forests worldwide (Pacala et al 1994). Additional evidence for a treatment effect was the larger neighborhood sizes estimated for both small and large regeneration richness in the large-gap treatment, which appears to be linked to greater harvest intensity in this treatment relative to the other two levels.

Results presented in this study are for a single snap-shot in time. In the case of gap-harvested stands, results represent associations after nearly a decade of recovery following harvest treatments. Within this 10-year period, associations between tree regeneration and the measured environmental variables may have changed with time since disturbance in gap-harvested stands. For example, the association between small regeneration richness and BFr in harvested stands may have switched from negative to positive immediately following the harvest and then back to negative with increasing cover of balsam fir regeneration. This scenario was suggested by results from Peterson and Pickett (1995) who found that seedling establishment decreased with time since disturbance and seedling cohorts established later had lower survivorship, which they attributed to increasing shade from the developing canopy. Furthermore, this dynamic interpretation is consistent with expected changes in seedling establishment during the initiation stage of stand development (Oliver and Larson 1990).

3.5.3. Influence of Correlates on Tree Recruitment and Diversity

Comparing results from small and large regeneration richness models provided some insight into the process of tree recruitment in this study. Sapling recruitment in forest understories is facilitated by increased resource availability (Pacala et al. 1994) and canopy disturbance is a primary driver of fluctuating resource availability in forests (Davis et al. 2000). This pattern of causation was suggested by changes in associations with two modeled correlates when scaling from small to large regeneration richness, specifically: 1) the shift from positive to negative associations with OBA and 2) the intensification (i.e., increasing magnitude and significance of slope parameters) of positive associations with %T.

This analysis found that the amount of variance accounted for was consistently higher for models of large regeneration richness than small regeneration richness. In part, the difference in variance accounted for appeared to be related to the inclusion of a neighborhood term, which was added more frequently into models of large regeneration richness. Recall neighborhood sizes were specified based on the pattern of positive and significant spatial autocorrelation detected in OLS residual variance (i.e., endogenous autocorrelation). According to Fortin and Dale (2005), an initial range of positive spatial autocorrelation can be interpreted as an indicator of patch size. Compared to models of small regeneration richness, the higher variance accounted for, stronger model fits, and the presence of fined-grained spatial structure in large regeneration richness models suggested that tree regeneration richness was more tightly correlated with the local environment as tree regeneration developed in the forest understories we measured.

3.6. CONCLUSIONS

Overall, modeling results from this investigation supported the conventional wisdom that forest regeneration is a complex ecological process. Despite the small amount of variance we were able to account for with the models tested, we were able to identify consistent associations across models. Negative associations between small regeneration richness and the density of balsam fir regeneration suggested that understory fir was negatively associated with species richness by excluding regeneration establishment, while positive associations between small and large regeneration richness and percent canopy transmittance appeared to reflect the importance of increasing the availability of limiting resources to the processes of seedling establishment and sapling recruitment in forest understories.

Our results also suggested that the mixedwood stands measured in this study support a relatively diverse species assemblage of tree species regardless of treatment, and that balsam fir regeneration was prolific under a wide range of local environmental conditions. Although our results indicated that increasing canopy openness increased the species richness of understory tree regeneration, the persistence of certain species may not be sustainable if the depleting effects of understory fir density on small regeneration richness continue. If establishing and maintaining a diverse array of tree species is a silvicultural objective in stands similar to those used in this study, then additional treatments, in conjunction with gap harvesting, may be needed to reduce understory fir density. For example, applying cleaning treatments within harvest gaps that simultaneously releases advance regeneration of desirable species while reducing enough fir cover to encourage establishment of new tree regeneration.

EPILOGUE

The work presented in this thesis analyzed factors affecting the spatial pattern of tree regeneration and the temporal development of forest composition to gain a deeper understanding about natural regeneration processes in managed forest stands of the Acadian Forest. Although this approach had several drawbacks, in particular the difficulty of integrating results, the methods used successfully revealed patterns of forest regeneration over a range of spatial and temporal scales.

In Chapter 1, I used five-decades (1953-2005) of permanent sample point data from a long-term experiment of traditional silvicultural methods to document compositional development at the plot level. Rather than averaging plot attributes within treatments over time to retrace treatment-level development, I treated plots as experimental units and subjected this population of plots to two separate rounds of cluster analysis in order to group plots based on: 1) pre-treatment tree species composition and 2) harvest disturbance history estimated at the plot level. Results from this analysis revealed that the USFS experiment initiated an array of developmental pathways at the plot level that were associated with both pre-treatment composition and harvest disturbance history. At the species level, northern white cedar declined, precipitously in some cases, in all clusters, while balsam fir largely maintained its pre-treatment dominance

In Chapters 2 and 3, I focused on the spatial aspects of forest regeneration for a single point in time over the AFERP expanding-gap experiment. In Chapter 2, I used univariate spatial patterning of tree regeneration density and canopy openness, and the

spatial associations between both factors, to serve as response variables for evaluating the effect of gap-harvesting on the spatial aspects of forest regeneration. Results from Chapter 2 indicated that the spatial responses of tree regeneration and canopy openness among treatments were related to a pre-existing spatial pattern, as well as the creation of new understory spatial patterns that were unique to each harvest treatment. In Chapter 3, I used spatial and non-spatial modeling approaches to identify small-scale ecological correlates affecting species richness of tree regeneration. Negative associations between the species richness of small regeneration and balsam fir regeneration density suggested that the presence of understory balsam fir reduces tree species richness of smaller established regeneration.

STRENGTHS AND LIMITATIONS

In cases where appropriate scales of ecological phenomena are not known *a priori*, Levin (1992) argued that research linking patterns in nature to a complex ecological process like forest regeneration should consider the patterns over a range of spatial and temporal scales. The two experiments used in this thesis provided a relatively wide range of spatial and temporal variability in forest regeneration patterns across scales. With over 50 years of experiment maintenance and continuous plot re-measurements, the USFS compartment management study provided me with an opportunity to take advantage of a longitudinal dataset capturing forest responses to over a half century of harvest treatments. The AFERP expanding-gap experiment, although much younger than the USFS study, generated within-treatment heterogeneity, which lended itself to methods that captured and identified explicit spatial patterns. Together,

these experiments provided an opportunity to investigate forest regeneration across a range of spatial and temporal scales using a diverse set of analytical tools.

From the outset of this investigation, I wanted to integrate analyses of spatial and temporal pattern to better understand natural forest regeneration. However, the inherent mismatch between the spatial and temporal data made it difficult to integrate both analyses seamlessly. Instead, the spatial and temporal aspects of forest regeneration covered in this thesis turned out to be separate investigations of forest regeneration in space and over time. Although I was unable to successfully integrate both spatial and temporal analyses, the results from this study still increased our understanding about each component of forest regeneration.

Cale et al. (1989) stressed that caution should be exercised when inferring process from pattern in nature, which was based on their premise that a single pattern can arise from multiple processes or that multiple patterns can result from a single process. This point has been argued for many types of ecological investigation, but especially for the study of univariate spatial pattern (Rossi et al 1992, Legendre 1993). To help connect pattern and process, I included potential drivers of spatial and temporal patterning in the analyses presented in this thesis. In Chapter 1, I attempted to "control" for the influences of pre-treatment tree species composition and harvest disturbance history to make inferences about the developmental pathways of forest composition in managed stands. In Chapter 2, I was able to relate pattern of tree regeneration not only to gap-harvest treatments, but also to resource heterogeneity as indicated by sub-stand variability of canopy openness. In Chapter 3, I included several local environmental factors and spatial autocorrelation as explanatory variables in models of tree regeneration richness. I believe

these approaches provided insights into the pattern and process of forest regeneration in the Acadian Forest. For example, the ability of balsam fir dominance to persist over a range of harvest intensities observed in Chapter 1 in conjunction with the negative effect of understory balsam fir density on small regeneration richness hypothesized in Chapter 3, suggests that additional silvicultural treatments may be needed to conserve tree species diversity in Acadian mixedwoods.

The work in Chapter 1 placed heavy emphasis on the results of two multivariate methods: cluster analysis and ordination. There are many pitfalls with relying too heavily on multivariate statistics due to the importance of user subjectivity in interpreting results. In particular, cluster analysis is often criticized in ecology since the user imposes classification on a group of objects that may vary continuously as a whole (McCune and Grace 2002). However, multivariate approaches are required when attempting to handle many variables simultaneously (Gauch 1982, McCune and Grace 2002); which was the case in Chapter 1. Since my aim in Chapter 1 was to use plots as experimental units for investigating temporal dynamics of forest composition at the plot level rather than at the treatment level, I believe the use of multivariate methods was appropriate.

Since traditional sampling methods are designed to detect treatment-averaged responses, they often are too coarse to adequately capture within-treatment heterogeneity. This problem is often the case when studying forest regeneration, which can exhibit substantial variability under fairly uniform conditions (Pardos et al. 2009). Therefore, the sampling design used in Chapters 2 and 3, which was able to capture and identify explicit spatial pattern over a range of sub-stand scales, is a strength of this thesis. In Chapter 2, I was able to use spatial pattern as a response to harvesting, which surprisingly few studies

have attempted to do. In Chapter 3, I took advantage of both spatial autocorrelation, when present, and variations in the local environment captured by the spatial sampling design to model the distribution of tree regeneration richness. The spatial sampling design used in thesis allowed me to investigate the spatial aspects of forest regeneration that more traditional methods are generally unable to address.

As with most studies of complex ecological phenomena, the analysis presented in this thesis was data limited. In Chapter 1, I used BA reductions recorded from pre-and post-harvest inventories (i.e., harvest treatments only), since cause of tree mortality was not recorded until the late-1990s. Thus, information on BA reduction included all possible sources of mortality, not just tree losses to harvesting. This fact limited my ability to accurately estimate harvest disturbance history. Based on the relatively short time frame within which both pre- and post-harvest inventories were conducted (approximately 5 years of each harvest entry), however, one can argue that most of the losses were due to harvesting. There were two salient data limitations imposed by my spatial sampling design. The first is that transect length was likely too short, which placed the majority of plots within the "matrix" and too few within harvest gaps (applies only to harvest-treated stands). The second is that plot size (2 m x 2 m) was likely too small to appropriately sample sapling regeneration. Finally, the decision to exclude tree regeneration < 10 cm tall in plot tallies and data on micro-site conditions limited my ability to investigate patterns of regeneration prior to establishment, which, in some cases, has been shown to impact spatial patterning of larger, established regeneration (Montes and Cañellas 2007, Pardos et al. 2008) and is thought to be a determinant of the

abundance of certain Acadian Forest tree species (balsam fir and red spruce; Greenwood et al. 2008).

Lastly, the results from this investigation, which were based largely on plot-to-plot variation, fit well within the emerging theory of neighborhood dynamics (see Canham and Uriarte [2006] for a review of neighborhood dynamics). Historically, most regeneration studies in managed forests focus on treatment-averaged patterns to make inferences at a scale relevant to land managers. Although this approach has provided valuable information on forest management practices, the application of treatment-averaging to evaluate regeneration pattern eliminates small-scale heterogeneity and, therefore, potentially important information regarding effects of silvicultural practices on forest regeneration. Because of this, I feel the emphasis placed on neighborhood-scale patterning of forest regeneration was a strength of this thesis.

RECOMMENDATIONS

Increasingly, biodiversity conservation is being viewed as an important consideration in natural resource management (Wilson 1988). Based on results from this thesis, my primary recommendation for forest managers working with similar stands is to consider applying treatments to reduce the abundance of balsam fir regeneration as a way to increase recruitment opportunities for other species. Although this recommendation applies to a variety of forestland types (e.g., public, industrial, small woodlot), it is largely infeasible for broad application across a managed landscape. However, small woodlot owners with biodiversity objectives in managing their forest may be willing to incur the out-of-pocket cost of controlling fir regeneration.

As mentioned above, I was unable to integrate my analyses of spatial and temporal components of forest regeneration, which largely was a result of incompatibility between the spatial and temporal data that were available. After reflecting on this issue, I can recommend two analytical approaches for future investigations that seek to integrate spatial and temporal patterns. First, where long-term, repeated-measures datasets exist, the methods developed by Saunders (2006) can be used to reconstruct spatial and temporal dynamics of forest regeneration. However, this approach may be data limited since many long-term studies employ sampling methods designed to capture treatment-averaged responses rather than within-treatment heterogeneity over a range of spatial scales. Second, another way to integrate spatial and temporal patterns is to use methods that capture and analyze variation in space and time simultaneously across scales (i.e., spatio-temporal analysis). Spatio-temporal methods have recently been developed that can combine spatially continuous and discrete data over time into a single analytical framework, which has been useful for relating changes in regeneration density estimated from sample plots to the point pattern of larger trees (see Montes and Cañellas [2007] for a review of the $K_{rx}(d)$ function). A major drawback of spatio-temporal analysis, however, is the relatively large time and financial investment necessary to establish, implement, and maintain spatio-temporal sampling designs. I believe that either one of these solutions are suitable for integrating space and time, but the choice will largely depend on data availability, project budget, and, of course, specific research objectives.

In future investigations of forest regeneration, I recommend implementing pluralistic approaches to research that combine diverse data and analytical tools designed to address pattern, process, and scale across a spectrum of stages in the lives tree species,

but with emphasis on development during early and juvenile stages. I feel that adopting a pluralistic approach will advance our understanding of forest regeneration, which could help inform improvements of models that forecast future forest composition.

FUTURE DIRECTIONS

In managed forests where regeneration is almost exclusively natural, forecasting future species composition remains a major challenge for forest management (Taylor et al. 2009). The root cause of this uncertainty is our inability to accurately predict developmental outcomes of trees during early life stages (Miina and Heinonen 2008), which, in my opinion, is partly related to the paucity of research addressing spatio-temporal dynamics of regeneration in managed forests. However, analytical tools are now available to help reduce this uncertainty. Bayesian inference, in particular, is well-suited to dealing with complex ecological phenomena and has steadily gained acceptance as an analytical tool in ecology (McCarthy 2007). A major strength of Bayesian inference is the inclusion of prior information, which may be derived from expert opinion on system behavior or results of previous studies. To date, there is a wealth of information available from past research on forest regeneration that can be used to build "priors" into a Bayesian framework and several software packages are now available that can handle Bayesian statistics. By building on what is already known about forest regeneration, I feel that integrating a Bayesian approach into spatio-temporal investigations of forest regeneration from silviculture experiments will advance regeneration modeling efforts and, therefore, is a fruitful avenue of research to pursue.

BIBLIOGRAPHY

Abrams, M.D., D. Sprugel, and D.I. Dickmann. 1985. Multiple successional pathways of on recently disturbed jack pine sites. Forest Ecology and Management. 10:31-48.

Bahn, V., R.J. O'Connor, W.B. Krohn. 2006. Importance of spatial autocorrelation in modeling bird distributions. Ecography. 29:835-844.

Barnes, B.V., D.R. Zak, S.R. Denton, and S.H. Spurr. 1998. Forest ecology, 4th edition. John Wiley and Sons Inc., New York

Battaglia, M.A., P. Mou, B. Palik, and R.J. Mitchell. 2002. The effect of spatially variable overstory on the understory light environment of an open-canopied longleaf pine forest. Canadian Journal of Forest Research. 32:1984-1991.

Beckage, B. and J.S. Clark. 2003. Seedling survival and growth of three forest tree species: the role of spatial heterogeneity. Ecology. 84(7):1849-1861.

Brissette, J.C. 1996. Effects of intensity and frequency of harvesting on abundance, stocking, and composition, of natural regeneration in the Acadian Forest of Eastern North America. Silva Fennica. 30(2-3):301-314.

Burnham, K.P. and D.R. Anderson. 1998. Model selection and inference: A practical information-theoretic approach. Springer-Verlag, New York. 353 p.

Cale, W.G., G.M. Henebry, and J.A. Leakley. 1989. Inferring process from pattern in natural communities: can we understand what we see? BioScience. 39:600-605.

Canham, C.D., A.C. Finzi, S.W. Pacala, and D.H. Burbank. 1994. Causes and consequences of resource heterogeneity in forests: interspecific variation in light transmission by canopy trees. Canadian Journal of Forest Research. 24:337-349.

Canham, C.D. and M. Uriarte. 2006. Analysis of neighborhood dynamics of forest ecosystems using likelihood methods and modeling. Ecological Applications. 16:62-73.

Castonguay, M. 2006. Metadata: me_wtbl_cm. Forest Watershed Research Centre, University of New Brunswick, Canada

Chokkalingam, U. and A.S. White. 2001. Structure and spatial patterns of trees in old-growth northern hardwood and mixed forests of northern Maine. Plant Ecology. 156:139-160.

Clark, J.S., E. Macklin, and L. Wood. 1998. Stages and spatial scales of recruitment limitation in southern Appalachian forests. Ecological Monographs. 68: 213-235.

Colwell, R.K. 2000. EstimateS: Statistical estimation of species richness and shared species from samples (software and user's guide), Version 7.5.2. http://viceroy.eeb.uconn.edu/estimates

Cooper, D.W. 1968. The significance level in multiple tests made simultaneously. Heredity. 23:614-617.

Cressie, N.A.C. 1993. Statistics for spatial data. John Wiley and Sons Inc., New York, New York. 900p.

Curtis, J.T. and R.P. McIntosh. 1951. An upland forest continuum in the prarie-forest border region of Wisconsin. Ecology. 32:467-496.

Davis, M.A., J.P. Grime, and K. Thompson. 2000. Fluctuating resources in plant communities: a general theory of invisibility. Journal of Ecology. 88:528-534.

Davis, W.C. 1991. The role of advance growth in regeneration of red spruce and balsam fir in east central Maine. P. 157-168 in Proceeding of the conference on natural regeneration management, C.M. Simpson (ed.). Forestry Canada – Maritimes Region. 261 p.

Dent, C.L. and N.B. Grimm. 1999. Spatial heterogeneity of stream water nutrient concentrations over successional time. Ecology. 80:2283-2298.

Dovčiak, M., R.B. Reich, and L.E. Frelich. 2003. Seed rain, safe sites, competing vegetation, and soil resources spatially structure white pine regeneration and recruitment. Canadian Journal of Forest Research. 33:1892-1904.

Egler, F.E. 1954. Vegetation science concepts I: initial floristic composition, a factor in old-field vegetation development. Plant Ecology. 4:412-417.

Ehrenfeld, J.G., X. Han, W.F.J. Parsons, and W. Zhu. 1997. On the nature of environmental gradients: temporal and spatial variability of soils and vegetation in the New Jersey Pinelands. Journal of Ecology. 85:785-798.

Fortin, M.-J. and M.R.T. Dale. 2005. Spatial analysis: a guide for ecologists. Cambridge University Press, Cambridge, N.Y. 365 p.

Frank, R.M. 1990. *Abies balsamea* (L.) Mill. – Balsam fir. P. 26-35 in Silvics of North America, Vol. 1. Conifers, R. Burns and B.H. Honkala (Tech. Coors.). USDA Agriculture Handbook #654.

Fraver, S. and A.S. White. 2005. Disturbance dynamics of old-growth Picea rubens forests of northern Maine. Journal of Vegetation Science. 16:597-610.

Frelich, L.E., S. Sugita, P.R. Reich, M.B. Davis, and S.K. Friedman. 1998. Neighborhood effects in forests: implications for within-stand patch structure. Journal of Ecology. 86:149-161.

Frelich, L.E. 2002. Forest dynamics and disturbance regimes. Cambridge University Press, Cambridge, UK.

Gauch, H.G., Jr. 1982. Multivariate analysis in community ecology. Cambridge University Press.

Gotelli, N.J. and Colwell, R.K. 2001. Quantifying biodiversity: procedures and pitfalls in the measure and comparison of species richness. Ecology Letters. 4:379-391.

Gould, P.J., K.C. Steiner, J.C. Finley, and M.E. McDill. 2005. Developmental pathways following the harvest of oak-dominated stands. Forest Science. 51(1):76-90.

Gratzer, G., C. Canham, U. Dieckmann, A. Fischer, Y. Iwasa, R. Law, M.J. Lexer, H. Sandmann, T.A. Spies, B.E. Splechtna, and J. Szwagrzyk. 2004. Spatio-temporal development of forests – current trends in field methods and models. Oikos. 107:3-15.

Grime. J.P. 1998. Benefits of plant diversity to ecosystems: immediate, filter, and founder effects. Journal of Ecology. 86:902-910.

Grubb, P.J. 1977. The maintenance of species richness in plant communities: the importance of the regeneration niche. Biological Reviews. 52:107-145.

Halliday, W.E.D. 1937. A forest classification for Canada. Department of Mines and Resources, Forest Service, Ottawa, Ontario. Bulletin 89.

Hansen, A.J., T.A. Spies, F.J. Swanson, and J.L. Ohmann. 1991. Conserving biodiversity in managed forests – lessons from natural forests. BioScience. 41:382-392.

Harper J.L., J.N. Clatworthy, I.H. McNaughton, and G.R. Sagar. 1961. The evolution of closely related species living in the same area. Evolution. 15:209-227.

Hoffmeyer, P.V., L.S. Kenefic, and R.S. Seymour. 2009. Northern white-cedar ecology and silviculture in the northeastern United States and southeastern Canada: a synthesis of knowledge.

Houle, G. 1994. Spatio-temporal patterns in the components of regeneration of four sympatric tree species – Acer rubrum, A. saccharum, Betula alleghaniensis and Fagus grandifolia. Journal of Ecology. 82:39-53.

Irland, L.C., A.E. Camp, J.C. Brissette, and Z.R. Donohew. 2006. Long-term silvicultural and ecological studies: results for science and management. Yale University, School of Forestry and Environmental Studies, Global Institute of Sustainable Forestry, GISF Research Paper 005.

Jacobson, G.L., Jr., T. Webb, and E.C. Grimm. 1987. Pattern and rates of vegetation change during the deglaciation of northeastern North America. P. 277-288 in The Geology of North America, v. K-3. North America and adjacent oceans during the last glaciation, W.F. Ruddiman and H.E. Wright, Jr. (eds.). Geological Society of America, Boulder, CO.

Johnson, J.B. and K.S. Omland. 2004. Model selection in ecology and evolution. Trends in Ecology and Evolution. 19:101-108.

Kenefic, L.S., P.E. Sendak, and J.C. Brissette. 2006. Turning data into knowledge for over 50 years: USDA Forest Service research on the Penobscot Experimental Forest. P. 26-33 in Long-term silvicultural and ecological studies: results for science and management, Irland, L.C., A.E. Camp, J.C. Brissette, and Z.R. Donohew (eds.). Yale University, School of Forestry and Environmental Studies, Global Institute of Sustainable Forestry, GISF Research Paper 005.

Kohn, K.A. and J.F. Franklin. 1997. Creating a forestry for the 21st century: the science of ecosystem management. Island Press, Washington D.C.

Leduc, A., P. Drapeau, Y. Bergeron, and P. Legendre. 1992. Study of spatial components of forest cover using partial Mantel tests and path analysis. Journal of Vegetation Science. 3:69-78.

Legendre, P. 1993. Spatial autocorrelation: trouble or new paradigm? Ecology. 74:1659-1673.

Legendre, P. and M.-J. Fortin. 1989. Spatial pattern and ecological analysis. Plant Ecology. 80(2):107-138.

Legendre, P. and L. Legendre. 1998. Numerical ecology, 2nd English edition. Elsevier Science, Amsterdam.

Levin, S. 1992. The problem with pattern and scale. Ecology. 73:1943-1967.

Lichstein, J.W., G.H. Ricardo, and A. Roxana. 2004. Recruitment limitation in secondary forests dominated by an exotic tree. Journal of Vegetation Science. 15:721-728.

LI-COR Inc. 1992. LAI-2000 Plant Canopy Analyzer Operating Manual. LI-COR, Inc. Lincoln, NE.

Lorimer, C.G. 1977. The presettlement forest and natural disturbance cycle of northeastern Maine. Ecology. 58:139-148.

McClure, J.W. and T.D. Lee. 1993. Small-scale disturbance in a northern hardwood forest: effects on tree species abundance and distribution. Canadian Journal of Forest Research. 23:1347-1360.

McCune, B. and M.J. Mefford. 1999. PC-ORD, Multivariate analysis of ecological data Version 4.07, MjM Software Design, Gleneden Beach, Oregon, USA.

McCune, B. and J.B. Grace. 2002. Analysis of ecological communities. MjM Software Design, Gleneden Beach, Oregon, USA.

McDonald, R.I., R.K. Peet, and D.L. Urban. 2003. Spatial pattern of Quercus regeneration limitation and Acer rubrum invasion in a Piedmont forest. Journal of Vegetation Science. 14:441-450.

McCarthy, M.A. 2007. Bayesian methods for ecology. Cambridge University Press.

McEuen, A.B. and L.M. Curran. 2004. Seed dispersal and recruitment limitation across spatial scales in temperate forest fragments. Ecology. 85:507-518.

McInnis, B. and M. Roberts. 1991. The effects of full-tree and tree-length harvests on natural regeneration and seedling microenvironments. P. 85-102 in Proceeding of the conference on natural regeneration management, C.M. Simpson (ed.). Forestry Canada – Maritimes Region. 261 p.

Messaoud, Y. and G. Houle. 2006. Spatial pattern of tree seedling establishment and their relationship to environmental variables in a cold-temperate deciduous forest of eastern North America. Plant Ecology. 185:319-331.

Miina, J. and J. Heinonen. 2008. Stochastic simulation of forest regeneration establishment using a multilevel multivariate model. Forest Science. 54:206-219.

Miller, T.F., D.J. Mladenoff, and M.K. Clayton. 2002. Spatial autocorrelation and patterns of understory vegetation and environment in old-growth northern hardwood forests. Ecological Monographs 72:487-503.

Montes, F. and Cañellas, I. 2007. Analysis of the spatial relationship between the remaining trees from the previous crop and the establishment o and development of saplings in Pinus sylvestris L. stands in Spain. Applied Journal of Vegetation Science. 10:151-160.

Moores, A.R., R.S. Seymour, and L.S. Kenefic. 2007. Height development of shade-tolerant conifer species in multiaged Acadian forest stands. Canadian Journal of Forest Research. 37:2715-2723.

Nagelkerke, N.J.D. 1991. A note on a general definition of the coefficient of determination. Biometrika. 78:691-692.

Nicotra, A.B, R.L. Chazdon, and S.V.B. Iriarte. 1999. Spatial heterogeneity of light and woody seedling regeneration in tropical wet forests. Ecology. 80:1908-1926.

Oliver and Larson. 1996. Forest stand dynamics, updated edition. John Wiley and Sons Inc., New York.

Pacala, S.W., C.D. Canham, J.A. Silander, and R.K. Kobe. 1994. Sapling growth as a function of resources in a north temperate forest. Canadian Journal of Forest Research. 24:2172-2183.

Pardos, M., F. Montes, and I. Cañellas. 2008. Spatial dynamics of natural regeneration in two differently managed Pinus sylvestris stands before and after silvicultural intervention using replicated spatial point patterns. Forest Science. 54:260-272.

Peterson, C.J. and S.T.A. Pickett. 1995. Forest reorganization: a case study in an old-growth forest catastrophic blowdown. Ecology. 76:763-774.

Pickett, S.T.A. and P.S. White. 1985. The ecology of natural disturbance and patch dynamics. Academic Press, Orlando, Florida, USA.

Puettmann, K.J. and P.B. Reich. 1995. The differential sensitivity of red pine and quaking aspen to competition. Canadian Journal of Forest Research. 25:1721-1737.

Puettmann, K.J., K.D. Coates, and C. Messier. 2009. A critique of silviculture: managing for complexity. Island Press, Washington D.C., USA.

R Development Core Team. 2008. R: a language and environment for statistical computing. Version 2.8.1. R Foundation for Statistical Computing, Vienna, Austria. http://www.R-project.org

Rees, M., R. Condit, M. Crawley, S. Pacala, and D. Tilman. 2001. Long-term studies of vegetation dynamics. Science. 293(July 27):650-655.

Rossi, R.E., D.J. Mulla, A.G. Journel, and E.H. Franz. 1992. Geostatistical tools for modeling and interpreting ecological spatial dependence. Ecological Monographs. 62(2):277-314.

Rowe, J.S. 1972. Forest regions of Canada. Pub. 1300. Dept. of the Environment, Canadian Forestry Service, Ottawa. 172 p.

Royo, A.A. and W.P. Carson. 2006. On the formation of dense understory layers in forests worldwide: consequences and implications for forest dynamics, biodiversity, and succession. Canadian Journal of Forest Research. 36:1345-1362.

Runkle, J.R. and T.C. Yetter. 1987. Treefalls revisited: gap dynamics in the southern Appalachians. Ecology. 68:417-424.

Runkle, J.R. 1998. Changes in southern Appalachian canopy tree gaps sampled thrice. Ecology.79:1768-1780.

Saunders, M.R., and Wagner, R.G. 2005. Ten-year results of the Forest Ecosystem Research Program (FERP) – successes and challenges. pp. 147-153. In: Peterson, C.E., and Maguire, D.A. (Eds.) Balancing ecosystem values: innovative experiments for sustainable forestry. U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR. Gen. Tech. Rep. PNW-GTR-635.

Saunders, M.R. and B.G. Wagner. 2008. Long-term spatial and structural dynamics in Acadian mixedwoods stands managed under various silvicultural systems. Canadian Journal of Forest Research. 38:498-517.

Scheller, R.M. and D.J. Mladenoff. 2002. Understory species patterns and diversity in old-growth and managed northern hardwood forests. Ecological Applications. 12(5):1329-1343.

Schnurr, J.L., C.D. Canham, R.S. Ostfeld, and R.S. Inouye. 2004. Neighborhood analyses of small-mammal dynamics: impacts on seed predation and seedling establishment. Ecology. 85:741-755.

Schofield, D.A. 2003. Vegetation dynamics and tree radial growth response in harvest gaps, natural gaps, and closed canopy conditions in Maine's Acadian forest. M.S. Thesis. University of Maine, Orono.

Sendak, P.E., J.C. Brissette, and R.M. Frank. 2003. Silviculture affects composition, growth, and yield in mixed northern conifers: 40-year results from the Penobscot Experimental Forest. Canadian Journal of Forest Research. 33:2116-2128.

Seymour, R.S. 1992. The red spruce and balsam fir forest of Maine: evolution of silvicultural practices in response to stand development patterns and disturbances. P. 217-244 in The ecology and silviculture of mixed-species forests, M.J. Kelty, B.C. Larson, and C.D. Oliver (eds.). Kluwer Academic Publishers.

Seymour, R.S. 1995. The northeastern region. P. 13-79 in Regional silviculture of the United States, 3rd edition. J.W. Barrett (ed.). John Wiley and Sons Inc., New York.

Seymour, R.S., A.S. White, and P.G. deMaynadier. 2002. Natural disturbance regimes in northeastern North America – evaluating and formulating silvicultural systems using natural scales and frequencies. Forest Ecology and Management. 155:357-367.

Seymour, R.S. 2005. Integrating disturbance parameters into conventional silvicultural systems: Experience from the Acadian forest of northeastern North America. P. 41-48 in Balancing ecosystem values: innovative experiments for sustainable forestry, Peterson, C.E., and D.A. Maguire, eds. PNW-GTR-635. USDA Forest Service, Pacific Northwest Research Station, Portland, OR.

Seymour, R.S., Guldin, J., Marshall, D, and Palik, B. 2006. Large-scale, long-term silvicultural experiments in the United States. Allg. Forst. Jagdztg. 177: 104-112.

Smith, D.M., B.G. Larson, M.J. Kelty, and P.M.S. Ashton. 1997. The practice of silviculture: applied forest ecology. 9th ed. John Wiley and Sons Inc., New York.

Sokal, R.R. 1979. Ecological parameters inferred from spatial correlograms. P. 167-196 in G.P. Patil and M.L. Rosenzweig, eds. Contemporary quantitative ecology and related ecometrics. International Co-operative Publ. House, Fairland, MD.

Taylor, A.R., H.Y.H. Chen, and L. VanDamme. 2009. A review of forest succession models and their suitability for forest management planning. Forest Science. 51:23-36.

Urban, D.L. 2003. Spatial analysis in ecology: Mantel's test. URL, http:///www.nceas.ucsb.edu/scicomp/Dloads/SpatialAnalysisEcologists/SpatialEcologyMantelTest.pdf

Wagner, R.G. and S.J. Colombo. 2001. Regenerating the Canadian forest: principles and practices for Ontario. Fitzhenry and Whiteside Limited.

Ward, J.S. and G.R. Parker. 1989. Spatial dispersion of woody regeneration in an old-growth forest. Ecology. 70(5):1279-1285.

Watt, A.S. 1947. Pattern and process in the plant community. Journal of Ecology. 35:1-22.

Westveld, M. 1931. Reproduction on the pulpwood lands in the northeast. USDA Technical Bulletin 223.

Westveld, M. 1953. Ecology and silviculture of the spruce-fir forests of eastern North America. Journal of Forestry. 51:422-430.

Wills, C., R. Condit, R.B. Foster, and S.P. Hubbell. 1997. Strong density- and diversity-related effects help to maintain tree species diversity in a neotropical forest. Proceedings of the National Academy of Sciences. 94:1252-1257.

Wilson, E.O. 1988. Biodiversity. National Academy Press.

Wyckoff, P.H. and S.L. Webb. 1996. Understory influence of the invasive Norway maple (*Acer platanoides*). Bulletin of the Torrey Botanical Club. 123:197-205.

APPENDICES

APPENDIX A

DISTRIBUTION OF SAMPLE PLOTS BY HDC

Table A.1. Percent breakdown of harvest disturbance clusters (HDC) according to plot membership by silvicultural treatments. Number in parentheses is the actual number of plots used to derive percentages.

Silviculture* Treatment	Percentage of HDC Type (n)		
	Low	Medium	High
FDL	2.5 (2)	5.7 (4)	17.5 (14)
MDL	6.2 (5)	14.3 (10)	16.3 (13)
CCC	0	0	25 (20)
SW2	0	14.3 (10)	25 (20)
SW3	0	17.2 (12)	6.3 (5)
SW3p	0	5.7 (4)	8.7 (7)
S20	19.8 (16)	28.6 (20)	1.2 (1)
S10	37.0 (30)	7.1 (5)	0
S05	34.6 (28)	7.1 (5)	0

* Silviculture treatment abbreviations are: FDL = fixed diameter-limit; MDL = modified diameter-limit; CCC = commercial clear-cutting; SW2 = two-stage shelterwood; SW3 = 3-stage shelterwood; SW3p = three-stage shelterwood with precommerical thinning; S20 = 20-year single-tree selection; S10 = 10-year single-tree selection; S05 = 5-year single-tree selection

APPENDIX B

VARIOGRAPHY RESULTS

Variography started with fitting theoretical variogram models to the semi-variance structures (empirical variogram) of tree regeneration density and canopy openness individually for each stand (Table B.1). I then selected the theoretical model with the lowest mean sum of squares to represent spatial dependence. Spatial dependence was estimated by nugget, sill, and range parameters from "best fit" theoretical models (Table B.2).

Table B.1. Comparison of spherical (Sph), exponential (Exp), and Gaussian (Gau) theoretical variogram model fits (mean sum of squares; MSS) estimated by weighted least squares for both tree regeneration density and canopy openness across eight stands. Bold and underlined MSS values correspond to theoretical models used to estimate spatial dependence.

Stand	**Tree Regeneration Density**			**Canopy Openness**		
	Sph	**Exp**	**Gau**	**Sph**	**Exp**	**Gau**
RA4w	**19.5**	23.2	21.6	102.9	**41.5**	53.1
RA4e	32.8	**30.3**	32.5	131.3	**79.2**	93.8
RA5	78.5	**59.8**	79.8	**135.3**	274.8	241.6
RA6	33.8	**29.3**	33.6	**44.4**	116.1	107.7
RA7n	**42.8**	51.1	45.6	**63.9**	98.3	100.1
RA7s	**43.5**	44.4	54.7	140.3	**116.0**	151.3
RA8	72.3	**62.2**	76.4	**66.0**	91.1	75.8
RA9	35.1	**33.1**	38.9	55.6	**50.3**	75.0

Table B.2. Nugget, sill, and range parameters from best fit theoretical variogram models estimated by weighted least squares for both tree regeneration density and canopy openness across eight stands.

Stand	**Tree Regeneration Density**			**Canopy Openness**		
	Nugget	**Sill**	**Range**	**Nugget**	**Sill**	**Range**
RA4w	0.9	2.93	13.3	0.001	0.007	11.25
RA4e	0.76	2.33	8.58	0.002	0.009	16.5
RA5	0.63	2.03	18.78	0.001	0.01	21.1
RA6	0.7	1.37	34	0.005	0.015	31.0
RA7n	0.46	2.61	20.93	0.002	0.011	16.99
RA7s	0.42	1.99	22.94	0.006	0.015	25.31
RA8	0.73	3.12	14.95	0.006	0.014	12.9
RA9	0.69	1.66	25.69	0.006	0.014	28.01

APPENDIX C

DISTRIBUTION OF BALSAM FIR REGENERATION

Table C.1. Mean density (Mean BFr) and percent stocking (Freq BFr) of balsam fir regeneration centered on 7-quantile means of overstory basal area (OBA), depth-to-watertable (DWT), and percent canopy transmittance (%T).

Correlate		**7-quantile bin**						
		1st	**2nd**	**3rd**	**4th**	**5th**	**6th**	**7th**
a) OBA	Mean ($m^2 \bullet ha^{-1}$)	20	24	32	32	36	40	48
Mean BFr (stems $\bullet$ ha^{-1})		14,131	11,337	12,325	10,334	10,307	10,000	8,441
Freq BFr (%)		82.4	81.8	81.3	77.5	73.8	78.6	66.7
b) DWT	Mean (m)	0.89	1.65	2.34	2.85	3.51	4.85	6.99
Mean BFr (stems $\bullet$ ha^{-1})		12,072	10,709	8,333	9,104	11,484	12,164	10,000
Freq BFr (%)		80.7	74.9	70.6	67.4	75.4	84.0	79.1
c) %T	Mean (%)	2.02	2.83	3.89	4.54	5.36	8.02	13.77
Mean BFr (stems $\bullet$ ha^{-1})		11,230	8,831	9,572	10,615	11,283	11,815	10,521
Freq BFr (%)		71.1	72.1	71.1	81.8	81.8	78.1	75.9

APPENDIX D

MODEL SELECTION RESULTS

Table D.1. AIC for models of (A) small and (B) large regeneration richness estimated by LME (all RAs) and OLS (each RA separately) regression.

	Candidate Model	All RAs (LME)	RA4w	RA4e	RA8	RA5	RA7n	RA7s	RA6	RA9
A)	Global	3850.8	389.19	363.66	484.85	793.62	70.74	276.17	658.16	557.84
	BFr + OBA + DWT	3862.9	389.01	365.71	485.63	802.90	68.82	274.35	656.40	556.48
	BFr + OBA + %T	3842.4	387.19	361.75	483.10	794.56	70.99	276.28	656.18	558.20
	BFr + DWT + %T	3850.0	388.29	361.67	482.85	797.77	69.40	274.47	662.61	567.46
	OBA + DWT + %T	3892.7	394.84	361.85	491.21	801.12	68.74	275.97	671.45	567.00
	BFr + OBA	3854.5	387.01	363.94	483.85	802.64	69.12	274.28	654.42	556.01
	BFr + DWT	3857.0	388.52	364.31	483.65	802.45	67.45	272.67	660.71	565.83
	BFr + %T	3840.6	386.29	359.77	481.10	799.51	69.99	274.32	660.81	567.15
	OBA + DWT	3906.9	393.96	363.72	490.82	810.88	66.82	274.06	671.69	567.56
	OBA + %T	3885.3	392.84	359.96	489.23	805.91	69.06	278.00	669.77	569.09
	DWT + %T	3903.8	393.60	359.86	489.78	809.75	67.43	274.01	680.68	579.96
	BFr	3848.7	386.52	362.55	481.86	802.73	68.21	272.12	658.92	565.39
	OBA	3899.4	391.96	361.96	488.84	813.77	67.18	276.13	670.14	570.54
	DWT	3910.2	393.01	362.31	489.22	813.06	65.47	272.13	679.04	578.39
	%T	3896.7	391.60	357.97	487.82	817.63	67.97	276.30	680.93	581.28
B)	Global	4388.1	358.21	342.37	581.74	529.45	287.37	424.48	731.73	733.81
	BFr + OBA + DWT	4581.5	375.83	348.07	593.12	578.57	293.09	431.38	758.39	774.94
	BFr + OBA + %T	4374.1	356.21	343.26	584.47	532.77	285.38	425.27	735.81	744.03
	BFr + DWT + %T	4379.8	358.57	342.27	580.01	527.48	286.56	427.58	729.82	735.80
	OBA + DWT + %T	4387.0	358.15	340.41	580.70	534.72	285.78	422.63	732.33	731.83
	BFr + OBA	4576.0	373.84	348.08	595.28	578.38	292.35	435.59	762.33	785.89
	BFr + DWT	4596.6	377.85	351.28	591.61	583.81	292.59	434.43	758.28	772.94
	BFr + %T	4373.6	356.57	343.13	582.55	530.93	284.57	430.59	734.56	745.06
	OBA + DWT	4577.3	376.72	346.50	593.17	581.20	292.19	429.79	756.43	778.39
	OBA + %T	4381.4	356.16	340.37	584.01	535.32	283.81	424.30	735.20	742.48
	DWT + %T	4381.1	358.72	341.32	579.36	532.96	285.44	426.54	730.33	734.01
	BFr	4590.8	375.86	351.04	593.51	582.67	292.35	442.06	761.02	784.00
	OBA	4572.2	374.73	346.60	596.06	579.82	291.18	436.42	760.40	792.69
	DWT	4597.4	379.13	349.89	592.47	589.99	292.57	433.95	756.45	776.41
	%T	4381.9	356.72	341.24	582.41	533.36	283.45	432.99	733.52	743.94

APPENDIX E

SPECIES LIST OF TREE REGENERATION

Table E.1. List of tree species tallied in understory plots along with stands (RAs) and treatments where they were found for the AFERP study.

Scientific name	Common name	RAs found	Treatments found [1]
Abies balsamea	Balsam fir	All	All
Acer pensylvanicum	Striped maple	5,6,8	All
Acer rubrum	Red maple	All	All
Acer saccharum	Sugar maple	4,6,8,9	C,LG
Betula alleghaniensis	Yellow birch	4,5,6,8	All
Betula papyrifera	Paper birch	All	All
Betula populifolia	Grey birch	6,9	LG
Fagus grandifolia	American beech	All	All
Fraxinus americana	White ash	All	All
Fraxinus nigra	Black ash	4,5,7,9	All
Ostrya virginiana	Eastern hophornbeam	4,5,6,8	All
Picea glauca	White spruce	All	All
Picea rubens	Red spruce	All	All
Pinus resinosa	Red pine	5,6,9	SG,LG
Pinus strobus	White pine	All	All
Populus grandidentata	Bigtooth aspen	All	All
Populus tremuloides	Trembling aspen	All	All
Prunus pensylvanica	Red cherry	6,7,9	SG,LG
Prunus serotina	Black cherry	4,6,7,9	All
Quercus rubra	Red oak	All	All
Thuja occidentalis	Northern white-cedar	All	All
Tilia americana	American basswood	4,6	C,LG
Tsuga canadensis	Eastern hemlock	All	All
Ulmus americana	American elm	5	SG

[1] Treatments codes are: C = control, SG = small gap, and LG = large gap.

APPENDIX F

LME SLOPES FOR NESTED STAND-LEVEL MODELS

The degree of overlap seen among 95 % confidence intervals for slopes and intercepts over stand-level models nested within small and large regeneration LME models suggested minimal deviance in parameters across stands and treatments (Figure E.1); Therefore, there was weak evidence of treatment effects on model parameters.

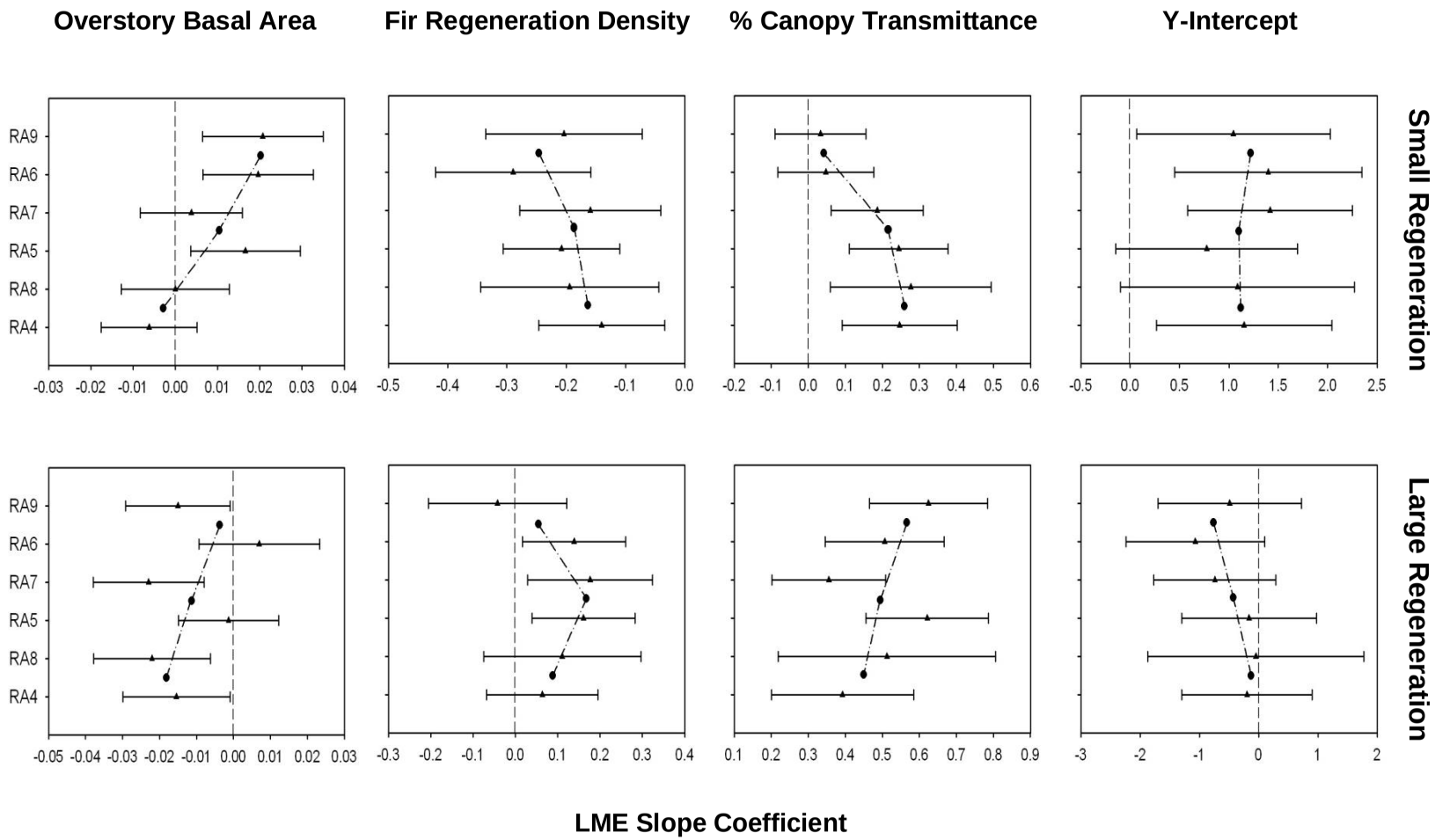

Figure F.1. Estimated slope coefficients ± two standard deviations from LME models presented over the treatment gradient of the AFERP study.

BIOGRAPHY OF THE AUTHOR

Matt was born in North Jersey on August 18, 1977. At age 5 or so, Matt and his family made the long trip down the Turnpike from exit 11 to 7 where they re-settled in South Jersey. This event, more than anything, probably had the greatest influence on Matt's interest in the outdoors.

The Olson family relocated to Medford; a town in the Pine Barrens of southern New Jersey. As a kid, Matt spent much of his free time in the woods near the family home. Before long, the NJ Pine Barrens became a passion of Matt's. In fact, much of Matt's spare time was spent exploring the Pine Barrens and, of course, removing ticks from skin and clothes.

While scanning colleges, Matt discovered a small school called Paul Smith's College that offered programs in forestry, which, until that point, he didn't consider a career option. That following fall (1997), Matt enrolled in the forestry program at "Smittys" and quickly become interested in forest ecology and silviculture. From PSC, Matt transferred to UMaine in 1999 to continue his forestry education. After receiving his Bachelor's degree from UMaine, Matt went to the UTenn to study oak regeneration and silviculture for his Master's degree. From there, Matt spent time at Penn State working towards a Ph.D., but wound up transferring back to UMaine in pursuit of his doctorate. Since then, Matt has been engaged in many research projects at UMaine.

Matt currently lives with his wife, Liz, and dog, Hank Williams, in a house on the University Forest known affectionately as "the deer pen". He is a candidate for the Doctor of Philosophy Degree in Forest Resources from UMaine in December 2009.